

WHEN IS EVERY RATIONAL HODGE CLASS ALGEBRAIC?

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ABSTRACT. We determine which combinations of the standard statements around the rational Hodge conjecture imply it. Every such combination contains the conjecture modulo abelian varieties, unless it contains both the Lefschetz standard conjecture and André’s conjecture that every Hodge class is motivated, a pair equivalent to the Hodge conjecture itself. Granting the recently claimed Hodge conjecture for CM abelian varieties, we prove it for every Fermat variety and reduce the abelian case to propagation from CM points. For Weil classes we prove a criterion by bounded degree at CM points, and we extract from Schoen’s cycles one subvariety whose semiregularity would settle the split Weil families over $\mathbb{Q}(\sqrt{-3})$ in every dimension. Without any hypothesis, we prove the Hodge conjecture for Fermat fourfolds of every odd degree; for degrees prime to six this rests on a classification of their Hodge characters that is also checked in Lean. The Hodge conjecture itself is not proved.

1. INTRODUCTION

Let X be a smooth complex projective variety. A class in $H^{2p}(X, \mathbb{Q})$ is a *Hodge class* if its image in $H^{2p}(X, \mathbb{C})$ lies in $H^{p,p}(X)$, and it is *algebraic* if it is a rational linear combination of the classes of subvarieties of codimension p . Algebraic classes are Hodge classes. Hodge asked for the converse [Hod52].

Conjecture 1.1 (Hodge). *For every smooth complex projective variety X and every $p \geq 0$, every Hodge class in $H^{2p}(X, \mathbb{Q})$ is algebraic.*

We write (HC) for this statement. It holds for $p \leq 1$ and $p \geq \dim X - 1$, so for every variety of dimension at most three, and it is open in general. The question in the title has a precise form: *from which of the statements that the literature has isolated does (HC) follow?* This paper answers that question, proves the conditional theorems that the answer allows, and stops at the first statement on each route that we cannot prove. We do not prove the Hodge conjecture, and no statement below should be read as a claim that it holds.¹

The statements. The following statements are made precise in Definition 2.7.

- (F2) The Hodge conjecture for abelian varieties.
- (F3') The Hodge conjecture modulo abelian varieties: every Hodge class on every X is a sum of classes $\Gamma_*\gamma$, where γ is a Hodge class on an abelian variety B and Γ is an algebraic cycle on $B \times X$.

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Source, figures and verification code: <https://github.com/creelie/e84hc>, archived on Zenodo under the concept DOI 10.5281/zenodo.23234362; release v1.2.0, the first with Theorem F for every odd degree, has the version DOI 10.5281/zenodo.23272739, and release v1.3.0 has 10.5281/zenodo.23272887.

¹Here and below a statement is called *proved* when it is proved in this paper or in refereed work. Results that are available only as preprints are marked as such where they are used. The one preprint on which a theorem of this paper depends, [Ope26b], enters only as an explicit hypothesis.

- (L) The Lefschetz standard conjecture $B(X)$ of Grothendieck [Gro69], for every X .
- (M) Every Hodge class is motivated in the sense of André [And96].
- (V) The variational Hodge conjecture: in a smooth projective family over a smooth connected base, a flat family of rational classes that is algebraic on one fibre is algebraic on every fibre.
- (CM) The Hodge conjecture for abelian varieties of CM type.
- (IP) Propagation from CM points: (V) for polarized abelian schemes and classes that are algebraic on a fibre of CM type.

The statement (CM) is the main theorem claimed in the preprint [Ope26b], which has not been refereed. We have not verified its proof, and Section 4.1 says which parts of it we have checked.

Theorem A. *The following are equivalent:*

- (i) (HC);
- (ii) (F2) and (F3');
- (iii) (L) and (F3');
- (iv) (V) and (F3');
- (v) (CM), (IP) and (F3');
- (vi) (L) and (M).

The implications from (ii)–(vi) to (i) combine known theorems of André, Deligne, Abdulali and Milne with two observations of this paper: (F3') implies (M) (Proposition 2.9), and (CM) together with (IP) implies (F2) (Proposition 4.4). The theorem is proved in Section 4.3. The next theorem says that these are all the ways in which the conjecture follows from the statements above, together with the other statements the paper meets: the semiregularity of Schoen's subvariety (Question 5.14), and the Hodge conjecture for Fermat varieties and for the Weil classes of the split Weil families over $\mathbb{Q}(\sqrt{-3})$.

Theorem B (The routes). *Let S be a set of the statements of Table 1 that does not contain (HC), and suppose that (HC) follows from S by the implications of Table 1. Then*

- (a) *either S contains (F3') together with one of (F2), (L), (V), or both of (CM) and (IP);*
- (b) *or S contains both (L) and (M).*

Each of the five minimal sets so described implies (HC), and each implies every statement of Table 1 except the semiregularity of Schoen's subvariety.

So there are two ways in. One passes through abelian varieties and needs (F3') together with a statement that implies (F2); the other avoids abelian varieties and assumes (L) and (M), which by Theorem A is the Hodge conjecture in another form. In particular (F2) and (F3') can be avoided as hypotheses, but never as consequences, and Schoen's subvariety lies on no route at all (Corollary 6.2). The routes are drawn in Figure 1.

The remaining results are conditional on one statement, or prove one case. Let $X_m^n \subseteq \mathbb{P}^{n+1}$ be the Fermat variety $x_0^m + \cdots + x_{n+1}^m = 0$.

Theorem C. *Assume (CM). Then the Hodge conjecture holds for every Fermat variety X_m^n , for every product of Fermat varieties of any degrees and dimensions, and for every smooth projective variety onto which such a product maps surjectively. Without any hypothesis, (F3') holds for all of these varieties.*

Fermat varieties are the first varieties on which the conjecture was studied in every dimension [Ran80, Shi79], and unconditional proofs are known only for special degrees (Remark 3.7); Theorem F below adds the fourfolds of odd degree. The proof of Theorem C uses the inductive structure of Shioda and Katsura [SK79] to dominate every Fermat variety by a product of

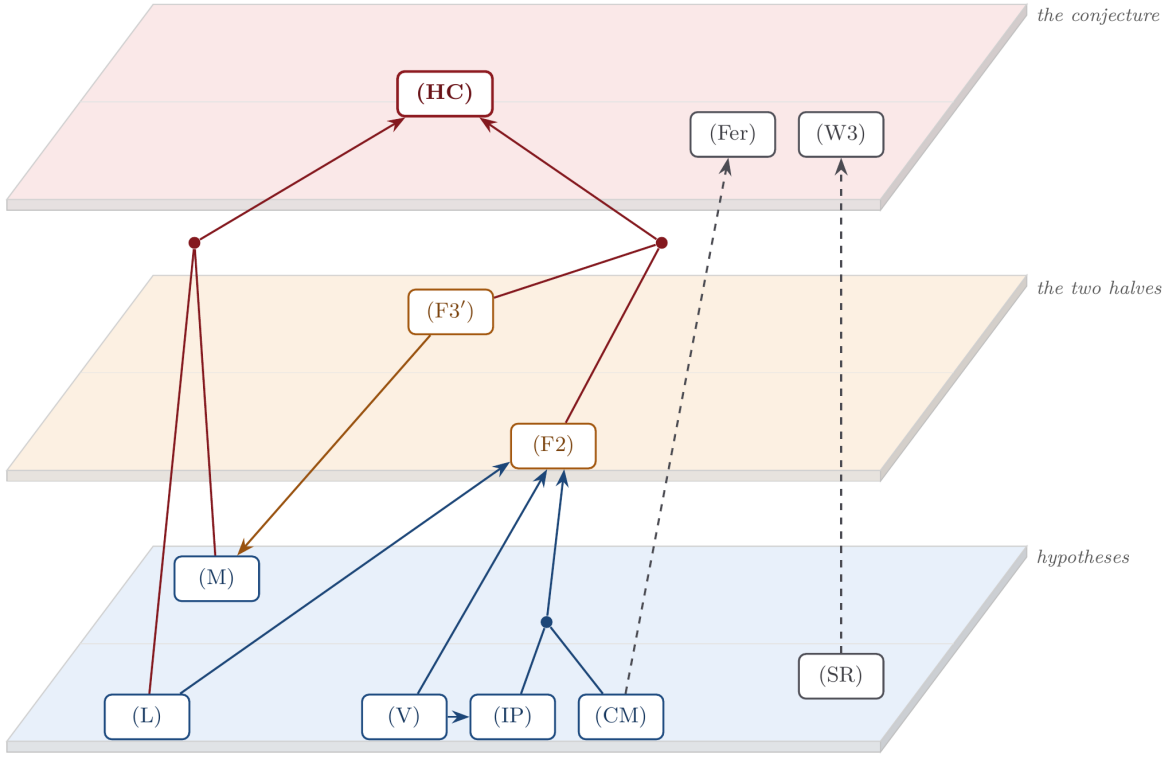


FIGURE 1. The routes of Theorem B. Hypotheses lie on the bottom layer, the two halves (F2) and (F3') in the middle, and (HC) on top. Arrows are implications of Table 1. At each dot two lines combine into a conjunction: $(CM) \wedge (IP) \Rightarrow (F2)$, $(L) \wedge (M) \Rightarrow (HC)$ and $(F2) \wedge (F3') \Rightarrow (HC)$. The arrows from (HC) and (F2) to the statements they contain are not drawn. The dashed arrows on the right lead to the special cases (Fer) and (W3) of Section 6, and not to (HC).

Fermat curves, blown up along smooth centres, and the fact that Fermat curves have Jacobians of CM type.

Abelian varieties of Weil type are where (F2) is hardest. On the split Weil family of an imaginary quadratic field over its Shimura variety S (Section 4.4), write $\Sigma \subseteq S$ for the set of members on which the Weil classes are algebraic, and $\Sigma_d \subseteq \Sigma$ for the set where they have algebraic representatives of degree at most d (Definition 4.5).

Theorem D. *Each Σ_d is Zariski closed in S , and $\Sigma = \bigcup_d \Sigma_d$. The following are equivalent:*

- (i) $\Sigma = S$;
- (ii) $\Sigma_d = S$ for some d ;
- (iii) for some d , the CM points in Σ_d are not contained in any finite union of proper special subvarieties of S .

Under (CM) every CM point of S lies in Σ .

So under (CM) the Weil classes are algebraic on the whole family exactly when, at the CM points, they have algebraic representatives of uniformly bounded degree on a set that is not trapped in finitely many proper Shimura subvarieties. The equivalence of (ii) and (iii) is where the André–Oort conjecture, proved by Pila, Shankar and Tsimerman [PST21], enters.

The last result concerns the only cycles known to represent Weil classes at members with maximal Hodge group in every dimension, those constructed by Schoen [Sch88]. Let X be a curve of genus $n + 1 \geq 3$, let $C \rightarrow X$ be the étale cyclic triple cover given by a line bundle of order three, and let B be its Prym variety, an abelian variety of dimension $2n$ of Weil type for $\mathbb{Q}(\sqrt{-3})$ with polarization class η .

Theorem E. *There is an irreducible subvariety $Y \subseteq B$ of dimension n , built from one of the three components into which Schoen's cover splits over the canonical system of X , whose class pairs nontrivially with the Weil classes of B . If the Hodge group of B is maximal, which holds for very general (X, C) , then*

$$\mathrm{cl}(Y) = c\eta^n + w, \quad c \in \mathbb{Q}_{>0}, \quad 0 \neq w \text{ a Weil class.}$$

If Y is semiregular for one such B , the Weil classes are algebraic on every member of the split Weil family of $\mathbb{Q}(\sqrt{-3})$ in dimension $2n$.

For $n \leq 3$ this family is already covered by Schoen's work [Sch88, Sch98]. For $n \geq 4$ it is not: the Prym locus has dimension $3n$ and the family has dimension n^2 , so Y would have to deform in $n^2 - 3n$ further directions. The proof of the pairing in Theorem E is a short computation with twisted coefficients on the symmetric product of X , and it also gives a proof of Schoen's theorem that B carries algebraic Weil classes (Corollary 5.7). We give the evidence on both sides of Question 5.14 in Section 5.6, including a precise analogy with the Abel–Jacobi curve, which is not semiregular, a description of Y as a Brill–Noether locus of the Prym variety, smooth of dimension n at its general point, whose analogue for double covers is Mumford's Prym theta divisor, and a question of Markman about a weak form of the semiregularity theorem.

The last result is unconditional.

Theorem F. *Let m be odd. Then the Hodge conjecture holds for the Fermat fourfold X_m^4 .*

By Proposition 3.6 the Hodge classes of X_m^4 are spanned by lines indexed by sextuples of nonzero residues modulo m . When m is prime to 3, we show that each such sextuple contains two entries a and $-a$, or is, up to order, $(x, x + m/5, x + 2m/5, x + 3m/5, x + 4m/5, -5x)$ (Theorem 3.9). The classes of the first kind are algebraic by the inductive structure of Shioda and Ran, and those of the second kind by the cycles of Aoki [Aok87], pulled back along a covering of Fermat fourfolds. The proof of the classification uses the Dirichlet characters modulo the largest order of an entry, and its only analytic input is the classical fact that $B_{1,\chi} \neq 0$ for odd primitive χ . When $3 \mid m$ the classification fails, and we argue by induction instead (Section 3.7): a sextuple without such a pair either has a *move*, which trades it, by means of one cycle of Aoki, for a quadruple or for a sextuple of smaller orders, or it lies in one of three explicit orbits, in degrees 21, 33 and 39, whose classes we obtain on the Fermat fourfolds of degrees 21, 66 and 78. When m is prime to 15, Theorem F follows from Aoki's theorem that every Hodge character then splits into pairs [Aok83]; it is also known when m is a power of a prime [Aok87] and when $m = 3^a 5^b 7^c$ [Aok00, Pet26]. In the remaining degrees it is new for $m > 199$, apart from a statement of Kang whose proof has a gap (Remark 3.21); smaller degrees were covered by computer searches (Remark 3.7). The classification for m prime to 6, together with the fact that $B_{1,\chi} \neq 0$, has been checked in Lean 4 with Mathlib (Appendix A).

What is not proved. None of the statements (F2), (F3'), (L), (M), (V), (IP) is proved here, and (CM) is used only as a hypothesis. The Hodge conjecture is open, and by Corollary 6.2 every set of these statements from which it follows also implies (F2) and (F3'). Section 7 lists what is proved and what is not, and names the first open cases: for (F2), the Weil classes of nonsplit abelian sixfolds and of abelian eightfolds, and the exceptional classes on the square of

a Mumford fourfold; for (F3'), the square of a K3 surface with real multiplication. Theorem F does not reach Fermat fourfolds of even degree: there the method breaks down, because $m/2$ is its own negative and the subgroups K_2 of Section 3.6 are too small, and Hodge characters of further kinds occur [Aok83].

Organization. Section 2 proves the reductions that hold for every variety and defines the statements. Section 3 proves Theorem C and Theorem F. Section 4 treats (CM) and (IP), proves Theorem A, and proves Theorem D. Section 5 proves Theorem E and the transfer theorem for Weil classes. Section 6 proves Theorem B. Section 7 collects what is proved and what is not, and Appendix A describes the machine checks of the finite steps. No proof in this paper depends on a computer.

Conventions. Varieties are complex, smooth, projective and, unless said otherwise, connected. Cohomology is singular cohomology with rational coefficients. For a variety X we write $\text{Hdg}^p(X) \subseteq H^{2p}(X, \mathbb{Q})$ for the Hodge classes and $\text{Alg}^p(X) \subseteq \text{Hdg}^p(X)$ for the algebraic classes. Our Tate twist is $V(r)^{a,b} = V^{a+r,b+r}$, so that $V(r)$ has weight $\text{wt}(V) - 2r$.

2. REDUCTIONS THAT HOLD FOR EVERY VARIETY

2.1. Correspondences. Let B and X be varieties, possibly disconnected, with B of pure dimension b . For an algebraic cycle Γ of codimension c on $B \times X$ with rational coefficients, the map

$$\Gamma_*: H^k(B, \mathbb{Q}) \rightarrow H^{k+2c-2b}(X, \mathbb{Q}), \quad y \mapsto \text{pr}_{X*}(\text{cl}(\Gamma) \cup \text{pr}_B^* y),$$

is a morphism of Hodge structures $H^k(B)(b-c) \rightarrow H^{k+2c-2b}(X)$. It carries algebraic classes to algebraic classes, the composite of two such maps is the map of the composite correspondence, and for a morphism $f: B \rightarrow X$ the graph Γ_f and its transpose give f_* and f^* [Ful98, Chapter 16]. For a cycle with components of several codimensions we add the maps of the components.

A pure rational Hodge structure V of weight w is *polarizable* if it carries a bilinear form Q such that $Q(Cx, \bar{x}) > 0$ for $x \neq 0$, where C acts on $V^{a,b}$ by i^{a-b} . The cohomology of a variety and its Tate twists are polarizable, by the Hodge–Riemann relations [Voi02, Chapter 6]. Polarizable Hodge structures are semisimple.²

Lemma 2.1. *Let $\varphi: V \rightarrow W$ be a surjective morphism of polarizable Hodge structures of weight $2p$. Then φ maps the Hodge classes of V , the rational classes of type (p, p) , onto those of W .*

Proof. By semisimplicity the kernel of φ has a complement V' that is a sub-Hodge structure, and φ restricts to an isomorphism of Hodge structures $V' \rightarrow W$. Its inverse carries a Hodge class of W to a Hodge class of $V' \subseteq V$. $\square$

Lemma 2.2 (Dominated cohomology). *Let X be a variety, and let $B_1, \dots, B_r$ be varieties with algebraic cycles Γ_j on $B_j \times X$, such that the images of the maps Γ_{j*} span $H^{2p}(X, \mathbb{Q})$. Then every Hodge class of degree $2p$ on X is a sum $\sum_j \Gamma_{j*} \gamma_j$ with γ_j Hodge classes on B_j . In particular, if the Hodge conjecture holds for every B_j , it holds for X in degree $2p$.*

Proof. Write $\Gamma_j = \sum_c \Gamma_{j,c}$ with $\Gamma_{j,c}$ of pure codimension c , and $b_j = \dim B_j$ (if B_j is not equidimensional, treat its components separately). The component $\Gamma_{j,c}$ maps into degree $2p$ exactly from $H^k(B_j)$ with $k = 2p + 2b_j - 2c$, which is even. The sum of the maps

$$\bigoplus_{j,c} H^{2p+2b_j-2c}(B_j, \mathbb{Q})(b_j - c) \longrightarrow H^{2p}(X, \mathbb{Q})$$

²If $W \subseteq V$ is a sub-Hodge structure, the restriction of Q to W is again a polarization, hence nondegenerate, so $V = W \oplus W^\perp$. The orthogonal $W^\perp$ is a sub-Hodge structure, because the decomposition $V_{\mathbb{C}} = \bigoplus V^{a,b}$ is orthogonal for the hermitian form $Q(Cx, \bar{y})$ and $W_{\mathbb{C}}$ is the sum of its intersections with the $V^{a,b}$.

is a surjective morphism of polarizable Hodge structures of weight $2p$. A Hodge class in $H^{2p+2b_j-2c}(B_j)(b_j - c)$ is a Hodge class of B_j . Apply Lemma 2.1. Algebraic classes go to algebraic classes under correspondences, which gives the last assertion. $\square$

2.2. Degree two and hard Lefschetz.

Theorem 2.3 (Lefschetz [Lef24], Kodaira–Spencer [KS53]). *Every Hodge class of degree two is algebraic.*

Proof. The exponential sequence gives an exact sequence

$$H^1(X, \mathcal{O}_X^\times) \rightarrow H^2(X, \mathbb{Z}) \rightarrow H^2(X, \mathcal{O}_X),$$

whose second map is the composite of $H^2(X, \mathbb{Z}) \rightarrow H^2(X, \mathbb{C})$ with the projection to $H^{0,2}(X)$. An integral class of type $(1, 1)$ therefore is the first Chern class of a holomorphic line bundle, which is algebraic by GAGA, and its first Chern class is the class of a divisor. A rational Hodge class has an integral multiple. $\square$

Let h be the class of a hyperplane section of X , of dimension n , and L the cup product with h . By the hard Lefschetz theorem $L^{n-k}: H^k(X, \mathbb{Q}) \rightarrow H^{2n-k}(X, \mathbb{Q})$ is an isomorphism for $k \leq n$ [Voi02, Theorem 6.25].

Proposition 2.4. *For $2p < n$, if every Hodge class of degree $2p$ on X is algebraic, so is every Hodge class of degree $2n - 2p$. Every Hodge class of degree $0, 2, 2n - 2$ or $2n$ is algebraic, and (HC) holds for every variety of dimension at most three.*

Proof. L^{n-2p} maps $H^{p,p}$ onto $H^{n-p,n-p}$ and rational classes to rational classes, so it maps $\text{Hdg}^p(X)$ onto $\text{Hdg}^{n-p}(X)$; and it maps the class of a subvariety Z to the class of $Z \cap H_1 \cap \cdots \cap H_{n-2p}$ for general hyperplanes H_i . Hence $\text{Hdg}^{n-p}(X) = L^{n-2p} \text{Alg}^p(X) \subseteq \text{Alg}^{n-p}(X)$. Degrees 0 and $2n$ are spanned by the fundamental class and the class of a point; degree 2 is Theorem 2.3, and degree $2n - 2$ follows from it. For $n \leq 3$ every even degree is one of these. $\square$

2.3. Surjective morphisms and blow-ups.

Proposition 2.5. *Let $f: Z \rightarrow X$ be a surjective morphism, with X connected, and let $r = \dim Z - \dim X$. Let h_Z be an ample class on Z , and $d > 0$ the degree of h_Z^r on a general fibre. Then $\alpha = d^{-1} f_*(h_Z^r \cup f^* \alpha)$ for every $\alpha \in H^*(X, \mathbb{Q})$. In particular the correspondence $\Gamma = d^{-1}(h_Z^r \times 1) \cdot \Gamma_f$ from Z to X satisfies $\Gamma_* f^* = \text{id}$, so Γ_* is surjective, and (HC) for Z implies (HC) for X .*

Proof. By the projection formula $f_*(h_Z^r \cup f^* \alpha) = f_*(h_Z^r) \cup \alpha$, and $f_*(h_Z^r) \in H^0(X, \mathbb{Q}) = \mathbb{Q}$ is the degree d , which is positive because h_Z is ample on the fibres. The last claim follows from Lemma 2.2. $\square$

Proposition 2.6. *Let $Y \subseteq X$ be a smooth closed subvariety, possibly disconnected, of codimension $c \geq 2$ in every component, let $\pi: \tilde{X} \rightarrow X$ be the blow-up along Y , let $j: E \rightarrow \tilde{X}$ be the exceptional divisor and $q: E \rightarrow Y$ the projective bundle, and let $\xi = c_1(\mathcal{O}_E(1))$. Then*

$$H^k(\tilde{X}, \mathbb{Q}) = \pi^* H^k(X, \mathbb{Q}) \oplus \bigoplus_{i=0}^{c-2} j_*(\xi^i \cup q^* H^{k-2i-2}(Y, \mathbb{Q})),$$

and each summand is the image of a correspondence: π^* from X , and $y \mapsto j_*(\xi^i \cup q^* y)$ from Y .

Proof. The decomposition is the blow-up formula [Voi02, Theorem 7.31]. The map $y \mapsto j_*(\xi^i \cup q^* y)$ is the correspondence given by the image in $Y \times \tilde{X}$ of a cycle on E representing ξ^i , pushed forward along $(q, j): E \rightarrow Y \times \tilde{X}$. $\square$

2.4. The statements.

Definition 2.7. Let X range over all varieties and p over all integers.

- (1) (F2): the Hodge conjecture for every complex abelian variety.³
- (2) (F3'): $\text{Hdg}^p(X) = \text{Ab}^p(X)$, where $\text{Ab}^p(X)$ is the span of the classes $\Gamma_*\gamma$ of degree $2p$ with B an abelian variety (the point included), γ a Hodge class on B and Γ an algebraic cycle on $B \times X$ [BMB26, Definition 20.34].
- (3) (L): for every X and every ample class on X , the Lefschetz involution $*_L$ defined below is induced by an algebraic cycle on $X \times X$. This is Grothendieck's standard conjecture $B(X)$ for every X [Gro69]: by [Kle68, 1.4.4] and [And96, Proposition 1.2], the $\mathbb{Q}$ -algebra of operators generated by L and $*_L$ is the one generated by L and the operator Λ of $B(X)$.
- (4) (M): every Hodge class is motivated in the sense of [And96, §1].
- (5) (V): for every smooth projective morphism $f: \mathcal{X} \rightarrow S$ over a smooth connected quasi-projective variety, and every global section ξ of $R^{2p}f_*\mathbb{Q}$ over S^{an} , if ξ_s is algebraic for one $s \in S$ then ξ_t is algebraic for every $t \in S$.
- (6) (CM): the Hodge conjecture for every abelian variety of CM type.
- (7) (IP): the case of (V) in which f is a polarized abelian scheme and $\mathcal{X}_s$ is of CM type.

An abelian variety A is *of CM type* if $\text{End}(A) \otimes \mathbb{Q}$ contains a commutative semisimple $\mathbb{Q}$ -algebra of dimension $2 \dim A$; the point is of CM type, and products of abelian varieties of CM type are of CM type.

Each $\Gamma_*\gamma$ in (F3') is a Hodge class, so $\text{Ab}^p(X) \subseteq \text{Hdg}^p(X)$; taking B a point shows that $\text{Ab}^p(X)$ contains $\text{Alg}^p(X)$. Thus (F3') asks only that Hodge classes be algebraic *modulo* what abelian varieties supply. By Proposition 2.5, applied to $A \times \mathbb{P}^1 \rightarrow A$, the conjecture for varieties that are not abelian already contains the conjecture for abelian varieties; this is why the half of the conjecture that concerns other varieties has to be stated modulo abelian varieties to be a separate problem.

2.5. The Lefschetz standard conjecture and motivated classes. Let X have dimension n and ample class h . Every class in $H^k(X)$ is uniquely $\sum_j L^j x_j$ with x_j primitive of degree $k - 2j$, and the *Lefschetz involution* is defined by $*_L(L^j x_j) = L^{n-k+j} x_j$ [And96, §1.1]. On H^k with $k \leq n$ it is L^{n-k} , and on H^{2n-k} it is the inverse of L^{n-k} . A class $\alpha \in H^{2p}(X, \mathbb{Q})$ is *motivated* [And96, Définition 1] if there are a variety Y , ample classes h_X on X and h_Y on Y , and algebraic classes β, γ on $X \times Y$ with

$$\alpha = \text{pr}_{X*}(\beta \cup *_L \gamma),$$

where $*_L$ is the Lefschetz involution of $X \times Y$ for the ample class $\text{pr}_X^* h_X + \text{pr}_Y^* h_Y$. We use three facts from [And96]. (a) Motivated classes are Hodge classes, they contain the algebraic classes, and they form a graded $\mathbb{Q}$ -algebra that is stable under pull-back and push-forward along projections [And96, Proposition 2.1 and §2.5]; hence a correspondence given by an algebraic cycle maps motivated classes to motivated classes.⁴ (b) Every Hodge class on an abelian variety is motivated [And96, Théorème 0.6.2]. (c) If the Lefschetz involution of every variety, for every ample class, is induced by an algebraic cycle, then every motivated class is algebraic [And96, §2, after Définition 1]. Fact (c) is immediate from the definition: $*_L \gamma$ is then the image of an algebraic class under an algebraic correspondence.

³In [BMB26] the name (F2) denotes a statement about the Hodge classes not generated by divisors and Weil classes, and it is proved there to be equivalent to the Hodge conjecture for abelian varieties. We use the simpler form.

⁴For a correspondence, $\Gamma_*\gamma = \text{pr}_{X*}(\text{cl}(\Gamma) \cup \text{pr}_B^* \gamma)$ is built from γ by a pull-back along a projection, a product with an algebraic class and a push-forward along a projection.

Proposition 2.8. (L) and (M) together imply (HC).

Proof. By (M) every Hodge class is motivated, and by (L) and fact (c) every motivated class is algebraic. $\square$

Proposition 2.9. (F3') implies (M).

Proof. Under (F3') every Hodge class is a sum of classes $\Gamma_*\gamma$ with γ a Hodge class on an abelian variety. Such a γ is motivated by fact (b), and $\Gamma_*\gamma$ is motivated by fact (a). $\square$

2.6. Families.

Lemma 2.10 (Theorem of the fixed part). *Let $f: \mathcal{X} \rightarrow S$ be a smooth projective morphism over a smooth connected quasi-projective variety, and ξ a global section of $R^kf_*\mathbb{Q}$. If ξ_s is a Hodge class for one $s \in S$, then ξ_t is a Hodge class for every $t \in S$.*

Proof. By Deligne's theorem of the fixed part [Del71, Corollaire 4.1.2] the space $H^0(S, R^kf_*\mathbb{Q})$ carries a Hodge structure for which each restriction $r_t: H^0(S, R^kf_*\mathbb{Q}) \rightarrow H^k(\mathcal{X}_t, \mathbb{Q})$ is an injective morphism of Hodge structures.⁵ An injective morphism of Hodge structures that sends a class x to a class of type (p, p) forces x to be of type (p, p) , because it respects the Hodge decompositions. So ξ is a Hodge class of $H^0(S, R^kf_*\mathbb{Q})$, and $\xi_t = r_t(\xi)$ is a Hodge class for every t . $\square$

Lemma 2.11. (HC) implies each of (F2), (F3'), (L), (M), (V), (CM) and (IP).

Proof. (F2) and (CM) are special cases, and (F3') holds because $\text{Ab}^p(X)$ contains $\text{Alg}^p(X)$. For (L): on each $H^k(X)$ the involution $*_L$ is a morphism of Hodge structures $H^k(X) \rightarrow H^{2n-k}(X)(n-k)$, so under the Künneth decomposition and Poincaré duality each of its components is a Hodge class on $X \times X$,⁶ and (HC) makes it algebraic. Algebraic classes are motivated, which gives (M). For (V): if ξ_s is algebraic, it is a Hodge class, so ξ_t is a Hodge class for every t by Lemma 2.10, and (HC) makes it algebraic. (IP) is a case of (V). $\square$

3. FERMAT VARIETIES

3.1. Varieties dominated by CM abelian varieties.

Definition 3.1. Let $\mathcal{D}$ be the class of varieties X , possibly disconnected, such that $H^*(X, \mathbb{Q})$ is spanned by the images of maps $\Gamma_*: H^*(B, \mathbb{Q}) \rightarrow H^*(X, \mathbb{Q})$, where B runs over the abelian varieties of CM type, the point included, and Γ over the algebraic cycles on $B \times X$.

Proposition 3.2. (i) Every $X \in \mathcal{D}$ satisfies (F3'). If (CM) holds, every $X \in \mathcal{D}$ satisfies (HC).

(ii) $\mathcal{D}$ contains every abelian variety of CM type and every finite set of points, and every curve whose Jacobian is of CM type.

(iii) $\mathcal{D}$ is closed under finite disjoint unions, under taking connected components, and under products.

(iv) If $Z \in \mathcal{D}$ and $f: Z \rightarrow X$ is a surjective morphism, then $X \in \mathcal{D}$.

⁵Injective because a section of a local system over a connected base is determined by its value at one point. The Hodge structure is the image of $H^k(\bar{\mathcal{X}}, \mathbb{Q}) \rightarrow H^k(\mathcal{X}_t, \mathbb{Q})$ for a smooth compactification $\bar{\mathcal{X}}$ of $\mathcal{X}$, which by [Del71, 4.1.1] is the whole invariant part. A global section that is of type (p, q) at one point is of type (p, q) everywhere [Del71, 4.1.3.1].

⁶A $\mathbb{Q}$ -linear map $u: H^a(X) \rightarrow H^b(X)$ corresponds to an element of $H^a(X)^\vee \otimes H^b(X) \cong H^{2n-a}(X)(n) \otimes H^b(X)$, a summand of $H^{2n-a+b}(X \times X)(n)$, and the correspondence it defines acts as u . The map u is a morphism of Hodge structures $H^a(X) \rightarrow H^b(X)(r)$ exactly when this element is a Hodge class of the appropriate weight.

- (v) If $X \in \mathcal{D}$ and $Y \subseteq X$ is a smooth closed subvariety of codimension at least two in every component, with $Y \in \mathcal{D}$, then the blow-up of X along Y lies in $\mathcal{D}$.

Proof. (i) By Lemma 2.2, applied in each degree to finitely many Γ_* whose images span, every Hodge class on X is a sum of classes $\Gamma_*\gamma$ with γ a Hodge class on an abelian variety of CM type. Such a class lies in $\text{Ab}^p(X)$, and under (CM) it is algebraic.

(ii) For B of CM type take Γ the diagonal. For a finite set of points take B a point and Γ the points of X one at a time. For a curve C with Jacobian J of CM type: $H^0(C)$ and $H^2(C)$ are spanned by the fundamental class and the class of a point, which are images from the point, and $H^1(C) = \alpha^*H^1(J)$ for an Abel–Jacobi map $\alpha: C \rightarrow J$, and α^* is the correspondence given by the transpose of the graph of α .

(iii) A disjoint union: the cohomology is the direct sum, and a cycle on $B \times X_i$ is a cycle on $B \times \bigsqcup_i X_i$. A connected component X' of X : restriction $H^*(X) \rightarrow H^*(X')$ is surjective and is the correspondence given by the transpose of the graph of the inclusion, so composing gives correspondences from the B 's onto $H^*(X')$. A product: by the Künneth formula $H^*(X \times X')$ is spanned by the classes $\text{pr}^*x \cup \text{pr}'^*x'$, and if $x = \Gamma_*y$ and $x' = \Gamma'_*y'$ then $\text{pr}^*x \cup \text{pr}'^*x' = \pm(\Gamma \times \Gamma')_*(\text{pr}_B^*y \cup \text{pr}_{B'}^*y')$, where $\Gamma \times \Gamma'$ is a cycle on $(B \times B') \times (X \times X')$ and $B \times B'$ is of CM type.

(iv) By (iii) we may assume X connected and replace Z by a component that maps onto X . By Proposition 2.5 there is a correspondence Γ from Z to X with Γ_* surjective; composing it with the correspondences that dominate $H^*(Z)$ gives correspondences that dominate $H^*(X)$.

(v) By Proposition 2.6 the cohomology of the blow-up is spanned by the images of correspondences from X and from Y ; compose them with the correspondences that dominate $H^*(X)$ and $H^*(Y)$. $\square$

3.2. Fermat curves.

Lemma 3.3. *For every $m \geq 1$ the Jacobian of the Fermat curve X_m^1 is of CM type.*

Proof. For $m \leq 2$ the curve is rational and the Jacobian is a point. Let $m \geq 3$, let $g = (m-1)(m-2)/2$ be the genus, and use the affine model $u^m + v^m = 1$, which is isomorphic to X_m^1 over $\mathbb{C}$. The group $G = \mu_m \times \mu_m$ acts by $(\zeta, \zeta') \cdot (u, v) = (\zeta u, \zeta' v)$, and this action extends to the smooth projective curve. The forms

$$\omega_{r,s} = u^{r-1}v^{s-m} du, \quad r, s \geq 1, \quad r + s \leq m-1,$$

are holomorphic⁷, and there are $(m-1)(m-2)/2 = g$ of them, so they form a basis of $H^{1,0}$. The form $\omega_{r,s}$ is an eigenvector of G with character $(\zeta, \zeta') \mapsto \zeta^r \zeta'^s$, and distinct pairs (r, s) give distinct characters, since $1 \leq r, s \leq m-2$. The complex conjugates span $H^{0,1}$ and have the characters $(-r, -s) = (m-r, m-s)$; since $(m-r) + (m-s) \geq m+1$, none of these occurs in $H^{1,0}$. So $H^1(X_m^1, \mathbb{C})$ is the direct sum of $2g$ lines on which G acts by $2g$ distinct characters.

Let $E \subseteq \text{End}(H^1(X_m^1, \mathbb{Q}))$ be the image of the group algebra $\mathbb{Q}[G]$. It is commutative and semisimple, as a quotient of a commutative semisimple algebra, and $E \otimes \mathbb{C}$ is the image of $\mathbb{C}[G]$, which acts on each of the $2g$ lines through its own character; since the characters are distinct, $E \otimes \mathbb{C} \cong \mathbb{C}^{2g}$ and $\dim_{\mathbb{Q}} E = 2g$. The elements of G act on the Jacobian, and H^1 of the Jacobian is H^1 of the curve, so $E \subseteq \text{End}(J) \otimes \mathbb{Q}$. Thus J is of CM type. $\square$

⁷On the affine curve $v^{s-m}du = -v^{s-1}u^{1-m}dv$ shows that $\omega_{r,s}$ has no poles where u or v is finite. At the m points at infinity, u and v have simple poles and du has a double pole, so $\omega_{r,s}$ has order $-(r-1) - (s-m) - 2 = m-1-r-s \geq 0$ there.

The characters can be grouped into orbits of $(\mathbb{Z}/m)^\times$, and each orbit spans the cohomology of an isogeny factor of J with complex multiplication by a quotient of $\mathbb{Q}(\mu_m)$ [KR78]; this is the classical computation of Weil [Wei52].

3.3. The inductive structure. Fix $m \geq 1$ and write $X^n = X_m^n$, the Fermat variety $x_0^m + \cdots + x_{n+1}^m = 0$ in $\mathbb{P}^{n+1}$, for $n \geq 0$; thus X^0 is a set of m points. Fix ε with $\varepsilon^m = -1$.

Proposition 3.4 (Shioda–Katsura [SK79]). *Let $r, s \geq 1$, let $Y \subseteq X^r \times X^s$ be the subvariety $x_{r+1} = y_{s+1} = 0$, which is isomorphic to $X^{r-1} \times X^{s-1}$ and has codimension two, and let Z be the blow-up of $X^r \times X^s$ along Y . The rational map $\varphi: X^r \times X^s \dashrightarrow X^{r+s}$,*

$$\varphi(x, y) = [x_0 y_{s+1} : \cdots : x_r y_{s+1} : \varepsilon x_{r+1} y_0 : \cdots : \varepsilon x_{r+1} y_s],$$

is defined exactly off Y , and it extends to a surjective morphism $Z \rightarrow X^{r+s}$.

Proof. The image lies in X^{r+s} , because $y_{s+1}^m \sum_{i \leq r} x_i^m + \varepsilon^m x_{r+1}^m \sum_{j \leq s} y_j^m = -y_{s+1}^m x_{r+1}^m + x_{r+1}^m y_{s+1}^m = 0$. All coordinates vanish exactly on Y : if $y_{s+1} \neq 0$ they force $x_0 = \cdots = x_r = 0$ and then $x_{r+1}^m = -\sum_{i \leq r} x_i^m = 0$, which is impossible; so $y_{s+1} = 0$, and in the same way $x_{r+1} = 0$. On the blow-up, on the chart where $y_{s+1} = t x_{r+1}$, the map is $[tx_0 : \cdots : tx_r : \varepsilon y_0 : \cdots : \varepsilon y_s]$; here $y_0, \dots, y_s$ do not all vanish, because otherwise $y_{s+1}^m = 0$ as well. The other chart is symmetric. So φ extends to a morphism on Z [SK79, Lemma 1.2]. It is dominant, since from a general point of X^{r+s} one recovers $[x_0 : \cdots : x_r]$ and $[y_0 : \cdots : y_s]$, and then x_{r+1} and y_{s+1} up to finitely many choices; its image is closed and X^{r+s} is irreducible for $r + s \geq 2$, so it is surjective.⁸ $\square$

3.4. Proof of Theorem C.

Theorem 3.5. *Every Fermat variety X_m^n lies in $\mathcal{D}$. So do all products of Fermat varieties of any degrees and dimensions, and every variety onto which such a product maps surjectively.*

Proof. Fix m . We show $X^n \in \mathcal{D}$ by induction on n . The set of points X^0 lies in $\mathcal{D}$, and so does the curve X^1 , by Proposition 3.2(ii) and Lemma 3.3. Let $n \geq 2$ and suppose $X^k \in \mathcal{D}$ for $k < n$. Apply Proposition 3.4 with $r = 1$ and $s = n - 1$. The product $X^1 \times X^{n-1}$ and the centre $Y \cong X^0 \times X^{n-2}$ lie in $\mathcal{D}$ by Proposition 3.2(iii); the blow-up Z lies in $\mathcal{D}$ by Proposition 3.2(v); and X^n , the image of Z , lies in $\mathcal{D}$ by Proposition 3.2(iv). The remaining assertions are Proposition 3.2(iii) and (iv). $\square$

Proof of Theorem C. Combine Theorem 3.5 with Proposition 3.2(i). $\square$

The proof shows more than Theorem C states: every Hodge class on a Fermat variety is the image, under an explicit algebraic correspondence, of a Hodge class on a power of the Jacobians of Fermat curves of the same degree. Theorem C is therefore exactly as strong as (CM) for these powers.

3.5. The Hodge classes of a Fermat variety. The group $G_n = \mu_m^{n+2}/\mu_m$ acts on X_m^n by multiplying the coordinates, and its characters are the vectors $a = (a_0, \dots, a_{n+1}) \in (\mathbb{Z}/m)^{n+2}$ with $\sum a_i = 0$. Let $\mathfrak{A}_m^n$ be the set of characters with every $a_i \neq 0$. For $a \in \mathfrak{A}_m^n$ write $\langle a \rangle = \sum_i \bar{a}_i/m$, where $\bar{a}_i \in \{1, \dots, m-1\}$ represents a_i ; it is an integer between 1 and $n+1$.

⁸Shioda and Katsura show more: the morphism $Z \rightarrow X^{r+s}$ is the composite of the quotient by a cyclic group of order m and a birational morphism that contracts two disjoint subvarieties of the quotient [SK79, Theorem 1.7]. We use only that it is a surjective morphism.

Proposition 3.6. (i) *The primitive cohomology $H_{\text{prim}}^n(X_m^n, \mathbb{C})$ is the direct sum of lines $V(a)$, one for each $a \in \mathfrak{A}_m^n$, on which G_n acts by pull-back through the character a ; and $V(a) \subseteq H^{n-q,q}$ with $q = \langle a \rangle - 1$. Hence*

$$\dim H_{\text{prim}}^n = |\mathfrak{A}_m^n| = \frac{(m-1)^{n+2} + (-1)^n(m-1)}{m}, \quad h_{\text{prim}}^{n-q,q} = \#\{a : \langle a \rangle = q+1\}.$$

(ii) *Let n be even. The space of primitive Hodge classes, tensored with $\mathbb{C}$, is the sum of the lines $V(a)$ for which $\langle ta \rangle = n/2 + 1$ for every $t \in (\mathbb{Z}/m)^\times$.*

Proof. (i) By Griffiths' description of the cohomology of a hypersurface by residues [Gri69, §8], $H_{\text{prim}}^{n-q,q}(X_m^n)$ is isomorphic to the graded piece of degree $(q+1)m - (n+2)$ of the Jacobian ring $\mathbb{C}[x_0, \dots, x_{n+1}]/(x_0^{m-1}, \dots, x_{n+1}^{m-1})$, the class of a monomial x^b corresponding to the residue of $x^b \Omega / F^{q+1}$, where $F = \sum x_i^m$ and $\Omega = \sum (-1)^i x_i dx_0 \wedge \dots \widehat{dx_i} \dots \wedge dx_{n+1}$. That residue is an eigenvector of G_n with character $a = b + (1, \dots, 1)$, because Ω has character $(1, \dots, 1)$ and F is invariant. As b runs over the exponents with $0 \leq b_i \leq m-2$, the vector a runs once over the characters with $1 \leq \bar{a}_i \leq m-1$; the degree condition is $\sum \bar{a}_i = (q+1)m$, that is, $\langle a \rangle = q+1$, and it forces $\sum a_i = 0$ in $\mathbb{Z}/m$. The count of $\mathfrak{A}_m^n$ follows from the recursion $N_k = (m-1)^{k-1} - N_{k-1}$, $N_1 = 0$, for the number N_k of k -tuples of nonzero residues with sum zero.

(ii) The eigenspace decomposition is defined over $\mathbb{Q}(\mu_m)$, and the automorphism $\zeta \mapsto \zeta^t$ of $\mathbb{Q}(\mu_m)$ carries $V(a)$ to $V(ta)$. So for each orbit O of $(\mathbb{Z}/m)^\times$ on $\mathfrak{A}_m^n$ the sum $\bigoplus_{a \in O} V(a)$ is defined over $\mathbb{Q}$, these rational subspaces decompose $H_{\text{prim}}^n(X_m^n, \mathbb{Q})$, and the one for O consists of Hodge classes exactly when every $V(a)$, $a \in O$, has type $(n/2, n/2)$, that is, when $\langle a \rangle = n/2 + 1$ on O . A rational class with a component on some $V(a)$ has components on the whole orbit of a , so the Hodge classes are the sum of the rational subspaces of the orbits of this kind. $\square$

For a Fermat surface the Picard number is $\rho(X_m^2) = 1 + \#\{a \in \mathfrak{A}_m^2 : \langle ta \rangle = 2 \text{ for all } t \in (\mathbb{Z}/m)^\times\}$, and $h^{1,1} = (2m^3 - 6m^2 + 7m)/3$. Figure 2 compares the two for $m \leq 12$. For m prime to 6, $\rho = 3(m-1)(m-2) + 1$, and the Néron–Severi group is spanned by the $3m^2$ lines on the surface [AS83]; the quartic, quintic and sextic give $\rho = 20, 37$ and 86 .

Remark 3.7. The Hodge conjecture for X_m^n is known without hypothesis when m is prime or $m = 4$ [Ran80, Shi79], when $m \leq 20$ [Shi79], and when m is a power of a prime [Aok87, Corollary 2-3]; see [dSJ21, §2]. Aoki proves it for every abelian variety of Fermat type of degree m when $m = 2^a 3^b 5^c 7^d$ with $c = 0$ or $d = 0$, and when m is p^e or $2p^e$ [Aok00, Theorem 0.1]; as noted after the proof of Theorem C, this gives it for X_m^n . The preprint [Pet26] adds the degrees $2^a 3^b 5^c 7^d$ not divisible by 420. When every prime factor of m exceeds $n+2$, every Hodge character is a union of pairs [Aok83, Theorem A], and the classes are spanned by linear subspaces [Ran80, Shi79]; for fourfolds this covers the degrees whose prime factors are all at least 7. For fourfolds of degree $m \leq 100$ prime to 6 the conjecture is proved in [dSJ21, Proposition 3.1] by a computer search, and the preprint [Jum26] gives a computer-assisted proof for fourfolds of odd degree at most 199. Kang states it for every Fermat fourfold [Kan16, Corollary 3.2], and the preprint [MMRV26], which treats all dimensions in degree less than 65 except 44, 51 and 52, uses that statement; its proof has the gap described in Remark 3.21. So Theorem F is new for the odd degrees $m > 199$ that are divisible by 3 or by 5 and have a prime factor larger than 7, the first of which are 201, 205, 207, 213 and 215, and its proof, unlike those of [dSJ21, Jum26], does not use a computer. In all these proofs the Hodge classes are written as combinations of the classes of linear subspaces and of cycles built from them, characters a that split into pairs $a_i + a_j = 0$ corresponding to linear subspaces. Theorem C needs no such decomposition: it moves the whole question to the Jacobians of Fermat curves. The preprint

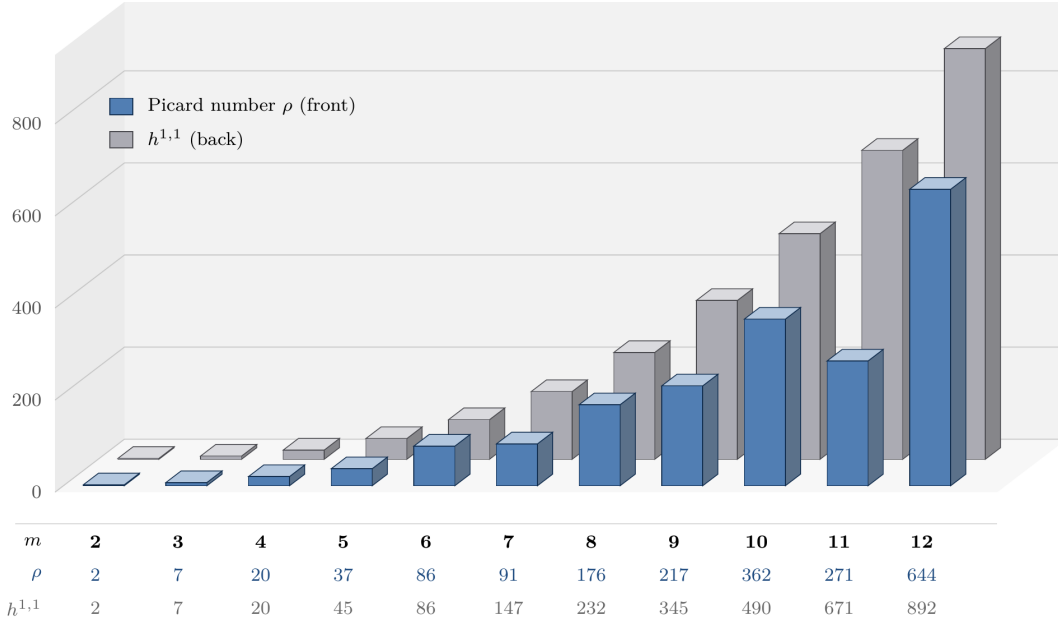


FIGURE 2. The Fermat surfaces X_m^2 , $2 \leq m \leq 12$: the Picard number ρ (front row) against $h^{1,1}$ (back row), both computed from Proposition 3.6. On a surface every Hodge class is algebraic (Theorem 2.3); the figure shows how the share of $H^{1,1}$ that is defined over $\mathbb{Q}$ varies with the arithmetic of m . The values of $h^{1,1}$ are recomputed from the Jacobian ring in Macaulay2 (Appendix A).

[Ope26a] announces the Hodge conjecture on every self-power of a very general member of the family of diagonal hypersurfaces; the Fermat varieties are special members of that family.

3.6. Fermat fourfolds of degree prime to six. In this subsection m is prime to 6, and we prove Theorem F for such m ; Section 3.7 treats the odd m divisible by 3. For $a \in \mathbb{Z}/m$ we write $\bar{a} \in \{0, \dots, m-1\}$ for its representative, extending the notation of Section 3.5.

Definition 3.8. Let A be a multiset of nonzero elements of $\mathbb{Z}/m$ whose sum is 0. It is *balanced* if the integer $\sum_{a \in A} \bar{t}a$ is the same for every $t \in (\mathbb{Z}/m)^\times$; we call this integer $h(A)$. It *contains a pair* if $a \in A$ and $-a \in A$ for some a ; since m is odd, $a \neq -a$, so these are two different members of A . When $5 \mid m$, the *fiber* of $x \in \mathbb{Z}/m$ is the set

$$\Phi(x) = \{x + jm/5 : 0 \leq j \leq 4\}$$

of the five solutions y of $5y = 5x$, and A is *5-standard* if $A = \Phi(x) + \{-5x\}$ for some x , as multisets.

For a balanced multiset of n elements, comparing t with $-t$ gives $h(A) = nm - h(A)$, so $h(A) = nm/2$. By Proposition 3.6(ii), a character $\alpha \in \mathfrak{A}_m^4$ spans a line of Hodge classes exactly when the multiset of its entries is balanced, and the same holds for the characters $\beta \in \mathfrak{A}_m^2$ of the Fermat surface. Aoki writes $\sigma_{5,x} = (x, x+d, x+2d, x+3d, x+4d, -5x)$, $d = m/5$, for the 5-standard characters [Aok87].

Theorem 3.9. *Let m be prime to 6.*

- (i) *Every balanced multiset of four nonzero elements of $\mathbb{Z}/m$ contains a pair.*
- (ii) *Every balanced multiset of six nonzero elements of $\mathbb{Z}/m$ contains a pair or is 5-standard.*

Part (i) is Aoki's theorem that the Hodge classes of a Fermat surface of degree prime to 6 are spanned by lines [Aok83, Theorem A]; we prove it again because the proof of (ii) uses it and runs on the same lines. When $5 \nmid m$, part (ii) says that every Hodge character of X_m^4 splits into three pairs, which is also contained in [Aok83, Theorem A], since then every prime factor of m exceeds 6. The case $5 \mid m$ is new. The theorem has been checked in Lean (Appendix A).

Proof of Theorem F for m prime to 6, given Theorem 3.9. Write $G = G_4$ for the group acting on $X = X_m^4$. By Proposition 3.6(ii), and since $H^4(X, \mathbb{Q})$ is the sum of the primitive part and the line spanned by the square of the hyperplane class, it is enough to show that every line $V(\alpha)$ with α balanced lies in $\text{Alg}^2(X) \otimes \mathbb{C}$. Then the Hodge classes and the algebraic classes span the same complex subspace, and since $\text{Alg}^2(X) \subseteq \text{Hdg}^2(X)$ are rational subspaces they are equal. A subvariety $Z \subseteq X$ represents α [Aok87] if the α -component

$$\frac{1}{|G|} \sum_{g \in G} \alpha(g)^{-1} g^*[Z] = \frac{1}{|G|} \sum_{g \in G} \alpha(g)^{-1} [g^{-1}Z]$$

of $[Z]$ is not zero; it lies in $V(\alpha)$ and in $\text{Alg}^2(X) \otimes \mathbb{C}$, so then $V(\alpha) \subseteq \text{Alg}^2(X) \otimes \mathbb{C}$. Permuting the coordinates is an automorphism of X that permutes the lines $V(\alpha)$ accordingly, so the order of the entries of α does not matter. By Theorem 3.9(ii) there are two cases.

α contains a pair. Order the entries as $\alpha = (\beta_0, \beta_1, \beta_2, \beta_3, a, -a)$. Since $\bar{t}a + \overline{-ta} = m$ for every unit t , the multiset of entries of $\beta = (\beta_0, \dots, \beta_3)$ is balanced, so β is a Hodge character of the Fermat surface X_m^2 , and $V(\beta)$ consists of algebraic classes by Theorem 2.3. The character $(a, -a)$ of the set X_m^0 of m points spans a line of $H^0(X_m^0, \mathbb{C})$, which is spanned by classes of points. By the inductive structure, in the form of Shioda and Ran stated in [Aok87, Theorem 1-4(i)], $V(\alpha)$ is then spanned by algebraic classes.

α is 5-standard. Then $5 \mid m$; write $d = m/5$ and $\alpha = (x, x + d, \dots, x + 4d, -5x)$. Aoki constructs a subvariety of codimension two representing α when $\gcd(\bar{x}, d) = 1$ [Aok87, Theorem 2-1]. In general let $g = \gcd(\bar{x}, d)$, $m' = m/g$ and $d' = d/g = m'/5$. Every entry of α is a multiple of g , so $\alpha = g\alpha'$ with

$$\alpha' = (x', x' + d', \dots, x' + 4d', -5x') \in (\mathbb{Z}/m')^6, \quad x' = \bar{x}/g, \quad \gcd(x', d') = 1,$$

where $g\alpha'$ means the image of α' under the injective homomorphism $\mathbb{Z}/m' \rightarrow \mathbb{Z}/m$, $y \mapsto gy$. The entries of α' are nonzero, because those of α are. Every unit t' modulo m' lifts to a unit t modulo m , and then $t'\alpha'_i/m' = \overline{t\alpha_i}/m$ for each i ; so α' is balanced, and it is a Hodge character of $X' = X_{m'}^4$. In particular $d' \geq 3$, since for $d' = 1$ one of $x', x' + 1, \dots, x' + 4$ would vanish modulo 5. By [Aok87, Theorem 2-1] there is a subvariety $Y' \subseteq X'$ of codimension two that represents α' . Let

$$\pi: X \rightarrow X', \quad [x_0 : \dots : x_5] \mapsto [x_0^g : \dots : x_5^g].$$

It is a finite surjective morphism, and it is equivariant for the homomorphism $G \rightarrow G'$, $\zeta \mapsto \zeta^g$, to the group G' acting on X' . For $\xi \in V(\alpha')$ and $\zeta \in G$ we get $\zeta^* \pi^* \xi = \pi^* (\zeta^g)^* \xi = \alpha'(\zeta^g) \pi^* \xi = \alpha(\zeta) \pi^* \xi$, so π^* maps $V(\beta')$ into $V(g\beta')$ for every character β' of G' , and distinct β' go to distinct $g\beta'$. As $\pi_* \pi^*$ is multiplication by $\deg \pi$, the map π^* is injective on cohomology. Hence the α -component of $\pi^*[Y']$, the class of the cycle π^*Y' , is the image under π^* of the α' -component of $[Y']$, which is not zero; so the cycle π^*Y' represents α , and again $V(\alpha) \subseteq \text{Alg}^2(X) \otimes \mathbb{C}$. $\square$

The rest of this subsection proves Theorem 3.9. We fix a balanced multiset A of $n \in \{4, 6\}$ nonzero elements of $\mathbb{Z}/m$ that contains no pair, and show that $n = 6$ and A is 5-standard.

Step 1: the counting function. Let e be the largest additive order of an element of A . It divides m , so it is prime to 6, and $e > 1$. The map $\kappa: \mathbb{Z}/e \rightarrow \mathbb{Z}/m$, $y \mapsto (m/e)y$, is an injective homomorphism, and the elements of $\mathbb{Z}/m$ of order exactly e are the $\kappa(z)$ with $z \in U = (\mathbb{Z}/e)^\times$. For $z \in U$ put

$$F(z) = \text{mult}_A(\kappa z) - \text{mult}_A(-\kappa z),$$

the difference of the multiplicities in A . Then $F(-z) = -F(z)$. If $a_0 = \kappa(z_0) \in A$ has order e , then $-a_0 \notin A$, so $F(z_0) \geq 1$. We call $\|F\| = \sum_{z \in U} |F(z)|$ the *mass* of F .

Lemma 3.10. $\|F\| \leq 2n$. If $\|F\| = 2n$, then every element of A has order e and $\sum_{z \in U} F(z)z = 0$ in $\mathbb{Z}/e$.

Proof. $\|F\| \leq \sum_z (\text{mult}_A(\kappa z) + \text{mult}_A(-\kappa z)) = 2N$, where $N \leq n$ is the number of elements of A of order e . If $\|F\| = 2n$, then $N = n$, and writing $a = \kappa(z_a)$ for $a \in A$,

$$\sum_z F(z)z = \sum_{a \in A} z_a - \sum_{a \in A} (-z_a) = 2 \sum_{a \in A} z_a,$$

while $\kappa(\sum_a z_a) = \sum_a a = 0$; as κ is injective, $\sum_a z_a = 0$. $\square$

Step 2: the characters. For a Dirichlet character χ modulo e let $B_{1,\chi} = e^{-1} \sum_{s=1}^e \chi(s)s$. If χ is odd and primitive, then $B_{1,\chi} \neq 0$. This is classical: by the relative class number formula [Was97, Theorem 4.17] the product of the numbers $-B_{1,\chi}/2$, over the odd characters of the Galois group of $\mathbb{Q}(\mu_e)$, is a nonzero rational multiple of the relative class number, a positive integer; equivalently, $B_{1,\chi}$ is a nonzero multiple of $L(1, \bar{\chi})$ [Was97, Theorem 4.9]. This is the only analytic input of the proof.

Lemma 3.11. $\sum_{z \in U} F(z)\chi(z)^{-1} = 0$ for every primitive Dirichlet character χ modulo e .

Proof. If χ is even, the sum vanishes because F is odd. Let χ be odd, and regard it as a character of $(\mathbb{Z}/m)^\times$ through reduction modulo e ; it is not trivial. Since A is balanced,

$$\sum_{a \in A} \sum_{t \in (\mathbb{Z}/m)^\times} \chi(t) \bar{t}a = h(A) \sum_t \chi(t) = 0.$$

Let $a \in A$ have order f . If $f \neq e$, then $e \nmid f$, since $f \leq e$. For a divisor f of m let $W_f \subseteq (\mathbb{Z}/m)^\times$ be the units congruent to 1 modulo f . If χ were trivial on W_f , it would be trivial on $W_f W_e = W_{\gcd(f,e)}$ (by the Chinese remainder theorem), so it would be induced from a character modulo the proper divisor $\gcd(f,e)$ of e , against primitivity. So χ is not trivial on W_f ; as $\bar{t}a$ depends only on t modulo f , the inner sum vanishes. If $f = e$, write $a = \kappa(z)$. For a unit t with image $s \in U$ we have $ta = \kappa(sz)$ and $\bar{t}a = (m/e) \bar{s}z$, where now $\bar{s}z \in \{0, \dots, e-1\}$. Reduction $(\mathbb{Z}/m)^\times \rightarrow U$ is onto, with fibres of size $\varphi(m)/\varphi(e)$, so

$$\sum_t \chi(t) \bar{t}a = \frac{m \varphi(m)}{e \varphi(e)} \sum_{s \in U} \chi(s) \bar{s}z = \frac{m \varphi(m)}{\varphi(e)} \chi(z)^{-1} B_{1,\chi},$$

after the substitution $s \mapsto sz^{-1}$. Adding over A and dividing by $m \varphi(m) B_{1,\chi} / \varphi(e) \neq 0$ gives $\sum_z \text{mult}_A(\kappa z) \chi(z)^{-1} = 0$. The substitution $z \mapsto -z$ turns $\sum_z \text{mult}_A(-\kappa z) \chi(z)^{-1}$ into $\chi(-1)^{-1}$ times the same sum, so it vanishes too, and so does their difference. $\square$

Step 3: splitting along the subgroups K_p . For a prime $p \mid e$ let $K_p \subseteq U$ be the subgroup of units congruent to 1 modulo e/p , the kernel of $U \rightarrow (\mathbb{Z}/(e/p))^\times$.

Lemma 3.12. (a) $|K_p| = p$ if $p^2 \mid e$, and $|K_p| = p-1$ if $p \parallel e$. So $|K_p| \geq 4$, with equality only for $p = 5 \parallel e$.

- (b) Let $p^v \parallel e$. The product of the K_q with $q \neq p$ consists of units congruent to 1 modulo p^v . Hence K_p meets it trivially, the product H of all the K_q is direct, and -1 does not lie in the product of the K_q with $q \neq p$.
- (c) A Dirichlet character modulo e that is not primitive is trivial on some K_p .

Proof. (a) $|K_p| = \varphi(e)/\varphi(e/p)$. (b) For $q \neq p$, p^v divides e/q . An element of K_p congruent to 1 modulo p^v is congruent to 1 modulo $\text{lcm}(e/p, p^v) = e$, and $-1 \not\equiv 1 \pmod{p}$ as p is odd. (c) A character induced from a proper divisor f of e is trivial on the units congruent to 1 modulo f , and f divides e/p for some $p \mid e$. $\square$

Definition 3.13. Let $K_1, \dots, K_r$ be subgroups of a finite abelian group, H their product and C a coset of H . A function $f: C \rightarrow \mathbb{Z}$ is *split along* $(K_1, \dots, K_r)$ if $f = f_1 + \dots + f_r$ with functions $f_i: C \rightarrow \mathbb{C}$ such that $f_i(zk) = f_i(z)$ for $z \in C$ and $k \in K_i$; for $r = 0$ this means $f = 0$. A *line along* K_i is a coset wK_i .

Lemma 3.14. *On every coset of H in U , the function F is split along the family $(K_p)_{p|e}$.*

Proof. By Fourier inversion on U , $F = \varphi(e)^{-1} \sum_{\chi} \hat{F}(\chi) \chi$ with $\hat{F}(\chi) = \sum_z F(z) \chi(z)^{-1}$. By Lemma 3.11 only characters that are not primitive contribute, and by Lemma 3.12(c) each of them is trivial on some K_p . Collect the terms according to such a p . $\square$

We use four simple properties of split functions, for f split along $(K_1, \dots, K_r)$ on C and $H' = K_2 \cdots K_r$.

- (S1) For $k \in K_1$ and every coset $S \subseteq C$ of H' , the function $z \mapsto f(z) - f(zk)$ on S is split along $(K_2, \dots, K_r)$: the terms f_1 cancel.
- (S2) If f vanishes on one coset $S_0 \subseteq C$ of H' , then f is split along $(K_2, \dots, K_r)$ on every coset $S \subseteq C$ of H' : choose $k \in K_1$ with $Sk = S_0$ and apply (S1).
- (S3) For $r = 2$ and $C = xK_1K_2$: $f(xij) = f(xi) + f(xj) - f(x)$ for $i \in K_1$ and $j \in K_2$.
- (S4) For $r = 1$, f is invariant under K_1 .

Step 4: three lemmas on split functions. In the next three lemmas, $K_1, \dots, K_r$ (or $K, K_2, \dots, K_r$) are subgroups of a finite abelian group whose product H is direct, and C is a coset of H . Fix $x \in C$ and put $H' = K_2 \cdots K_r$. For c in the first subgroup, the *slices* of C are the cosets $S_c = xH'$; they are pairwise distinct, and for a function f on C we write $\mu(c) = \sum_{z \in S_c} |f(z)|$.

Lemma 3.15 (grid). *Suppose $|K_i| \geq 4$ for every i , with equality for at most one i . Let $f: C \rightarrow \mathbb{Z}$ be split along $(K_1, \dots, K_r)$, not identically zero, with $\sum_C |f| \leq 6$. Then for some i with $|K_i| \leq 6$ there are a line $L = wK_i \subseteq C$ and $\sigma \in \{\pm 1\}$ such that $f = \sigma$ on L and $f = 0$ on $C \setminus L$.*

Proof. Induction on r . For $r = 0$ the function vanishes, which is excluded. Let $r \geq 1$ and $K = K_1$. First an observation.

(P) Let $c, c' \in K$ with $\mu(c) + \mu(c') \leq 6$, and suppose $g(z) = f(z) - f(zc^{-1}c')$ is not identically zero on S_c . Then $\mu(c) + \mu(c') \geq |K_i|$ for some $i \geq 2$. Indeed, g is split along $(K_2, \dots, K_r)$ on S_c by (S1), and $\sum_{S_c} |g| \leq \mu(c) + \mu(c')$ because $z \mapsto zc^{-1}c'$ maps S_c onto $S_{c'}$. By induction $|g|$ is 1 on a line along some K_i , $i \geq 2$, and 0 elsewhere, so $\sum_{S_c} |g| = |K_i|$.

Case A: $f(zk) = f(z)$ for all $z \in C$, $k \in K$. Let $f(z_0) \neq 0$. Then f equals $f(z_0)$ on the $|K| \geq 4$ points of z_0K , so $|f(z_0)| = 1$ and $|K| \leq 6$, and a nonzero value off z_0K would add at least $|K| \geq 4$ more. So $f = f(z_0)$ on z_0K and 0 elsewhere.

Case B1: $\mu(c_0) = 0$ for some c_0 . By (S2), f is split along $(K_2, \dots, K_r)$ on every slice. On a slice where it does not vanish, by induction it is ± 1 on a line along some K_i with $i \geq 2$ and

$|K_i| \leq 6$ and 0 elsewhere, which uses mass $|K_i| \geq 4$; a second such slice would bring the mass to 8. This is the claim.

Case B2: f is not K -invariant, and $\mu \geq 1$ on every slice. As $\sum \mu \leq 6 < 2|K|$, some c_s has $\mu(c_s) = 1$. Choose $c, k \in K$ such that $f(z) \neq f(zk)$ for some $z \in S_c$. By (P), applied to c and ck , or because $\mu(c) + \mu(ck) > 6$, one of them, c_1 , has $\mu(c_1) \geq 2$. Then $f(z) - f(zc_1^{-1}c_s)$ does not vanish identically on S_{c_1} , since translation carries S_{c_1} onto S_{c_s} and $\mu(c_1) \neq \mu(c_s)$. By (P), $\mu(c_1) + 1 \geq |K_i| \geq 4$ for some $i \geq 2$, so

$$6 \geq \sum_{c \in K} \mu(c) \geq \mu(c_1) + (|K| - 1) \geq 3 + 3.$$

All these are equalities: $|K| = 4$ and $|K_i| = \mu(c_1) + 1 = 4$, two subgroups of order four, which is excluded. $\square$

Lemma 3.16 (a large subgroup carrying the sign). *Let $|K| \geq 10$ and $|K_i| \geq 4$ for $i \geq 2$, with equality for at most one i . Let $f: C \rightarrow \mathbb{Z}$ be split along $(K, K_2, \dots, K_r)$, and suppose that for some $\varepsilon \in K \setminus \{1\}$ and $\varepsilon' \in H'$ we have $f(z\varepsilon\varepsilon') = -f(z)$ for all $z \in C$. If $\sum_C |f| \leq 12$ and $f(x) \geq 1$, then $f = 1$ on a line along some K_i , $i \geq 2$, with $|K_i| \leq 6$.*

Proof. Multiplication by $\varepsilon\varepsilon'$ carries S_c onto $S_{c\varepsilon}$ and preserves $|f|$, so $\mu(c\varepsilon) = \mu(c)$. If some slice has $\mu = 0$, then f is split along $(K_2, \dots, K_r)$ on $S_1 \ni x$ by (S2), and $2\mu(1) = \mu(1) + \mu(\varepsilon) \leq 12$; by Lemma 3.15 f is σ on a line along some K_i in S_1 and 0 elsewhere on S_1 , and $\sigma = 1$ because the line passes through x . Otherwise $\mu \geq 1$ everywhere, and as $\sum \mu \leq 12 < 2|K|$, some c has $\mu(c) = 1$, hence $\mu(c\varepsilon) = 1$. The function $f(z) - f(z\varepsilon)$ on S_c is split along $(K_2, \dots, K_r)$ by (S1) and has mass at most 2, so it vanishes by Lemma 3.15. Let z_0 be the point of S_c with $f(z_0) = \pm 1$. The points $z_0\varepsilon$ and $z_0\varepsilon\varepsilon'$ of $S_{c\varepsilon}$ carry the values $f(z_0)$ and $-f(z_0)$, both nonzero, so they coincide because $\mu(c\varepsilon) = 1$; then $f(z_0) = -f(z_0)$, a contradiction. $\square$

Lemma 3.17 (two subgroups). *Let $K \cap L = 1$, $|K| = 4$, $|L| = 6$, and let $\varepsilon \in K$ and $\varepsilon' \in L$ be different from 1, with $\varepsilon^2 = 1$. Let $f: xKL \rightarrow \mathbb{Z}$ be split along (K, L) , with $f(z\varepsilon\varepsilon') = -f(z)$, $\sum |f| \leq 12$ and $f(x) \geq 1$. Then*

- (a) $f(xi) = 1$ for all $i \in K$; or
- (b) $f(xj) = 1$ for all $j \in L$; or
- (c) $\sum |f| = 12$, and there are $u: K \rightarrow \{\pm 1\}$ and $v: L \rightarrow \{\pm 1\}$ with $u(i\varepsilon) = -u(i)$, $v(j\varepsilon') = -v(j)$ and $2f(xij) = u(i) + v(j)$.

Proof. By (S3), $f(xij) = f(xi) + f(xj) - f(x)$. Put $\delta = f(x) - f(x\varepsilon')$, $u(i) = 2f(xi) - \delta$ and $v(j) = 2f(xj) - 2f(x) + \delta$, so that $2f(xij) = u(i) + v(j)$. Comparing $f(xi\varepsilon \cdot \varepsilon') = -f(xi)$ with (S3) gives $f(xi\varepsilon) + f(xi) = \delta$, that is $u(i\varepsilon) = -u(i)$; for $i = 1$ it gives $f(x\varepsilon) = -f(x\varepsilon')$, and then the same comparison for $x\varepsilon \cdot j\varepsilon'$ gives $v(j\varepsilon') = -v(j)$. Since $j \mapsto j\varepsilon'$ permutes L and changes the sign of v ,

$$2 \sum_j |u(i) + v(j)| = \sum_j (|u(i) + v(j)| + |u(i) - v(j)|) = 2 \sum_j \max(|u(i)|, |v(j)|),$$

so $T = \sum_{i,j} \max(|u(i)|, |v(j)|) = 2 \sum |f| \leq 24$. All $u(i)$ and $v(j)$ have the same parity, because $u(i) + v(j) = 2f(xij)$. Write $K = \{1, \varepsilon, a, a\varepsilon\}$, $V = \sum_j |v(j)|$ and $M(s) = \sum_j \max(s, |v(j)|)$; then $T = 2M(|u(1)|) + 2M(|u(a)|)$, $M(s) \geq \max(6s, V)$, and $V \geq 2|v(1)|$ because $|v(\varepsilon')| = |v(1)|$.

If the parity is odd, all 24 terms of T are at least 1, so all equal 1, and we are in case (c). If it is even and $u(1) \geq 1$, then $u(1) \geq 2$ and $24 \geq T \geq 24 + 2V$, so $u(1) = 2$, $v = 0$, and $f(xj) = (u(1) + v(j))/2 = 1$: case (b). If it is even and $u(1) \leq 0$, then $v(1) = 2f(x) - u(1) \geq 2$. If $u(1) \neq 0$, then $T \geq 2 \cdot 12 + 2V > 24$; so $u(1) = 0$, and in the same way $u(a) = 0$. Then $u = 0$

on K , $T = 4V \leq 24$, so $2v(1) \leq V \leq 6$ and $v(1) = 2$, and $f(xi) = (u(i) + v(1))/2 = 1$: case (a). $\square$

Step 5: the core dichotomy.

Proposition 3.18. *Let $e > 1$ be prime to 6, $U = (\mathbb{Z}/e)^\times$, and let $F: U \rightarrow \mathbb{Z}$ satisfy $F(-z) = -F(z)$, $\|F\| \leq 12$ and $F(x) \geq 1$ for some x , and be split along $(K_p)_{p|e}$ on every coset of their product H . Then*

- (1) $F \geq 1$ on a line wK_p for $p = 5$ or $p = 7$; or
- (2) the primes dividing e are 5 and 7, $|K_5| = 4$, $|K_7| = 6$, and there are $y \in U$, $\varepsilon \in K_5$ and $\varepsilon' \in K_7$ with $\varepsilon\varepsilon' = -1$, and functions $u: K_5 \rightarrow \{\pm 1\}$ and $v: K_7 \rightarrow \{\pm 1\}$ with $u(i\varepsilon) = -u(i)$, $v(j\varepsilon') = -v(j)$ and $2F(yij) = u(i) + v(j)$ for $i \in K_5$, $j \in K_7$; the mass of F on yK_5K_7 is 12.

Proof. By Lemma 3.12(a) every $|K_p|$ is at least 4, only K_5 can have order 4, and $|K_p| \leq 9$ forces $p \in \{5, 7\}$.

Case (a): $-1 \notin H$. The cosets xH and $-xH$ are distinct and carry the same mass, so $\sum_{xH} |F| \leq 6$. By Lemma 3.15, $F = \sigma$ on a line along some K_p with $|K_p| \leq 6$, so $p \in \{5, 7\}$, and $\sigma = 1$ because the line passes through x .

Case (b): $-1 \in H$ and $|K_p| \geq 10$ for some p . Write $-1 = \varepsilon\varepsilon'$ with $\varepsilon \in K_p$ and ε' in the product of the other K_q ; $\varepsilon \neq 1$ by Lemma 3.12(b). Lemma 3.16 gives (1).

Case (c): $-1 \in H$ and $|K_p| \leq 9$ for all p . Then every prime factor of e is 5 or 7. If only one prime p divides e , then $-1 \in K_p$ and F is K_p -invariant on xK_p by (S4), so $F(x) = F(-x) = -F(x)$, which is impossible. So $H = K_5K_7$, and $-1 = \varepsilon\varepsilon'$ with $\varepsilon \in K_5$, $\varepsilon' \in K_7$, both different from 1 by Lemma 3.12(b). From $\varepsilon^2\varepsilon'^2 = 1$ and $K_5 \cap K_7 = 1$ we get $\varepsilon^2 = \varepsilon'^2 = 1$, so $|K_5|$ and $|K_7|$ are even: $|K_5| = 4$ and $|K_7| = 6$. Lemma 3.17 on xK_5K_7 gives (1) or (2). $\square$

Step 6: the sum of the entries.

Lemma 3.19. *Let $p \mid e$ with $|K_p| \in \{4, 6\}$, and let $F: U \rightarrow \mathbb{Z}$ be odd with $F \geq 1$ on a line wK_p and $\|F\| \leq 2|K_p|$. Then $\|F\| = 2|K_p|$ and $\sum_z F(z)z \neq 0$ in $\mathbb{Z}/e$.*

Proof. If $e = p$ then $K_p = U \ni -1$, and $F(w) \geq 1$, $F(-w) \geq 1$ contradict oddness. So $e/p \geq 5$, $-1 \notin K_p$, and the lines wK_p and $-wK_p$ are disjoint. As $F \geq 1$ on the first and $F \leq -1$ on the second, $\|F\| = 2|K_p|$, $F = 1$ on wK_p , $F = -1$ on $-wK_p$, and $F = 0$ elsewhere. Then $\sum_z F(z)z = 2w \sum_{k \in K_p} k \equiv 2|K_p|w \pmod{e/p}$. If it vanished, e/p would divide $2|K_p| \in \{8, 12\}$, which is impossible for $e/p > 1$ prime to 6. $\square$

Lemma 3.20. *In case (2) of Proposition 3.18, if $\|F\| = 12$, then $\sum_z F(z)z \neq 0$ in $\mathbb{Z}/e$.*

Proof. The coset yK_5K_7 carries the whole mass, so $F = 0$ off it. Let $\psi: \mathbb{Z}/e \rightarrow \mathbb{Z}/5$ be reduction. Since $5 \mid e/7$, every $j \in K_7$ has $\psi(j) = 1$. Since $|K_5| = 4$, $5 \nmid e$, and ψ is injective on K_5 : an element of K_5 congruent to 1 modulo 5 is congruent to 1 modulo $\text{lcm}(5, e/5) = e$. From $\varepsilon\varepsilon' = -1$, $\psi(\varepsilon) = -1$. Write $K_5 = \{1, \varepsilon, a, a\varepsilon\}$; then $\psi(a) \neq \pm 1$. As $v(j\varepsilon') = -v(j)$, $\sum_j v(j) = 0$. Hence

$$\begin{aligned} \psi\left(2 \sum_z F(z)z\right) &= \sum_{i \in K_5} \sum_{j \in K_7} (u(i) + v(j)) \psi(y) \psi(i) \\ &= 6 \psi(y) \sum_i u(i) \psi(i) = 12 \psi(y) (u(1) + u(a) \psi(a)), \end{aligned}$$

using $u(\varepsilon) = -u(1)$ and $u(a\varepsilon) = -u(a)$. Here 12 and $\psi(y)$ are units of $\mathbb{Z}/5$. If $\sum_z F(z)z$ vanished, then $u(1) + u(a)\psi(a) = 0$, and $\psi(a) = \pm 1$, a contradiction. $\square$

Step 7: end of the proof. By Lemma 3.10, Lemma 3.14 and the remarks in Step 1, F satisfies the hypotheses of Proposition 3.18 with $\|F\| \leq 2n \leq 12$. In case (2) the mass is 12, so $n = 6$ and $\|F\| = 12 = 2n$, and Lemma 3.10 contradicts Lemma 3.20. So $F \geq 1$ on a line wK_p with $p \in \{5, 7\}$. Since F is odd, $-1 \notin K_p$, so wK_p and $-wK_p$ are disjoint and $\|F\| \geq 2|K_p|$.

If $p = 7$, then $|K_7| = 7$ is impossible as $14 > 12$, and $|K_7| = 6$ forces $n = 6$ and $\|F\| = 12 = 2n$, which Lemma 3.10 and Lemma 3.19 exclude. So $p = 5$. If $n = 4$, then $|K_5| = 5$ is impossible as $10 > 8$, and $|K_5| = 4$ forces $\|F\| = 8 = 2n$, excluded in the same way. This proves (i).

Let $n = 6$ and $p = 5$, and put $x = \kappa(w)$. For $k \in K_5$ we have $5k = 5$ in $\mathbb{Z}/e$, so $5\kappa(wk) = \kappa(5w) = 5x$; and $F(wk) \geq 1$ means $\kappa(wk) \in A$. So A contains the $|K_5|$ distinct elements of $L = \kappa(wK_5) \subseteq \Phi(x)$. Recall that $\sum_{y \in \Phi(x)} y = 5x$ and, writing $\bar{x} = qm/5 + r$ with $0 \leq r < m/5$,

$$\sum_{y \in \Phi(x)} \bar{y} = \sum_{j=0}^4 \left(((q+j) \bmod 5) \frac{m}{5} + r \right) = 2m + 5r = 2m + \bar{5x}.$$

Since $t\Phi(x) = \Phi(tx)$, also $\sum_{y \in \Phi(x)} t\bar{y} = 2m + \bar{5tx}$ for every unit t .

If $|K_5| = 5$, then $L = \Phi(x)$, and the sixth element b of A satisfies $5x + b = 0$; so $A = \Phi(x) + \{-5x\}$ is 5-standard.

If $|K_5| = 4$, then $L = \Phi(x) \setminus \{y_0\}$ for one y_0 , and $A = L + \{b, c\}$. Here $e \neq 5$, since otherwise $K_5 = U$ would contain -1 ; so $5x = \kappa(5w) \neq 0$. Hence $y_0 \neq 0$ (as $5y_0 = 5x$), and $y_0 \neq 5x$ (otherwise $20x = 0$, so $5x = 0$, as m is odd). The sum of A gives $5x - y_0 + b + c = 0$, and for every unit t ,

$$h(A) = 2m + \bar{5tx} - \bar{ty_0} + \bar{tb} + \bar{tc}.$$

So the multiset $Q = \{5x, b, c, -y_0\}$ of nonzero elements has sum 0, and $\sum_{q \in Q} \bar{tq} = \bar{5tx} + \bar{tb} + \bar{tc} + m - \bar{ty_0} = h(A) - m$ for every t : it is balanced. By (i) it contains a pair, formed by two of its four members. The pair $\{5x, -y_0\}$ is excluded by $y_0 \neq 5x$, and $\{b, c\}$ is excluded because A contains no pair. Each of the remaining four, $b = -5x$, $c = -5x$, $b = y_0$ or $c = y_0$, together with $5x - y_0 + b + c = 0$, gives $\{b, c\} = \{y_0, -5x\}$, so $A = \Phi(x) + \{-5x\}$ is 5-standard. This proves (ii). $\square$

Remark 3.21. Kang states the Hodge conjecture for every Fermat fourfold [Kan16, Corollary 3.2], and [MMRV26] rely on that statement. The proof bounds a refined motivic dimension of $X_m^1 \times X_m^{n-1}$ by [Kan15, Proposition 2.2(iv)], whose proof writes the level filtration of a product as $F^m(H \otimes H') = \bigoplus_{t+s=m} F^t H \otimes F^s H'$, where F^p is the largest rational sub-Hodge structure contained in the p -th step of the Hodge filtration. This decomposition fails already for $H = H' = H^1(E, \mathbb{Q})$, E an elliptic curve, and $m = 1$: the Künneth component of the diagonal of $E \times E$ in $H^1(E) \otimes H^1(E)$ is a nonzero rational class of type $(1, 1)$, so it lies in $F^1(H \otimes H')$, while $F^1 H^1(E) = 0$ and the right-hand side vanishes. We do not know whether the inequality of [Kan15, Proposition 2.2(iv)], or [Kan16, Corollary 3.2], holds; Theorem F does not use them.

3.7. Fermat fourfolds of odd degree divisible by three. In this subsection m is odd and divisible by 3, and we prove Theorem F for such m . Here Theorem 3.9 fails: the set $\{1, 4, 9, 15, 16, 18\}$ of residues modulo 21 is balanced, contains no pair and is not 5-standard. So we replace the classification by an induction. A balanced sextuple either has a *move*, which reduces it to a smaller one by means of one cycle of Aoki, or lies in one of three explicit orbits (Theorem 3.27), whose classes we write down directly (Lemma 3.26). We keep the notation of Section 3.6.

Algebraic multisets. In this paragraph $M \geq 2$ is any integer; the even levels $M = 66$ and $M = 78$ occur in Lemma 3.26. We use Definition 3.8 at level M ; when M is even, a *pair* is a

submultiset $\{a, -a\}$, and $\{M/2, M/2\}$ is one. Let A be a balanced multiset of an even number $n \geq 2$ of nonzero elements of $\mathbb{Z}/M$. Ordering its elements gives a character $\alpha \in \mathfrak{A}_M^{n-2}$, and by Proposition 3.6(ii) the line $V(\alpha)$ consists of Hodge classes; for $n = 2$, X_M^0 is a set of M points. Permuting the coordinates of X_M^{n-2} permutes these lines, so the following notion does not depend on the ordering: A is *algebraic* if $V(\alpha) \subseteq \text{Alg}^{(n-2)/2}(X_M^{n-2}) \otimes \mathbb{C}$. For an odd prime $\ell \mid M$ and $x \in \mathbb{Z}/M$ we put

$$\Phi_\ell(x) = \{x + jM/\ell : 0 \leq j < \ell\}, \quad \sigma_{\ell,x} = \Phi_\ell(x) + \{-\ell x\},$$

so that $\Phi_5(x)$ is the fiber $\Phi(x)$ of Definition 3.8.

Lemma 3.22. *Let A and B be balanced multisets of an even number of nonzero elements of $\mathbb{Z}/M$.*

- (a) *If $|A| \leq 4$, then A is algebraic.*
- (b) *If A and B are algebraic, so is $A + B$. If P is a union of pairs and $A + P$ is algebraic, then A is algebraic.*
- (c) *Let $t \in (\mathbb{Z}/M)^\times$. Then tA is balanced, and it is algebraic if and only if A is.*
- (d) *Let $g \mid M$, $M' = M/g$, let A' be a multiset of nonzero elements of $\mathbb{Z}/M'$ with sum 0, and let $gA' \subseteq \mathbb{Z}/M$ be its image under $y \mapsto gy$. Then gA' is balanced if and only if A' is, and in that case gA' is algebraic if and only if A' is.*
- (e) *Let M be odd, ℓ an odd prime dividing M , and $x \in \mathbb{Z}/M$ with $\ell x \neq 0$. Then $\sigma_{\ell,x}$ is a balanced multiset of $\ell + 1$ nonzero elements, and it is algebraic.*

Proof. (a) For $|A| = 2$, $A = \{a, -a\}$, and $H^0(X_M^0)$ is spanned by the classes of points. For $|A| = 4$, $V(\alpha)$ consists of Hodge classes on the surface X_M^2 , which are algebraic by Theorem 2.3.

(b) This is the inductive structure of Shioda and Ran, [Aok87, Theorem 1-4]: a union of pairs is a character of the kind denoted there by δ , whose line is spanned by the class of a linear subspace [Aok87, Theorem 1-1].

(c) The first assertion is clear. As in the proof of Proposition 3.6(ii), the automorphism $\zeta \mapsto \zeta^t$ of $\mathbb{Q}(\mu_M)$, acting on $H^{n-2}(X_M^{n-2}, \mathbb{Q}) \otimes \mathbb{Q}(\mu_M)$ through the second factor, carries the line $V(\alpha)$, which is defined over $\mathbb{Q}(\mu_M)$, onto $V(t\alpha)$, and it preserves $\text{Alg}^{(n-2)/2} \otimes \mathbb{Q}(\mu_M)$.

(d) Every unit modulo M' lifts to a unit modulo M , and for a unit t modulo M with image t' we have $\overline{tgy} = g\overline{t'y}$ for $y \in \mathbb{Z}/M'$, the bars denoting representatives modulo M and M' ; this gives the first assertion. Let $\pi: X_M^{n-2} \rightarrow X_{M'}^{n-2}$ be the morphism $[x_i] \mapsto [x_i^g]$ and α' an ordering of A' . In the proof of Theorem F for m prime to 6 we saw that π^* is injective and maps $V(\alpha')$ into $V(g\alpha')$; both are lines, so $\pi^*V(\alpha') = V(g\alpha')$. If A' is algebraic, then so is gA' , because π^* preserves algebraic classes. Conversely $V(\alpha') = (\deg \pi)^{-1} \pi_* \pi^* V(\alpha') = \pi_* V(g\alpha')$, and π_* preserves algebraic classes.

(e) Since $\ell x \neq 0$, $0 \notin \Phi_\ell(x)$. The sum of $\sigma_{\ell,x}$ is $\ell x + \binom{\ell}{2} M/\ell - \ell x = 0$, as ℓ is odd. For a unit t we have $t\Phi_\ell(x) = \Phi_\ell(tx)$, and writing $\overline{tx} = qM/\ell + r$ with $0 \leq r < M/\ell$,

$$\sum_{y \in \Phi_\ell(tx)} \overline{y} = \sum_{j=0}^{\ell-1} \left(((q+j) \bmod \ell) \frac{M}{\ell} + r \right) = \frac{(\ell-1)M}{2} + \overline{\ell tx};$$

adding $-\overline{\ell tx} = M - \overline{\ell tx}$ gives $(\ell+1)M/2$. So $\sigma_{\ell,x}$ is balanced. For $\ell = 3$ it is algebraic by (a). For $\ell \geq 5$ the argument in the proof of Theorem F for m prime to 6, case α 5-standard, applies with 5 replaced by ℓ : with $g = \gcd(\bar{x}, M/\ell)$ we have $\sigma_{\ell,x} = g\sigma_{\ell,x'}$, where $\sigma_{\ell,x'}$ lives at the odd level $M' = M/g$ and $\gcd(x', M'/\ell) = 1$; moreover $M'/\ell \geq 3$, since for $M' = \ell$ the set $\Phi_\ell(x')$ would contain 0. Aoki constructs a subvariety of $X_{M'}^{\ell-1}$ of codimension $(\ell-1)/2$ that represents $\sigma_{\ell,x'}$ [Aok87, Theorem 2-1], so $\sigma_{\ell,x'}$ is algebraic, and so is $\sigma_{\ell,x}$ by (d). $\square$

Moves. From now on m is odd and divisible by 3. A *sextuple* is a balanced multiset of six nonzero elements of $\mathbb{Z}/m$. The *profile* $o(A)$ of a sextuple A is the sequence of the additive orders of its elements in nonincreasing order, and profiles are compared lexicographically.

Definition 3.23. Let A be a sextuple, ℓ an odd prime dividing m , and $x \in \mathbb{Z}/m$ with $\ell x \neq 0$. Let $S \subseteq A$ consist of one copy of each element of $\Phi_\ell(x)$ that lies in A , let $k = |S|$, and put

$$A'' = (A - S) + (-(\Phi_\ell(x) \setminus S)) + \{\ell x\},$$

a multiset of $7 + \ell - 2k$ elements. The pair (ℓ, x) is a *direct move* for A if $|A''| \leq 4$, and a *lowering move* if $|A''| = 6$ and $o(A'') < o(A)$.

Lemma 3.24. *In Definition 3.23, A'' is a balanced multiset of nonzero elements, and $A'' + \sigma_{\ell, x} = A + P$ for a union of pairs P . If A'' is algebraic, then so is A . In particular A is algebraic if it has a direct move.*

Proof. The elements of A'' are nonzero, because $0 \notin \Phi_\ell(x)$ and $\ell x \neq 0$. The identity holds with $P = (\Phi_\ell(x) \setminus S) + (-(\Phi_\ell(x) \setminus S)) + \{\ell x, -\ell x\}$. The sum and the function $t \mapsto \sum_a \bar{t}a$ are additive in the multiset, and A , P and $\sigma_{\ell, x}$ are balanced (Lemma 3.22(e)); so A'' is balanced. If A'' is algebraic, then $A'' + \sigma_{\ell, x} = A + P$ is algebraic by Lemma 3.22(e),(b), and so is A , by Lemma 3.22(b). After a direct move, A'' has at most four elements, so it is algebraic by Lemma 3.22(a). $\square$

Lemma 3.25. *Let A be a sextuple, $x \in \mathbb{Z}/m$ an element of order e , and $\ell \neq e$ an odd prime dividing e . If A contains at least $(\ell + 1)/2$ distinct elements of $\Phi_\ell(x)$ of order e , then (ℓ, x) is a direct or a lowering move for A .*

Proof. Here $\ell x \neq 0$, as $e \nmid \ell$. Since $\ell \mid e$, the element $m/\ell = (m/e)(e/\ell)$ lies in the subgroup of order e , and hence so does $\Phi_\ell(x)$; and ℓx has order e/ℓ . If $k \geq (\ell + 3)/2$, then $|A''| \leq 4$. Otherwise $k = (\ell + 1)/2$, $|A''| = 6$, and S consists of elements of order e . Passing from A to A'' removes these k elements and adds $\ell + 1 - k = k$ elements of order at most e , one of which, ℓx , has order e/ℓ . So A and A'' have the same elements of order larger than e , and A'' has fewer elements of order e ; hence $o(A'') < o(A)$. $\square$

Three exceptional orbits. Let

$$O_{21} = \{1, 4, 9, 15, 16, 18\}, \quad O_{33} = \{1, 4, 16, 22, 25, 31\}, \quad O_{39} = \{1, 7, 16, 22, 34, 37\},$$

subsets of $\mathbb{Z}/21$, $\mathbb{Z}/33$ and $\mathbb{Z}/39$. In terms of residues: O_{21} consists of the three elements congruent to 1 modulo 3 whose residue modulo 7 is a nonzero square, and the three elements $3v$ with v a nonsquare modulo 7; O_{33} consists of the five elements congruent to 1 modulo 3 whose residue modulo 11 is a nonzero square, and 22; and O_{39} consists of the six elements congruent to 1 modulo 3 whose residue modulo 13 lies in $\{1, 3, 9\} \cup \{7, 8, 11\}$, the union of two cosets of the subgroup generated by 3 in $(\mathbb{Z}/13)^\times$.

Lemma 3.26. *The sets O_{21} , O_{33} and O_{39} are balanced and algebraic.*

Proof. That they are balanced, that is, $\sum_a \bar{t}a = 3m$ for every unit t , is a direct check of 6, 10 and 12 sums, as t and $-t$ give $3m$ together. For an even level $M = 2M'$ and $x \in \mathbb{Z}/M$ with $2x \neq 0$, the multiset $T(x) = \{x, x + M', -2x, M'\}$ of nonzero elements has sum 0, and it is balanced: for a unit t , which is odd, put $y = \bar{t}x \neq M'$; then $tM' = M'$, and the representatives of the entries of $tT(x)$ are $y, y + M', M - 2y, M'$ if $y < M'$, and $y, y - M', 2M - 2y, M'$ if $y > M'$, adding up to $2M$ in both cases. These are Aoki's 2-standard elements [Aok87, §1].

O_{21} . With $\sigma_{3,2} = \{2, 9, 16\} + \{15\}$ and $Q = \{1, 4, 18, 19\}$ we have

$$O_{21} + \{2, 19\} = \sigma_{3,2} + Q.$$

As O_{21} , the pair and $\sigma_{3,2}$ are balanced, so is Q . By Lemma 3.22(a) $\sigma_{3,2}$ and Q are algebraic, so $O_{21} + \{2, 19\}$ is algebraic by Lemma 3.22(b), and so is O_{21} .

O_{33} . By Lemma 3.22(d) it suffices to show that $a = 2O_{33} = \{2, 8, 32, 44, 50, 62\} \subseteq \mathbb{Z}/66$ is algebraic; it is balanced because O_{33} is. In $\mathbb{Z}/66$,

$$a + \{1, 65\} + \{25, 41\} + \{33, 33\} = Q + T(32) + T(8),$$

with $Q = \{1, 25, 44, 62\}$, $T(32) = \{2, 32, 33, 65\}$ and $T(8) = \{8, 33, 41, 50\}$. As before Q is balanced, all three quadruples on the right are algebraic, so is their sum, and a is algebraic by Lemma 3.22(b).

O_{39} . In the same way, $a = 2O_{39} = \{2, 14, 32, 44, 68, 74\} \subseteq \mathbb{Z}/78$ satisfies

$$a + \{5, 73\} + \{7, 71\} + \{39, 39\} = Q + T(32) + T(5),$$

with $Q = \{2, 7, 73, 74\}$, $T(32) = \{14, 32, 39, 71\}$ and $T(5) = \{5, 39, 44, 68\}$, and we conclude as for O_{33} . $\square$

None of the three sets has a direct or a lowering move; this is a finite check, which the programs of Appendix A also carry out. For $m = 21$ Lemma 3.26 is contained in [dSJ21], and for $m = 33$ and $m = 39$ in [Jum26]: the identity used for O_{33} is [Jum26, Theorem 1.4], while [Jum26] treats O_{39} through the correspondence of Shioda and Katsura [Jum26, Theorem 1.2]. We have not found the identity at level 78 in the literature.

The reduction.

Theorem 3.27. *Let m be odd and divisible by 3, and let A be a sextuple that contains no pair and generates $\mathbb{Z}/m$. Then A has a direct or a lowering move, or $m \in \{21, 33, 39\}$ and $tA = O_m$ for some $t \in (\mathbb{Z}/m)^\times$.*

Proof of Theorem F for odd m divisible by 3, given Theorem 3.27. As in the proof for m prime to 6, it suffices to show that every sextuple A is algebraic. We argue by induction on m and, for fixed m , on $o(A)$; at the odd levels prime to 6 every sextuple is algebraic by the proof in Section 3.6. If A contains a pair $\{a, -a\}$, then $A - \{a, -a\}$ is balanced, so A is algebraic by Lemma 3.22(a),(b). If A lies in the subgroup $g\mathbb{Z}/m$ for some $g > 1$ dividing m , then $A = gA'$ for a sextuple A' at the odd level $m/g < m$, which is algebraic, and so is A , by Lemma 3.22(d). Otherwise Theorem 3.27 applies. If A has a direct move, it is algebraic by Lemma 3.24. If it has a lowering move, then A'' is a sextuple with $o(A'') < o(A)$, hence algebraic, and A is algebraic by Lemma 3.24. In the remaining case tA is algebraic by Lemma 3.26, and so is A , by Lemma 3.22(c). $\square$

The rest of this subsection proves Theorem 3.27. We fix A as there and assume that A has no direct and no lowering move.

Step 1: maximal orders and the character relation. An order e of an element of A is *maximal* if no element of A has an order that is a proper multiple of e . As A generates $\mathbb{Z}/m$, m is the least common multiple of the maximal orders. For a maximal order e we use the objects of Step 1 of Section 3.6: the embedding $\kappa: \mathbb{Z}/e \rightarrow \mathbb{Z}/m$, $U = (\mathbb{Z}/e)^\times$, and the odd function F on U ; we let N_e be the number of elements of A of order e . As A contains no pair, $|F(z)| = \text{mult}_A(\kappa z) + \text{mult}_A(-\kappa z)$, so $\|F\| = 2N_e \leq 12$. Lemma 3.10 holds with the same proof. For a prime $p \mid e$ the subgroup $K_p \subseteq U$ and the product H are defined as in Step 3 there; Lemma 3.12(b),(c) hold with the same proofs, and by Lemma 3.12(a), $|K_p| \geq 4$ for $p \geq 5$, with equality only for $p = 5 \parallel e$, while $|K_3| = 3$ if $9 \mid e$ and $|K_3| = 2$ if $3 \parallel e$.

Lemma 3.28. *Let B be a balanced multiset of nonzero elements of $\mathbb{Z}/m$. For $a \in B$ let f_a be its order, and write $a = (m/f_a)z_a$ with $z_a \in (\mathbb{Z}/f_a)^\times$. Let χ be an odd primitive Dirichlet character of conductor $c \mid m$. Then*

$$(1) \quad \sum_{a \in B, c \mid f_a} \frac{\bar{\chi}(z_a)}{\varphi(f_a)} \prod_{\ell \mid f_a, \ell \nmid c} (1 - \chi(\ell)) = 0,$$

where ℓ runs over primes and $\bar{\chi}(z_a)$ is the value at the residue of z_a modulo c .

Proof. First, for a multiple f of c ,

$$(2) \quad \sum_{\substack{1 \leq s \leq f \\ (s, f) = 1}} \chi(s) s = f B_{1, \chi} \prod_{\ell \mid f, \ell \nmid c} (1 - \chi(\ell)).$$

Indeed, for N divisible by c , $\sum_{s=1}^N \chi(s) s = \sum_{r=1}^c \chi(r) \sum_{j=0}^{N/c-1} (r + jc) = N B_{1, \chi}$, since $\sum_r \chi(r) = 0$. Let f_0 be the product of the primes dividing f but not c . As $\chi(s) = 0$ unless $(s, c) = 1$, and then $(s, f) = 1$ if and only if $(s, f_0) = 1$, Möbius inversion gives

$$\sum_{(s, f) = 1} \chi(s) s = \sum_{d \mid f_0} \mu(d) \chi(d) d \sum_{s'=1}^{f/d} \chi(s') s' = f B_{1, \chi} \sum_{d \mid f_0} \mu(d) \chi(d),$$

which is (2). Now regard χ as a character of $(\mathbb{Z}/m)^\times$ through reduction modulo c . Since B is balanced, $\sum_{a \in B} \sum_t \chi(t) \overline{ta} = h(B) \sum_t \chi(t) = 0$. If $c \nmid f_a$, the inner sum vanishes, as in the proof of Lemma 3.11: χ is not trivial on the units congruent to 1 modulo f_a , since otherwise it would be induced from a character modulo the proper divisor $\gcd(f_a, c)$ of c . If $c \mid f_a$, then $\overline{ta} = (m/f_a) \overline{tz_a}$, with $\overline{tz_a} \in \{0, \dots, f_a - 1\}$, and $\chi(t)$ depends only on t modulo f_a ; so, substituting $s \mapsto sz_a^{-1}$ and using (2),

$$\sum_t \chi(t) \overline{ta} = \frac{m \varphi(m)}{f_a \varphi(f_a)} \bar{\chi}(z_a) \sum_{s \in (\mathbb{Z}/f_a)^\times} \chi(s) \bar{s} = \frac{m \varphi(m) B_{1, \chi}}{\varphi(f_a)} \bar{\chi}(z_a) \prod_{\ell \mid f_a, \ell \nmid c} (1 - \chi(\ell)).$$

Adding over a and dividing by $m \varphi(m) B_{1, \chi} \neq 0$ gives (1). $\square$

For a maximal order e and $c = e$, the only elements in (1) are those of order e , and (1) becomes the relation $\sum_z \text{mult}_A(\kappa z) \chi(z)^{-1} = 0$ of the proof of Lemma 3.11. So Lemma 3.11 holds for every maximal order e , and with it Lemma 3.14: F is split along $(K_p)_{p|e}$ on every coset of H .

Lemma 3.29. *Let e be a maximal order, p a prime dividing e and $w \in U$. Then $\kappa(wK_p) \subseteq \Phi_p(\kappa w)$, with equality if $p^2 \mid e$. If $e \neq p$, $|K_p| \geq (p+1)/2$ and $F \geq 1$ on wK_p , then A has a direct or a lowering move.*

Proof. An element $k \in K_p$ is congruent to 1 modulo e/p , so $\kappa(wk) - \kappa(w) = (m/e)w(k-1)$ lies in $(m/e)(e/p)\mathbb{Z} = (m/p)\mathbb{Z}$. If $p^2 \mid e$, both sides have p elements. If $F \geq 1$ on wK_p , then the $|K_p|$ elements $\kappa(wk)$ of order e lie in A , and Lemma 3.25 applies. $\square$

Step 2: every maximal order is divisible by 3. Let e be a maximal order prime to 3; then e is prime to 6. The function F is odd, $\|F\| \leq 12$, $F(z_0) \geq 1$ when $\kappa(z_0) \in A$ has order e , and F is split, so Proposition 3.18 applies. In its case (2), $\|F\| = 12$, so all six elements of A have order e , and Lemma 3.10 gives $\sum_z F(z)z = 0$, against Lemma 3.20. In case (1), $F \geq 1$ on a line wK_p with $p \in \{5, 7\}$. As F is odd, $-1 \notin K_p$, so $e \neq p$, and $|K_p| \geq 4 \geq (p+1)/2$; by Lemma 3.29 A has a move, which we excluded.

Step 3: no maximal order is divisible by 9. Let e be a maximal order with $9 \mid e$. Each K_p consists of units congruent to 1 modulo e/p , and $3 \mid e/p$; so H consists of units congruent to 1 modulo 3, and $-1 \notin H$. Let $F(x) \geq 1$ and $C = xH$. The cosets C and $-C$ are distinct and carry the same mass, so $\sum_C |F| \leq 6$. Write $K_3 = \{1, k, k^2\}$ and $H' = \prod_{p \neq 3} K_p$, and let $S_c = xcH'$, for $c \in K_3$, be the slices of C , with $\mu(c) = \sum_{S_c} |F|$, as in Step 4 of Section 3.6.

(i) *No line zK_3 contains two points $z_1 \neq z_2$ with $F(z_1)F(z_2) > 0$.* Otherwise, replacing z_1, z_2 by $-z_1, -z_2$ if necessary, $F(z_1), F(z_2) \geq 1$, and $\kappa(z_1), \kappa(z_2)$ are two elements of A of order e in $\Phi_3(\kappa z_1)$ (Lemma 3.29); as $e \neq 3$, Lemma 3.25 gives a move.

(ii) $\mu(c) \geq 1$ for every c . Otherwise, by (S2), F is split along $(K_p)_{p \neq 3}$ on $S_1 \ni x$, where its mass is at most 6. These subgroups have orders at least 4, with equality for at most one of them, so by Lemma 3.15 $F = 1$ on a line xK_p with $p \in \{5, 7\}$ and $|K_p| \leq 6$, and Lemma 3.29 gives a move.

(iii) For $c \neq c'$ in K_3 the function $G(z) = F(z) - F(zc^{-1}c')$ on S_c is split along $(K_p)_{p \neq 3}$ by (S1), and $\sum_{S_c} |G| \leq \mu(c) + \mu(c') \leq 5$ by (ii). It is not zero: by (ii) some $z \in S_c$ has $F(z) \neq 0$, and $F(zc^{-1}c') = F(z)$ would contradict (i), as z and $zc^{-1}c'$ lie on the line zK_3 . By Lemma 3.15, $|G| = 1$ on a line along some K_p with $|K_p| \leq 5$ and $G = 0$ elsewhere, so $p = 5$ and $\mu(c) + \mu(c') \geq |K_5| \geq 4$. Adding over the three pairs $\{c, c'\}$ gives $2 \sum_c \mu(c) \geq 12$. As $\sum_c \mu(c) \leq 6$, we get $|K_5| = 4$ and $\mu(c) = 2$ for every c .

Now take $c = 1$ and $c' = k$: $G = \sigma$ on a line $L \subseteq S_1$ along K_5 and $G = 0$ elsewhere, for a sign σ . By (i), $F(z)F(zk) \leq 0$ for all z . Off L , $F(z) = F(zk)$, so both vanish; on L , $F(z) - F(zk) = \sigma$, so $(F(z), F(zk))$ is $(\sigma, 0)$ or $(0, -\sigma)$. Since $\mu(1) = 2$, $F = \sigma$ at two points of L , and $F = 0$ at the other two points z of L , where $F(zk) = -\sigma$. The same argument with $c' = k^2$ gives a line $L' \subseteq S_1$ and a sign σ' such that F takes values in $\{0, \sigma'\}$ on S_1 , with support in L' ; so $L' = L$ and $\sigma' = \sigma$, and $F(zk^2) = -\sigma$ at the two points $z \in L$ with $F(z) = 0$. For such a z we get $F(zk) = F(zk^2) = -\sigma$, against (i).

Step 4: the case $3 \parallel e$. By Steps 2 and 3, every maximal order e satisfies $3 \parallel e$; write $e = 3e^*$. Then $e^* > 1$: for $e = 3$, $U = K_3 = \{\pm 1\}$, and F would be K_3 -invariant by (S4), so $F(1) = F(-1) = -F(1)$, while $F \neq 0$. Fix a maximal order e . By the Chinese remainder theorem $U = (\mathbb{Z}/3)^\times \times U^*$ with $U^* = (\mathbb{Z}/e^*)^\times$, and we write $z = (s, u)$ and $\kappa(s, u)$. Then $K_3 = (\mathbb{Z}/3)^\times \times \{1\}$, and for $p \mid e^*$, $K_p = \{1\} \times K_p^*$, where $K_p^* \subseteq U^*$ consists of the units congruent to 1 modulo e^*/p ; so $H = K_3 \times H^*$ with $H^* = \prod_{p \mid e^*} K_p^*$.

For $u \in U^*$ the points $(1, u)$ and $(-1, u)$ form a line along K_3 . If $F(1, u)F(-1, u) > 0$, then A has a move, as in (i) of Step 3. So $F(1, u)F(-1, u) \leq 0$ for all u , and we define $D: U^* \rightarrow \mathbb{Z}$ by $D(u) = F(1, u) - F(-1, u)$. The elements of A of order e are its *tops* at e .

Lemma 3.30. (a) D is even, $\|D\| = 2N_e$, and D is split along $(K_p^*)_{p \mid e^*}$ on every coset of H^* .

(b) For $u \in U^*$, the tops of the form $\kappa(s, \pm u)$ are $|D(u)|$ in number, and each of them has $s = \text{sign } D(u)$.

Proof. (a) Since F is odd, $D(-u) = F(1, -u) - F(-1, -u) = -F(-1, u) + F(1, u) = D(u)$. As $F(1, u)F(-1, u) \leq 0$, $|D(u)| = |F(1, u)| + |F(-1, u)|$, and these add up to $\|F\| = 2N_e$. Applying (S1) with the element $(-1, 1)$ of K_3 to the cosets $(1, u)(\{1\} \times H^*)$ shows that D is split. (b) If $D(u) > 0$, then $F(1, u) \geq 0 \geq F(-1, u)$; as A contains no pair, it contains $\kappa(1, u)$ with multiplicity $F(1, u)$, $\kappa(1, -u) = -\kappa(-1, u)$ with multiplicity $-F(-1, u)$, and neither $\kappa(-1, u)$ nor $\kappa(-1, -u)$. The case $D(u) < 0$ is symmetric, and if $D(u) = 0$ there are no such tops. $\square$

Step 5: even split functions.

Lemma 3.31. *Let $e^* > 1$ be prime to 6, and let $D: U^* \rightarrow \mathbb{Z}$ be even, not zero, with $\|D\| \leq 12$, and split along $(K_p^*)_{p|e^*}$ on every coset of H^* . Then one of the following holds.*

- (E1) $D = \sigma(\mathbf{1}_L + \mathbf{1}_{-L})$ for a sign σ and a line L along K_5^* or K_7^* with $L \cap (-L) = \emptyset$.
- (E2) e^* is a prime $q \leq 13$ and D is constant.
- (E3) $e^* = 35$, and $D(u) = g(u \bmod 5) + h(u \bmod 7)$ with even functions g on $(\mathbb{Z}/5)^\times$ and h on $(\mathbb{Z}/7)^\times$.

Proof. By Lemma 3.12(a), $|K_p^*| \geq 4$, with equality only for $p = 5 \parallel e^*$, and $|K_p^*| \leq 6$ only for $p \in \{5, 7\}$.

e^* is not squarefree, say $p^2 \mid e^*$. Every K_q^* consists of units congruent to 1 modulo e^*/q , hence modulo p ; so $-1 \notin H^*$. For a coset C of H^* , the cosets C and $-C$ are distinct and carry the same mass, so $\sum_C |D| \leq 6$. By Lemma 3.15, on each coset C where $D \neq 0$, $D = \sigma_C$ on a line L_C along K_5^* or K_7^* and $D = 0$ elsewhere on C ; as D is even, D is then σ_C on $-L_C \subseteq -C$. Each such pair of cosets carries mass at least 8, so there is only one, and (E1) holds.

$e^* = q$ is prime. Then $U^* = K_q^*$, so D is constant by (S4), and $\|D\| = |D(1)|(q-1) \leq 12$ gives $q \leq 13$: (E2).

e^* is squarefree with at least two prime factors, one of them $p \geq 11$. Now $H^* = U^*$, by the Chinese remainder theorem. Put $H^* = \prod_{q \neq p} K_q^*$, and consider the slices $S_c = cH'^*$, $c \in K_p^*$, with masses $\mu(c)$. Let $\varepsilon_p \in K_p^*$ be congruent to -1 modulo p ; then $-1 = \varepsilon_p \varepsilon'$ with $1 \neq \varepsilon' \in H'^*$, so $-1 \notin K_p^*$, and $-S_c = S_{c\varepsilon_p}$, so $\mu(c\varepsilon_p) = \mu(c)$, with $c\varepsilon_p \neq c$. Suppose that every slice carries mass. As $|K_p^*| \geq 10$ and $\sum \mu \leq 12$, at most one slice has $\mu \geq 3$, hence none, by the pairing; and some c_0 has $\mu(c_0) = 1$. For every c , the function $D(z) - D(zc_0^{-1}c)$ on S_{c_0} is split along $(K_q^*)_{q \neq p}$ by (S1) and has mass at most 3, so it vanishes by Lemma 3.15. Hence D is constant on $z_0 K_p^*$, where z_0 is the point of S_{c_0} with $D(z_0) \neq 0$, and vanishes elsewhere. As $D(-z_0) = D(z_0) \neq 0$, this gives $-1 \in K_p^*$, a contradiction. So some slice has $\mu = 0$, and by (S2) D is split along $(K_q^*)_{q \neq p}$ on every slice. As $\mu(c) = \mu(c\varepsilon_p)$, every slice has mass at most 6, and we conclude by Lemma 3.15 as in the first case, with slices in place of cosets: (E1).

Otherwise e^* is squarefree and its prime factors are 5 and 7, so $e^* = 35$ and $U^* = K_5^* K_7^*$. By (S3), $D(ij) = D(i) + D(j) - D(1)$ for $i \in K_5^*$, $j \in K_7^*$; this gives $D(u) = g(u \bmod 5) + h(u \bmod 7)$, with $g(i) = D(i) - D(1)$ and $h(j) = D(j)$, as i and j are determined by $u \bmod 5$ and $u \bmod 7$. Since D is even, $g(-i) - g(i) = h(j) - h(-j)$ for all i, j ; both sides are a constant γ , and replacing i by $-i$ shows $\gamma = -\gamma$. So g and h are even: (E3). $\square$

Step 6: two lemmas.

Lemma 3.32. *Let $M' > 1$ be prime to 6. There are primitive Dirichlet characters modulo M' of both parities. Among those of a given parity, the values at 3 are not all equal, unless $M' = 5$ and the parity is even.*

Proof. A character modulo M' is a product of characters ψ_p modulo the prime powers $p^a \parallel M'$, and it is primitive if and only if every ψ_p is. The characters modulo p^a are the powers θ^j of a generator θ , which is faithful; so $\theta(-1) = -1$, and $\theta(9) \neq 1$ because $9 \not\equiv 1 \pmod{p}$. The character θ^j has parity $(-1)^j$, and it is primitive if and only if $(p-1) \nmid j$ when $a = 1$, and $p \nmid j$ when $a \geq 2$. Both conditions hold for $j = 1, 2$, so both parities occur. Replacing $\psi_p = \theta^j$ by $\theta^{j \pm 2}$ keeps the parity and multiplies the value at 3 by $\theta(9)^{\pm 1} \neq 1$. One of θ^{j+2} and θ^{j-2} is primitive, since p cannot divide both $j+2$ and $j-2$, and $p-1$ divides both only if $p = 5$ and $j \equiv 2 \pmod{4}$. So the values at 3 in a parity class are constant only if every ψ_p is the quadratic character modulo $p^a = 5$, that is, only if $M' = 5$ and the parity is even. $\square$

Lemma 3.33. *Let $e = 3e^*$ be a maximal order and $b \in A$ an element whose order $f \neq e$ is a multiple of e^* . Then $f = e^*$.*

Proof. The order f divides some maximal order e' , and $3 \mid e'$ by Step 2; so $e = 3e^*$ divides e' , and $e' = e$ by the maximality of e . Hence $e^* \mid f \mid e$ and $f \neq e$, so $f = e^*$. $\square$

Step 7: the cases (E1) and (E3) do not occur. Let $e = 3e^*$ be a maximal order. By Lemma 3.30(a), Lemma 3.31 applies to D .

Case (E1). Let $L = wK_p^*$ with $p \in \{5, 7\}$. By Lemma 3.30(b), $N_e = |L|$, and the tops are the elements $\kappa(\sigma, \varepsilon_u u)$, $u \in L$, one for each u , with signs ε_u . For a sign ε , the tops with $\varepsilon_u = \varepsilon$ lie in $\kappa((\sigma, \varepsilon w)K_p) \subseteq \Phi_p(\kappa(\sigma, \varepsilon w))$ by Lemma 3.29, and $e \neq p$; so by Lemma 3.25 fewer than $(p+1)/2$ of them have $\varepsilon_u = \varepsilon$. If $|L| = 5$, this is impossible. Since $2|L| \leq 12$, the remaining cases are $|L| = 4$, $p = 5$, with two tops of each sign, and $|L| = 6$, $p = 7$, with three of each sign.

Let $p = 5$ and $|L| = 4$, so $5 \parallel e^*$. Here $e^* \neq 5$, since $K_5^* = U^*$ would contain -1 ; so $e^* = 5e'$ with $e' > 1$ prime to 30, and reduction modulo 5 identifies K_5^* with $(\mathbb{Z}/5)^\times$. Put $\varepsilon'(k) = \varepsilon_{wk}$, a function on $(\mathbb{Z}/5)^\times$ with two values 1 and two values -1 . Let $\chi = \chi_5 \chi'$ be an odd primitive character modulo e^* , with χ_5 modulo 5 and χ' modulo e' . By Lemma 3.33, the elements $a \in A$ with $e^* \mid f_a$ are the tops and the elements of order e^* , and (1) with $c = e^*$, multiplied by $\varphi(e) = 2\varphi(e^*)$, reads

$$(1 - \chi(3)) \sum_{u \in L} \bar{\chi}(\varepsilon_u u) + 2 \sum_{f_b = e^*} \bar{\chi}(z_b) = 0.$$

As χ is odd and $\chi(k) = \chi_5(k)$ for $k \in K_5^*$, the first sum is $\bar{\chi}(w)E(\chi_5)$ with $E(\chi_5) = \sum_k \varepsilon'(k) \bar{\chi}_5(k)$. Since $\sum_k \varepsilon'(k) = 0$, E vanishes at the trivial character, and, with λ_5 the quadratic character, $E(\lambda_5) = 2(\varepsilon'(1) + \varepsilon'(4))$. So either $\varepsilon' = \pm \lambda_5$, and then E vanishes at the two characters of order 4; or $E(\lambda_5) = 0$, and then E does not vanish at some character of order 4, as $\varepsilon' \neq 0$. If no element of A has order e^* , choose χ_5 with $E(\chi_5) \neq 0$, and by Lemma 3.32 (note $e' \neq 5$) a primitive χ' with $\chi'(-1) = -\chi_5(-1)$ and $\chi'(3) \neq \chi_5(3)^{-1}$; the relation then fails. If exactly one element b has order e^* , choose $\chi_5 \neq 1$ with $E(\chi_5) = 0$ and a primitive χ' with $\chi'(-1) = -\chi_5(-1)$; the relation becomes $2\bar{\chi}(z_b) = 0$, which is absurd. If two elements have order e^* , then all six elements of A lie in $\kappa(\mathbb{Z}/e)$; writing them as $\kappa(y)$, we get $\sum y = 0$ in $\mathbb{Z}/e$, while the residues of the y modulo 3 are σ for the four tops and 0 for the other two, so $4\sigma \equiv 0 \pmod{3}$, which is false.

Let $p = 7$ and $|L| = 6$. Then $N_e = 6$, so A consists of tops, $m = e$, and $\sum_{u \in L} \varepsilon_u u = 0$ in $\mathbb{Z}/e^*$, because $\sum_{a \in A} a = 0$. Here $7 \parallel e^*$ and, as before, $e^* = 7e'$ with $e' > 1$ prime to 42. Put $\varepsilon'(k) = \varepsilon_{wk}$ for $k \in K_7^*$, identified with $(\mathbb{Z}/7)^\times$, and let P be the set where $\varepsilon' = 1$. Reducing modulo 7 gives $w \sum_k \varepsilon'(k)k \equiv 0$, and $\sum_k \varepsilon'(k)k = 2 \sum_P k - 21$, so $\sum_P k \equiv 0 \pmod{7}$. The only three-element subsets of $\{1, \dots, 6\}$ with this property are $\{1, 2, 4\}$ and $\{3, 5, 6\}$; so $\varepsilon' = \pm \lambda_7$, the Legendre symbol modulo 7, and $\varepsilon_u = \tau \lambda_7(u)$ for a sign τ . If $e' \neq 5$, let ψ be an even primitive character modulo e' with $\psi(3) \neq -1$ (Lemma 3.32) and $\chi = \lambda_7 \psi$, which is odd and primitive modulo e^* . Then $\chi(3) = -\psi(3) \neq 1$, and $\bar{\chi}(\varepsilon_u u) = \varepsilon_u \lambda_7(u) \bar{\psi}(u) = \tau \bar{\psi}(w)$ for $u \in L$, so the left side of (1) with $c = e^*$ is a nonzero multiple of $6\tau \bar{\psi}(w) \neq 0$. If $e' = 5$, then $m = 105$; take $\chi = \lambda_7$, of conductor 7. As 3 and 5 are nonsquares modulo 7, (1), multiplied by $\varphi(105)$, reads $(1 - \lambda_7(3))(1 - \lambda_7(5)) \sum_u \lambda_7(\varepsilon_u u) = 4 \cdot 6\tau = 0$, which is false.

Case (E3), not (E1). Here $e^* = 35$. The function D is constant on the six classes $\{(\pm i, \pm j)\}$ of four elements each, $i \in (\mathbb{Z}/5)^\times$, $j \in (\mathbb{Z}/7)^\times$; write its values as a matrix $(M_{\alpha\beta})$ with $\alpha \in \{1, 2\}$ and $\beta \in \{1, 2, 3\}$, so that $M_{\alpha\beta} = g_\alpha + h_\beta$ and $4 \sum |M_{\alpha\beta}| = \|D\| \leq 12$. The difference $\delta = M_{1\beta} - M_{2\beta}$ does not depend on β . If $\delta = 0$, the two rows agree, and $\sum |M_{\alpha\beta}| \leq 3$ leaves one column with both entries σ and zeros elsewhere: this is (E1), with a line along K_5^* . If $|\delta| \geq 2$, every column contributes at least 2, which is too much. If $|\delta| = 1$, every column is $(\delta, 0)$ or $(0, -\delta)$. If all three columns agree we are in (E1), with a line along K_7^* . Otherwise $\|D\| = 12$, so A consists of tops, and D takes one sign on eight elements of U^* and the other on four. By

Lemma 3.30(b) four tops have residue ± 1 modulo 3 and two have the opposite residue, so the residues of the z_a modulo 3 add up to ± 2 . But $\sum_a z_a = 0$ in $\mathbb{Z}/e$, because $\sum_a a = 0$ and κ is injective.

Step 8: the case (E2), and the end of the proof. By Step 7, every maximal order e is in case (E2): $e = 3q$ with a prime $q \in \{5, 7, 11, 13\}$, and $D = c_0$ is constant. By Lemma 3.30(b) all tops at e have the same residue $\sigma = \text{sign } c_0$ modulo 3, there are $|c_0|$ of them over each pair $\pm u$ of residues modulo q , and $N_e = |c_0|(q-1)/2$. For an odd character ψ modulo q , $\psi(3) = 1$ only if ψ is trivial on the subgroup generated by 3.

$q = 13$. Then $|c_0| = 1$ and $N_e = 6$, so A consists of tops and $m = 39$. Let $P \subseteq (\mathbb{Z}/13)^\times$ be the set of residues modulo 13 of the elements of A , one from each pair $\pm u$, and $\pi = \mathbf{1}_P - \mathbf{1}_{-P}$. For odd ψ modulo 13, (1) with $c = 13$ gives $(1 - \psi(3)) \sum_P \bar{\psi} = 0$, and $\sum_v \pi(v) \bar{\psi}(v) = 2 \sum_P \bar{\psi}$; for even ψ that sum vanishes because π is odd. So the Fourier transform of π is supported on the characters trivial on $\langle 3 \rangle = \{1, 3, 9\}$, and π is invariant under $\langle 3 \rangle$. The cosets of $\langle 3 \rangle$ are $C_j = 2^j \langle 3 \rangle$, $j \in \mathbb{Z}/4$, with $-C_j = C_{j+2}$; hence $P = C_j \cup C_{j+1}$ for some j . A unit t with $t \equiv \sigma \pmod{3}$ and $t \equiv 2^{3-j} \pmod{13}$ gives $tA = O_{39}$, since $C_3 \cup C_0 = \{7, 8, 11\} \cup \{1, 3, 9\}$.

$q = 11$. Then $|c_0| = 1$ and $N_e = 5$. The sixth element $b = -\sum (\text{tops})$ lies in $\kappa(\mathbb{Z}/33)$, so $m = 33$; its residue modulo 3 is $-5\sigma = \sigma \neq 0$, so $3 \mid f_b$, and $f_b = 3$ as $f_b \neq 33$; so $b \equiv 0 \pmod{11}$. With P and π as before, (1) with $c = 11$ involves only the tops, and shows that the Fourier transform of π is supported on the odd characters trivial on $\langle 3 \rangle$, the subgroup of squares. The only such character is the Legendre symbol λ_{11} , which is odd; so $\pi = \pm \lambda_{11}$, and P is the set of nonzero squares or of nonsquares. A unit $t \equiv \sigma \pmod{3}$, $t \equiv \pm 1 \pmod{11}$ gives $tA = O_{33}$.

So we may assume that every maximal order is 15 or 21.

$q = 7$. If $|c_0| = 2$, then $N_e = 6$, A consists of tops, and $m = 21$. Let $n(v)$ be the number of tops with residue v modulo 7. Since 3 generates $(\mathbb{Z}/7)^\times$, $\psi(3) \neq 1$ for odd ψ , and (1) gives $\sum_v n(v) \bar{\psi}(v) = 0$; so the odd function $n(v) - n(-v)$ vanishes, and $n(v) = 1$ for every v . Then A contains all six elements $\kappa(\sigma, v)$, which lie in $\Phi_7(\kappa(\sigma, 1))$ (Lemma 3.29), and Lemma 3.25 gives a move. So $|c_0| = 1$ and $N_{21} = 3$.

$q = 5$. If $|c_0| = 3$, then A consists of tops and $m = 15$; as $\psi(3) = \pm i \neq 1$ for the odd characters ψ modulo 5, (1) gives $n(v) = n(-v)$ as before, which is impossible since $n(v) + n(-v) = 3$. So $N_{15} \in \{2, 4\}$.

Both 15 and 21 are maximal. Then $m = 105$. As $N_{21} = 3$ and $N_{15} + N_{21} \leq 6$, $N_{15} = 2$, and there is one further element b . Let ρ_3 and ρ_5 be the reductions of $\mathbb{Z}/105$ modulo 3 and 5. The three tops at 21 have equal ρ_3 , which therefore add up to 0, and $\rho_5 = 0$. The two tops at 15 have ρ_3 equal to the same nonzero value, and their ρ_5 are $2v_1$ and $2v_2$ with $v_1 \in \{\pm 1\}$ and $v_2 \in \{\pm 2\}$ (one top over each pair), whose sum $2(v_1 + v_2)$ is not 0 modulo 5. From $\rho_3(\sum_a a) = 0$ we get $\rho_3(b) \neq 0$, so $3 \mid f_b$; as f_b divides 15 or 21 and is neither, $f_b = 3$ and $\rho_5(b) = 0$. Then $\rho_5(\sum_a a) \neq 0$, a contradiction.

Only 15 is maximal, so $m = 15$, and the elements of A other than the $N = N_{15}$ tops have order 3 or 5. For a of order 3 write $a = 5z_a$; as A contains no pair, its r elements of order 3 are equal. The character χ_3 modulo 3 in (1) gives $(1 - \chi_3(5))N\sigma/8 + r\chi_3(z)/2 = 0$, that is $N\sigma = -2r\chi_3(z)$. If $N = 4$, then $r = 2$: the other two elements have order 3, and (1) for the odd characters modulo 5 involves only the tops, so $n(v) = n(-v)$ as before and $n(v) = 1$ for each $v \in (\mathbb{Z}/5)^\times$. The four tops then lie in $\Phi_5(\kappa(\sigma, 1))$, and Lemma 3.25 gives a move. If $N = 2$, then $r = 1$, and the other three elements $a = 3z_a$ have order 5, with no two of the z_a opposite. Let ψ be the odd character modulo 5 with $\psi(2) = i$, and $z_1 \in \{\pm 1\}$, $z_2 \in \{\pm 2\}$ the

residues modulo 5 of the two tops. Then (1), multiplied by 8, reads

$$(1 - \psi(3))(\bar{\psi}(z_1) + \bar{\psi}(z_2)) + 2 \sum_{f_a=5} \bar{\psi}(z_a) = 0.$$

The first term has absolute value $|1 + i| \cdot |\pm 1 \pm i| = 2$. In the sum, the n_1 elements with $z_a \in \{\pm 1\}$ contribute $\pm n_1$ and the n_2 with $z_a \in \{\pm 2\}$ contribute $\pm i n_2$, with $n_1 + n_2 = 3$; so its absolute value is $(n_1^2 + n_2^2)^{1/2} \geq \sqrt{5} > 1$, a contradiction.

Only 21 is maximal, so $m = 21$, and the elements of A other than the three tops have order 3 or 7. Let $W \subseteq (\mathbb{Z}/7)^\times$ be the set of residues modulo 7 of the tops, one from each pair $\pm u$. For the character χ_3 the product in (1) vanishes for the tops, as $\chi_3(7) = 1$, so the elements of order 3, which are equal, have $\sum \chi_3(z_a) = 0$: there are none. So the other three elements are $a = 3z_a$ of order 7, and the multiset Z of the z_a contains no two opposite residues. Let $E = \mathbf{1}_W - \mathbf{1}_{-W}$ and $B(v) = \text{mult}_Z(v) - \text{mult}_Z(-v)$; both are odd, $|E| = 1$, and $\sum_v |B(v)| = 6$. For odd ψ modulo 7, (1), multiplied by 24, reads $(1 - \psi(3))\hat{E}(\psi) + 2\hat{B}(\psi) = 0$, where $\hat{E}(\psi) = \sum_v E(v)\bar{\psi}(v)$; as $\psi(3)\hat{E}(\psi)$ is the transform of $v \mapsto E(3v)$, and all three functions are odd, $2B(v) = E(3v) - E(v)$ for every v . So $|B| \leq 1$, hence $|B| = 1$ everywhere, and $E(3v) = -E(v)$. As 3 generates $(\mathbb{Z}/7)^\times$, $E = E(1)\lambda_7$ and $B = -E$: W is the set of nonzero squares or of nonsquares, and $Z = -W$. A unit $t \equiv \sigma \pmod{3}$, $t \equiv \pm 1 \pmod{7}$, with tW the set of squares, gives $tA = O_{21}$.

This proves Theorem 3.27, and with it Theorem F. $\square$

4. ABELIAN VARIETIES: CM POINTS AND PROPAGATION

4.1. The statement (CM). Let A be an abelian variety of CM type. Its Hodge group is a torus, and after extending scalars to a Galois CM field E that splits it, $H^1(A)$ is a sum of lines indexed by embeddings, each of type $(1, 0)$ or $(0, 1)$. Deligne and André showed that every Hodge class on A is a sum of pull-backs, along homomorphisms $A \rightarrow A_\Delta$, of Weil classes on abelian varieties A_Δ of split Weil type [Mil20, Theorem 1]. So (CM) is equivalent to the algebraicity of these split Weil classes, and it is open in the refereed literature.

The preprint [Ope26b] claims (CM) [Ope26b, Theorem 1.1]. Its proof has four steps. (1) The theorem is reduced [Ope26b, Proposition 2.6] to the algebraicity of a line of type $(2, 2)$ on a product of four CM abelian varieties $B(v_1), \dots, B(v_4)$, attached to odd sign functions $v_i: \text{Gal}(E/\mathbb{Q}) \rightarrow \{\pm 1\}$ with $v_1 + v_2 = v_3 + v_4$ [Ope26b, Theorem 2.5]. (2) That line is shown to be algebraic as soon as four degree-one classes on a surface, coming from the four CM Hodge structures, have a nonzero mixed period [Ope26b, Proposition 3.1]. (3) Such a period is produced by theta forms on compact quotients of the complex two-ball [Ope26b, §4]. (4) The theta classes are shown to come from the prescribed CM Hodge structures, through Hecke operators and a Frobenius relation on the reductions of the ball quotients [Ope26b, Theorem 5.1 and Corollary 6.7].

We have checked the combinatorial part of step (1): its core is Lemma 4.1 below, and its contraction formula [Ope26b, (2.6)] carries the sign $(-1)^{p(p-1)/2}$, which we recompute in Appendix A. We have not checked steps (2) to (4), and the claim rests on step (4). Accordingly, every result of this paper that uses (CM) states it as a hypothesis.

Lemma 4.1 (Two-row switches). *Let G be a finite set with a fixed-point-free involution c , let $s = |G|/2$, and call $v: G \rightarrow \{\pm 1\}$ odd if $v(cg) = -v(g)$. Let $x = (x_1, \dots, x_p)$ and $y = (y_1, \dots, y_p)$ be lists of odd functions with $\sum_i x_i = \sum_i y_i$. Then there is a chain of lists of odd functions $x = x^{(0)}, x^{(1)}, \dots, x^{(N)} = y$ with $N \leq s\lfloor p/2 \rfloor$, in which $x^{(k+1)}$ differs from $x^{(k)}$ in exactly two entries $i \neq j$, and $x_i^{(k+1)} + x_j^{(k+1)} = x_i^{(k)} + x_j^{(k)}$.*

Proof. Choose a set T of representatives of the orbits of c ; an odd function is determined by its values on T . Treat the elements $\tau \in T$ one at a time. Let P_τ be the set of indices i with $x_i(\tau) = 1$ and $y_i(\tau) = -1$, and N_τ the set with $x_i(\tau) = -1$ and $y_i(\tau) = 1$. Then $\sum_i x_i(\tau) - \sum_i y_i(\tau) = 2|P_\tau| - 2|N_\tau|$, so $|P_\tau| = |N_\tau| \leq \lfloor p/2 \rfloor$. Pair the two sets, and for each pair (i, j) exchange the values of the i -th and j -th functions at τ and at $c\tau$. Each exchange keeps the functions odd, changes exactly the entries i and j , and keeps their sum, since it only swaps two values at each of two points. After the pairs for τ are exchanged, the list agrees with y at τ and $c\tau$, and the later steps do not change these values. $\square$

In [Ope26b] the lists come from the types of the factors of a Hodge monomial, a two-row switch is exactly a relation $v_1 + v_2 = v_3 + v_4$, and the algebraic tensors of consecutive switches are composed by contracting the middle factors against an ample class [Ope26b, (2.6)].⁹ Nothing in this combinatorial part depends on the geometry of steps (2) to (4).

4.2. Propagation from CM points.

Lemma 4.2 (CM points are dense). *Let G be a connected reductive group over $\mathbb{Q}$, let $u_0: \mathrm{U}(1) \rightarrow G_{\mathbb{R}}$ be a homomorphism, and let $\mathcal{D}$ be its orbit under conjugation by $G(\mathbb{R})^+$. The points $u \in \mathcal{D}$ with $u(\mathrm{U}(1)) \subseteq T_{\mathbb{R}}$ for a maximal torus T of G defined over $\mathbb{Q}$ are dense in $\mathcal{D}$.*

Proof. Let $u_1 \in \mathcal{D}$ and let W be a neighbourhood of u_1 . A maximal torus T_1 of $G_{\mathbb{R}}$ contains the torus $u_1(\mathrm{U}(1))$; let $\mathfrak{t}_1$ be its Lie algebra and $Y_1 \in \mathfrak{t}_1$ a regular element. The map $\phi: G(\mathbb{R})^+ \times \mathfrak{t}_1 \rightarrow \mathfrak{g}_{\mathbb{R}}$, $(g, Y) \mapsto \mathrm{Ad}(g)Y$, has differential $(Z, Y') \mapsto [Z, Y_1] + Y'$ at $(1, Y_1)$, which is onto because $\mathfrak{g}_{\mathbb{R}} = \mathfrak{t}_1 \oplus [\mathfrak{g}_{\mathbb{R}}, Y_1]$ for a regular semisimple Y_1 . So for small neighbourhoods N_1 of 1, with $N_1 u_1 N_1^{-1} \subseteq W$, and N_2 of Y_1 consisting of regular elements, $\phi(N_1 \times N_2)$ contains a nonempty open set, and hence a rational element $Y = \mathrm{Ad}(g)Y'$ with $g \in N_1$, $Y' \in N_2$. The identity component T of the centralizer of Y is a maximal torus defined over $\mathbb{Q}$, and $T_{\mathbb{R}} = gT_1 g^{-1}$. Then $u = gu_1 g^{-1} \in W$ and $u(\mathrm{U}(1)) \subseteq T_{\mathbb{R}}$. $\square$

Proposition 4.3 (A CM member through every Hodge class). *Let A_0 be an abelian variety and γ a Hodge class of degree $2p$ on A_0 . There are a polarized abelian scheme $\pi: \mathcal{A} \rightarrow S$ over a smooth connected quasi-projective variety, a global section ξ of $R^{2p}\pi_*\mathbb{Q}$, and points $s_0, c \in S$, such that $\mathcal{A}_{s_0} \cong A_0$ with $\xi_{s_0} = \gamma$, ξ_s is a Hodge class for every s , and $\mathcal{A}_c$ is of CM type.*

Proof. Replacing γ by a multiple, we may assume it integral. Fix a polarization of A_0 with form ψ on $V = H_1(A_0, \mathbb{Q})$, and a level $N \geq 3$. The complex structures J on $V_{\mathbb{R}}$ that are compatible with ψ and positive for it form the Siegel space $\mathcal{H}$, and its quotient $\mathcal{S}$ by the principal congruence subgroup of level N in $\mathrm{Sp}(H_1(A_0, \mathbb{Z}), \psi)$ is a smooth quasi-projective variety with a universal polarized abelian scheme $f: \mathcal{A}_g \rightarrow \mathcal{S}$ [MFK94, Chapter 7]. Let $\mathbb{V} = R^{2p}f_*\mathbb{Z}$. By the theorem of Cattani, Deligne and Kaplan [CDK95, Theorem 1.1], the set of pairs (a, v) , with $a \in \mathcal{S}$ and $v \in \mathbb{V}_a$ an integral Hodge class with the same self-intersection as γ , is a closed algebraic subvariety of the total space of the vector bundle $\mathbb{V} \otimes \mathcal{O}_{\mathcal{S}}$, finite over $\mathcal{S}$; let $\mathcal{C}$ be its connected component through the point (a_0, γ) given by A_0 and a lift $J_0 \in \mathcal{H}$.

Let $G \subseteq \mathrm{Sp}(V, \psi)$ be the Hodge group of A_0 , the smallest algebraic subgroup over $\mathbb{Q}$ whose real points contain the image of the homomorphism $u_0: \mathrm{U}(1) \rightarrow \mathrm{GL}(V_{\mathbb{R}})$, $e^{i\theta} \mapsto \cos \theta + J_0 \sin \theta$. It is connected and reductive, and it fixes every Hodge class of A_0 , in particular γ [Del82, §3]. For $g \in G(\mathbb{R})^+$ the structure $J = gJ_0 g^{-1}$ lies in $\mathcal{H}$, and the Hodge group of A_J is contained in

⁹The contraction multiplies by $(-1)^{p(p-1)/2} \prod_i q_i$, where the sign comes from regrouping $\beta_1 \cdots \beta_p \gamma_1 \cdots \gamma_p$ as $\beta_1 \gamma_1 \cdots \beta_p \gamma_p$ for classes of degree one, and $q_i = \int_{B(b_i)} \beta_i \wedge \gamma_i \wedge \ell_i^{s-1}$ is nonzero by the Hodge–Riemann relations, because γ_i is a nonzero multiple of $\bar{\beta}_i$.

G , so γ is a Hodge class on A_J . The orbit $\mathcal{D} = \{gJ_0g^{-1}\}$ is connected, so its image $J \mapsto (a_J, \gamma)$ lies in $\mathcal{C}$. By Lemma 4.2 there are $J_k \in \mathcal{D}$ converging to J_0 such that $u_{J_k}(\mathrm{U}(1))$ lies in a torus defined over $\mathbb{Q}$; the Hodge group of A_{J_k} is then commutative, and A_{J_k} is of CM type.¹⁰ One of the finitely many irreducible components of $\mathcal{C}$, say $\mathcal{C}_0$, contains infinitely many of the points (a_{J_k}, γ) , and, being closed, also their limit (a_0, γ) .

Let $S \rightarrow \mathcal{C}_0$ be a resolution of singularities [Hir64], so that S is smooth, connected and quasi-projective, and let $\mathcal{A} \rightarrow S$ be the pull-back of $\mathcal{A}_g$. The total space of a local system is a covering space of the base, so $(a, v) \mapsto v$ is a flat section on $\mathcal{C}_0$, and its pull-back is a global section ξ of $R^{2p}\pi_*\mathbb{Z}$ that is a Hodge class at every point. Take s_0 over (a_0, γ) and c over some (a_{J_k}, γ) . $\square$

Proposition 4.4. (F2) holds if and only if both (CM) and (IP) hold.

Proof. (F2) contains (CM), and it implies (IP): if ξ_c is algebraic, then every ξ_s is a Hodge class by Lemma 2.10, hence algebraic. Conversely, let γ be a Hodge class on an abelian variety A_0 , and take π, ξ, s_0 and c from Proposition 4.3. Then ξ_c is a Hodge class on an abelian variety of CM type, so it is algebraic by (CM), and by (IP) the class $\gamma = \xi_{s_0}$ is algebraic. $\square$

The proposition separates the two inputs of the argument of Deligne and André as Milne records it [Mil20, §5]: the classes at CM points, and their transport along families. The variational conjecture (V) supplies both at once [Mil20, Theorem 3 and Remark 2], because the split Weil classes are algebraic at a suitable member of each family [Del82, 4.8]. (IP) supplies only the transport, and it does so only from CM members.

4.3. Proof of Theorem A. By Lemma 2.11, (i) implies each of (ii)–(vi). If (F2) holds, every γ in the definition of $\mathrm{Ab}^p(X)$ is algebraic, so $\mathrm{Ab}^p(X) = \mathrm{Alg}^p(X)$, and (F3') becomes (HC); so (ii) implies (i). (L) implies (F2) by the theorem of Abdulali and André [Mil20, Theorem 4]¹¹, (V) implies (F2) [Mil20, Theorem 3 and Remark 2], and (CM) with (IP) implies (F2) by Proposition 4.4; so (iii), (iv) and (v) imply (ii). Finally (vi) implies (i) by Proposition 2.8. $\square$

4.4. Weil families and degrees at CM points. Let K be an imaginary quadratic field and $n \geq 1$. An abelian variety A of dimension $2n$ with an action of K is of *Weil type* if both embeddings of K occur with multiplicity n on $H^{1,0}(A)$. Then the *Weil plane* $W(A) = \bigwedge_K^{2n} H^1(A, \mathbb{Q}) \subseteq H^{2n}(A, \mathbb{Q})$ is a $\mathbb{Q}$ -plane of Hodge classes [Wei79], [Del82, §4]. For a polarization whose Rosati involution induces complex conjugation on K , the Riemann form is $\mathrm{Tr}_{K/\mathbb{Q}}(fH)$ for a K -hermitian form H of signature (n, n) , and A is of *split Weil type* if H has a totally isotropic K -subspace of dimension n [Mil20, §2.1].

Fix such data and a level $N \geq 3$. The abelian varieties of split Weil type for (K, n) with this hermitian form and level structure are parametrized by a smooth connected quasi-projective variety S , a quotient $\Gamma \backslash \mathcal{D}$ of the hermitian symmetric domain $\mathcal{D}$ of $\mathrm{SU}(n, n)$, of dimension n^2 , by a neat arithmetic subgroup Γ of the unitary group; it is a connected Shimura variety of PEL type and carries a polarized abelian scheme $\pi: \mathcal{A} \rightarrow S$ with an action of $\mathcal{O}_K$ [Wei79, vG94].¹²

¹⁰The torus T lies in a maximal torus T' of $\mathrm{GL}(V)$ over $\mathbb{Q}$, the units of a commutative semisimple subalgebra $L \subseteq \mathrm{End}(V)$ of dimension $2g$. Since L commutes with $T' \supseteq \mathrm{Hg}(A_{J_k})$, it lies in the commutant of the Hodge group, which is $\mathrm{End}(A_{J_k}) \otimes \mathbb{Q}$.

¹¹Milne's hypothesis is the Lefschetz standard conjecture in the form that L^{d-2i} maps the algebraic classes of degree $2i$ onto those of degree $2d - 2i$. It follows from (L), since the inverse of L^{d-2i} is then induced by an algebraic cycle and maps algebraic classes to algebraic classes.

¹²The unitary group acts on the Weil plane through the determinant. For $\gamma \in \Gamma$ the determinant is a unit of $\mathcal{O}_K$ of norm one, hence a root of unity, and it is 1 because Γ is neat. So the Weil planes of the members form a trivial local system on S , and every Weil class on one member extends to a global section.

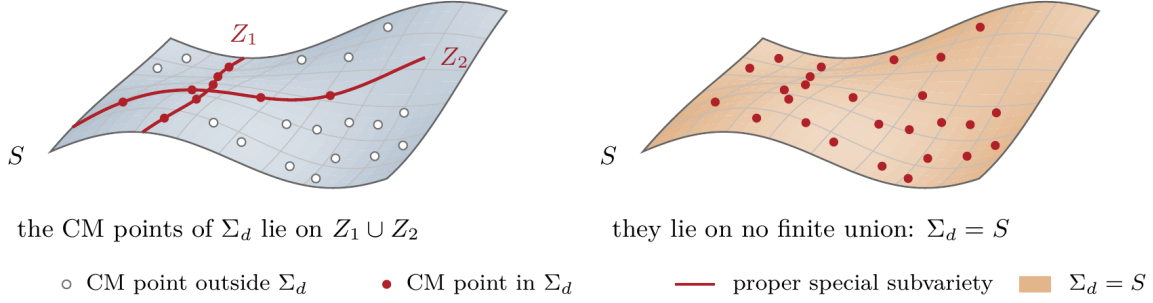


FIGURE 3. The criterion of Theorem D, drawn schematically (S has dimension n^2). CM points are dense in S . On the left the CM points of Σ_d lie on two proper special subvarieties Z_1 and Z_2 , and condition (iii) fails for this d . On the right they lie on no finite union of proper special subvarieties; by the André–Oort theorem their Zariski closure is then S , and $\Sigma_d = S$.

The special points of S are the members of CM type. Fix a nonzero global section ξ of the Weil planes. An element $\alpha \in \mathcal{O}_K$ acts on $\mathcal{A}_s$, and its pull-back multiplies the K -line $W(\mathcal{A}_s)$ by α^{2n} . Choose α with $\alpha^{2n} \notin \mathbb{Q}$.¹³ Then α^* moves any nonzero class of $W(\mathcal{A}_s)$ to a class that spans $W(\mathcal{A}_s)$ with it, and pull-back by an endomorphism preserves algebraic classes; so $W(\mathcal{A}_s)$ consists of algebraic classes exactly when ξ_s is algebraic.

Definition 4.5. Let $\Sigma \subseteq S$ be the set of s at which ξ_s is algebraic. For $d \geq 1$ let Σ_d be the set of s for which there are effective algebraic cycles Z^+, Z^- of codimension n on $\mathcal{A}_s$, each of degree at most d for the polarization, and an integer $1 \leq e \leq d$, with $e\xi_s = \text{cl}(Z^+) - \text{cl}(Z^-)$.

Lemma 4.6. *Each Σ_d is Zariski closed in S , $\Sigma_d \subseteq \Sigma_{d+1}$, and $\Sigma = \bigcup_d \Sigma_d$.*

Proof. The last two assertions are clear. Effective cycles of codimension n and degree at most d in the fibres of π are parametrized by finitely many relative Chow varieties $\mathcal{C}_i \rightarrow S$, which are projective over S [Kol96, Chapter I, §§3–4]. On a connected component of $\mathcal{C}_i \times_S \mathcal{C}_j$ the class $\text{cl}(Z^+) - \text{cl}(Z^-)$ of the fibre is a flat section of the pulled-back local system, and so is $e\xi$. Two flat sections over a connected base that agree at one point agree everywhere, so the set where they agree is a union of connected components. Its image in S is closed, because the projection is proper. Σ_d is the union of finitely many such images. $\square$

Proof of Theorem D. The first sentence is Lemma 4.6. (ii) implies (i) trivially. If (i) holds, S^{an} is the countable union of the closed sets Σ_d , so by the Baire category theorem one of them has an interior point; a Zariski closed subset of the irreducible variety S with an interior point is S , which gives (ii). If (ii) holds, the CM points in $\Sigma_d = S$ are dense in S^{an} by Lemma 4.2, so they lie in no finite union of proper closed subvarieties, which gives (iii). Assume (iii), and let Z be the Zariski closure of the set of CM points in Σ_d ; it lies in Σ_d . Every irreducible component of Z contains a Zariski dense set of special points, so it is a special subvariety, by the André–Oort conjecture, which is a theorem [PST21]. By (iii) one of these components is not proper, so $S \subseteq Z \subseteq \Sigma_d$, which is (ii). Finally, at a CM point ξ_s is a Hodge class on an abelian variety of CM type, so under (CM) it is algebraic. $\square$

¹³The $\alpha \in K \otimes \mathbb{R} = \mathbb{C}$ with α^{2n} real lie on $2n$ lines through 0, and the lattice $\mathcal{O}_K$ is not contained in finitely many lines.

Remark 4.7. Without any hypothesis Σ is dense. It contains the members K -isogenous to $E^n \times \bar{E}^n$, which is of split Weil type and whose Weil classes are polynomials in divisor classes (Lemma 5.10); and with every point it contains its Hecke orbit, which is dense. Algebraic representatives move along isogenies by push-forward and pull-back, and their degree grows with the degree of the isogeny. (CM) adds every CM point, but it gives no bound on the degrees, and by Theorem D a bound on a set of CM points that is not trapped in finitely many proper Shimura subvarieties is exactly what remains. Neither [Ope26b] nor any other source we know gives such a bound.

5. WEIL CLASSES AND SCHOEN'S SUBVARIETY

In this section K is an imaginary quadratic field with a fixed embedding $\chi: K \rightarrow \mathbb{C}$ and its conjugate $\bar{\chi}$. Let A be an abelian variety of Weil type for K of dimension $2n$ (Section 4.4) on which $\mathcal{O}_K$ acts, with a polarization class η whose Rosati involution induces complex conjugation on K . Endomorphisms act on cohomology by pull-back, which makes $V = H^1(A, \mathbb{Q})$ a K -vector space of dimension $2n$, and

$$V_{\mathbb{C}} = V \otimes_{\mathbb{Q}} \mathbb{C} = V_{\chi} \oplus V_{\bar{\chi}},$$

where $\alpha \in K$ acts on V_{χ} by $\chi(\alpha)$ and on $V_{\bar{\chi}}$ by $\bar{\chi}(\alpha)$. Each summand has dimension $2n$ and, because A is of Weil type, meets $H^{1,0}$ and $H^{0,1}$ in subspaces of dimension n . The Weil plane satisfies

$$W(A) \otimes \mathbb{C} = \wedge^{2n} V_{\chi} \oplus \wedge^{2n} V_{\bar{\chi}} \subseteq H^{2n}(A, \mathbb{C})$$

[Del82, §4], and both lines are of type (n, n) . We call A *Weil-algebraic* if every class in $W(A)$ is algebraic. Two facts are used repeatedly. First, an endomorphism $\alpha \in \mathcal{O}_K$ multiplies η by the norm $N(\alpha) = \chi(\alpha)\bar{\chi}(\alpha)$, since its Rosati adjoint is $\bar{\alpha}$; comparing with the action of α on $\wedge^2 V_{\mathbb{C}} = \wedge^2 V_{\chi} \oplus (V_{\chi} \otimes V_{\bar{\chi}}) \oplus \wedge^2 V_{\bar{\chi}}$ for any $\alpha \notin \mathbb{Q}$ shows that $\eta \in V_{\chi} \otimes V_{\bar{\chi}}$, and hence

$$(3) \quad \eta^k \in \wedge^k V_{\chi} \otimes \wedge^k V_{\bar{\chi}}.$$

Second, if $W(A)$ contains one nonzero algebraic class, then A is Weil-algebraic, by the argument given for ξ_s in Section 4.4.

The Hodge group $\mathrm{Hg}(A)$ commutes with K and preserves the Riemann form, so it lies in the unitary group $\mathrm{U}(V, H)$ of the hermitian form H of the polarization. It fixes the Hodge classes in $W(A)$, on which $\mathrm{U}(V, H)$ acts through the determinant, so $\mathrm{Hg}(A) \subseteq \mathrm{SU}(V, H)$. We say that A has *maximal Hodge group* if equality holds.

5.1. The Hodge ring of a general member.

Proposition 5.1. *If $\mathrm{Hg}(A) = \mathrm{SU}(V, H)$, then $\mathrm{Hdg}^k(A) = \mathbb{Q}\eta^k$ for $k \neq n$, and*

$$\mathrm{Hdg}^n(A) = \mathbb{Q}\eta^n \oplus W(A).$$

Moreover $\eta^n \cup w = 0$ for every $w \in W(A)$.

Proof. The Hodge classes in $H^{2k}(A, \mathbb{Q}) = \wedge^{2k} V$ are the invariants of $\mathrm{Hg}(A)$ [Del82, §3]. Over $\mathbb{C}$ the group $\mathrm{SU}(V, H)$ becomes $\mathrm{SL}(V_{\chi})$, acting on V_{χ} by the standard representation and on $V_{\bar{\chi}}$ by its dual, because the Riemann form pairs $V_{\bar{\chi}}$ perfectly with V_{χ} . The invariants in

$$\wedge^{2k} V_{\mathbb{C}} = \bigoplus_{a+b=2k} \wedge^a V_{\chi} \otimes \wedge^b V_{\chi}^{\vee} = \bigoplus_{a+b=2k} \mathrm{Hom}(\wedge^b V_{\chi}, \wedge^a V_{\chi})$$

form a line in each summand with $a = b$ or $\{a, b\} = \{0, 2n\}$, and vanish in the others, because the representations $\wedge^a V_{\chi}$, $0 \leq a \leq 2n$, are irreducible and pairwise non-isomorphic except for

$\bigwedge^0 \cong \bigwedge^{2n}$.¹⁴ By (3) the line for $a = b = k$ contains η^k , which is nonzero for $k \leq 2n$ because η^{2n} is a positive multiple of the orientation class. In degree $2n$ the two remaining lines span $W(A) \otimes \mathbb{C}$. This proves the first two assertions. By (3), $\eta^n \cup \bigwedge^{2n} V_\chi$ lies in $\bigwedge^{3n} V_\chi \otimes \bigwedge^n V_{\bar{\chi}} = 0$, and likewise for $V_{\bar{\chi}}$. $\square$

5.2. What is known. For $n = 1$ the Weil plane lies in H^2 and Theorem 2.3 applies. For $n = 2$ the hermitian form is split exactly when its discriminant is trivial, and Floccari and Fu prove that the Weil classes of every Weil fourfold of discriminant one are algebraic [FF26]. Schoen proves algebraicity for the Weil fourfolds of trivial discriminant for $\mathbb{Q}(\sqrt{-3})$ and for the Weil sixfolds of a family for $\mathbb{Q}(\sqrt{-3})$ in which his Prym varieties are dense [Sch88, Sch98], and for further families of fourfolds [Sch98]; Koike treats families for $\mathbb{Q}(\sqrt{-1})$ built in the same way [Koi04]. Markman's preprint proves algebraicity for every Weil fourfold and every split Weil sixfold [Mar25a]. Since the Hodge ring of an abelian variety of dimension at most five is generated by divisor classes and the Weil classes of its quotients [MZ99], the fourfold case gives (F2) in dimension at most five. For $2n \geq 8$ we know of no family of Weil type whose general member has maximal Hodge group and on which the Weil classes are known to be algebraic at every member.

5.3. Semiregularity. Let Z be a smooth projective variety of dimension N and $\mathcal{E}$ a coherent sheaf on Z , with Atiyah class $\text{at}(\mathcal{E}) \in \text{Ext}^1(\mathcal{E}, \mathcal{E} \otimes \Omega_Z^1)$. The *semiregularity map* of Buchweitz and Flenner is

$$\sigma_{\mathcal{E}}: \text{Ext}^2(\mathcal{E}, \mathcal{E}) \longrightarrow \bigoplus_{q=0}^{N-2} H^{q+2}(Z, \Omega_Z^q), \quad e \longmapsto \left(\text{tr}(\text{at}(\mathcal{E})^q \circ e) / q! \right)_q,$$

and $\mathcal{E}$ is *semiregular* if $\sigma_{\mathcal{E}}$ is injective [BF03]. A subvariety $Y \subseteq Z$ is semiregular if $\mathcal{O}_Y$ is.¹⁵ The point of the definition is the following theorem. If an infinitesimal deformation of Z keeps $\text{ch}(\mathcal{E})$ of Hodge type, the obstruction to deforming $\mathcal{E}$ along it is sent by $\sigma_{\mathcal{E}}$ to the obstruction for $\text{ch}(\mathcal{E})$ to stay of Hodge type, which vanishes; injectivity of $\sigma_{\mathcal{E}}$ then kills the first obstruction.

Theorem 5.2 (Buchweitz–Flenner [BF03, Theorem 1.5], after Bloch [Blo72]). *Let $f: \mathcal{Z} \rightarrow T$ be a smooth projective morphism to a connected complex manifold, let $0 \in T$, and let $\mathcal{E}$ be a semiregular coherent sheaf on $\mathcal{Z}_0$. Suppose that over a contractible neighbourhood of 0 the flat continuation of every component of $\text{ch}(\mathcal{E})$ is a Hodge class on every fibre. Then $\mathcal{E}$ extends to a coherent sheaf on $f^{-1}(U)$, flat over U , for some open neighbourhood U of 0.*

We return to the split Weil family $\pi: \mathcal{A} \rightarrow S$ of Section 4.4, with its global section ξ of Weil planes and the sets $\Sigma_d \subseteq \Sigma \subseteq S$ of Definition 4.5.

Proposition 5.3. *Let $s_0 \in S$ and let $\mathcal{E}$ be a semiregular coherent sheaf on $A = \mathcal{A}_{s_0}$ such that $\text{ch}_k(\mathcal{E}) \in \mathbb{Q}\eta^k$ for $k \neq n$ and $\text{ch}_n(\mathcal{E}) = a\eta^n + w$ with $a \in \mathbb{Q}$ and $0 \neq w \in W(A)$. Then $\Sigma = S$: the Weil classes are algebraic on every member of the family.*

Proof. The powers of η extend to flat sections that are Hodge classes on every member, and so does w , since the Weil planes form a trivial local system of Hodge classes on S (Section 4.4). So the hypothesis of Theorem 5.2 holds, and $\mathcal{E}$ extends to a sheaf on $\pi^{-1}(U)$, flat over U , for a contractible open neighbourhood U of s_0 in S^{an} . For $t \in U$ its restriction $\mathcal{E}_t$ is algebraic

¹⁴The representation $\bigwedge^a$ of SL_{2n} is irreducible with highest weight the a -th fundamental weight for $1 \leq a \leq 2n - 1$, and these weights are distinct; $\bigwedge^0$ and $\bigwedge^{2n}$ are trivial.

¹⁵Bloch introduced semiregularity for local complete intersection subvarieties, through a map $H^1(Y, N_{Y/Z}) \rightarrow H^{p+1}(Z, \Omega_Z^{p-1})$ in codimension p , and proved the deformation theorem below in that setting [Blo72]. Buchweitz and Flenner extend both the map and the theorem to coherent sheaves.

by GAGA and has a finite locally free resolution, so $\text{ch}(\mathcal{E}_t)$ is algebraic, and it is the flat continuation of $\text{ch}(\mathcal{E})$. Hence $w_t = \text{ch}_n(\mathcal{E}_t) - a\eta_t^n$ is a nonzero algebraic class in $W(\mathcal{A}_t)$, and $t \in \Sigma$. Thus $U \subseteq \Sigma = \bigcup_d \Sigma_d$. By the Baire category theorem some Σ_d contains a nonempty open subset of S^{an} ; it is Zariski closed by Lemma 4.6 and S is irreducible, so $\Sigma_d = S$. $\square$

5.4. Schoen's subvariety. From now on $K = \mathbb{Q}(\sqrt{-3})$, and $\zeta \in \mathcal{O}_K$ is a primitive cube root of unity. Let X be a curve of genus $g = n + 1 \geq 3$, let $L \in \text{Pic}^0(X)$ have exact order three, and let $\pi: C \rightarrow X$ be the étale cyclic triple cover defined by L , with covering automorphism σ ; thus $\pi_*\mathcal{O}_C = \mathcal{O}_X \oplus L \oplus L^2$, and C has genus $3g - 2$. In $J = J(C)$ put $N = 1 + \sigma + \sigma^2$ and $P = 3 - N$, and let $B = P(J)$, the *Prym variety* of the cover, with the polarization η induced by the principal polarization of J . Since $N^2 = 3N$, we have $NP = 0$ and $P^2 = 3P$; so P is multiplication by 3 on B , and σ acts on B with $\sigma^2 + \sigma + 1 = 0$, which lets ζ act as σ .

Lemma 5.4. (a) B is an abelian variety of Weil type for K of dimension $2n$.

(b) The hermitian form of η is split.

(c) For very general (X, L) , B has maximal Hodge group.

Proof. (a) By the decomposition of $\pi_*\mathcal{O}_C$, the space $H^0(C, K_C)$ is the sum of $H^0(X, K_X \otimes L^j)$, $j = 0, 1, 2$, on which σ^* acts by the three cube roots of unity. By Riemann–Roch these spaces have dimensions g , $g - 1$ and $g - 1$, since L^j is a nontrivial bundle of degree zero for $j = 1, 2$. Let $P': J \rightarrow B$ be P with its target restricted and $\iota: B \rightarrow J$ the inclusion. Since $P'\iota = 3$ and $P = \iota P'$, pull-back by P' identifies $H^{1,0}(B)$ with the image of $P^* = 3 - N^*$ on $H^{1,0}(J)$, which is the sum of the two nontrivial eigenspaces. So $\dim B = 2(g - 1) = 2n$, and both embeddings of K occur with multiplicity n .

(b) The Riemann form of η is the intersection form E on $H_1(B, \mathbb{Q}) = PH_1(C, \mathbb{Q})$, and a K -subspace is isotropic for the hermitian form exactly when it is isotropic for E . The cover corresponds to a surjection $\phi: H_1(X, \mathbb{Z}) \rightarrow \mathbb{Z}/3$. Since $\text{Sp}_{2g}(\mathbb{Z})$ maps onto $\text{Sp}_{2g}(\mathbb{Z}/3)$, which is transitive on nonzero vectors, there is a symplectic basis $a_1, b_1, \dots, a_g, b_g$ with $\phi(x) = \langle x, b_1 \rangle \bmod 3$; represent it by simple closed curves in standard position. For $i \geq 2$ the curves a_i and b_i have trivial monodromy, so each lifts to three disjoint closed curves, permuted by σ . Choose lifts $\tilde{a}_i$ and $\tilde{b}_i$ that meet. Then $E(\sigma^k \tilde{a}_i, \sigma^l \tilde{a}_j) = 0$ and $E(\sigma^k \tilde{a}_i, \sigma^l \tilde{b}_j) = \delta_{ij} \delta_{kl}$ for $i, j \geq 2$. The first relation shows that the $\mathbb{Z}[\sigma]$ -span of the classes $P\tilde{a}_i$, $2 \leq i \leq g$, is isotropic. These classes are K -linearly independent: if $\sum_i \alpha_i P\tilde{a}_i = 0$ with $\alpha_i = \sum_k c_{ik} \sigma^k \in \mathbb{Z}[\sigma]$, pairing with $\sigma^l \tilde{b}_j$ gives $3c_{jl} - \sum_k c_{jk} = 0$ for every l , so $c_{j0} = c_{j1} = c_{j2}$ and α_j is a multiple of N , which is zero on B . So they span an isotropic subspace of dimension $n = g - 1$.

(c) The monodromy of the Prym varieties over the moduli of pairs (X, L) is an arithmetic subgroup of the unitary group [Loo97], so by Borel density its Zariski closure contains $\text{SU}(V, H)$. By André's theorem [And92] the connected monodromy group is contained in the Hodge group of a very general member. Together with $\text{Hg}(B) \subseteq \text{SU}(V, H)$ this gives equality. $\square$

The pairs (X, L) of genus g have $3g - 3 = 3n$ moduli, while the split Weil family of K in dimension $2n$ has dimension n^2 . For $n \leq 3$ Schoen uses these Prym varieties to reach every member [Sch88, Sch98]. For $n \geq 4$ they form a proper subvariety (Figure 4).

The construction. Let $h = 2g - 2 = 2n$. The group $G = (\mathbb{Z}/3)^h \rtimes S_h$ acts on C^h by $(v, \tau) \cdot (c_i)_i = (\sigma^{v_i} c_{\tau^{-1}(i)})_i$, and the quotient is the symmetric product $X^{(h)}$; write $\mu: C^h \rightarrow X^{(h)}$. The map $\varepsilon(v, \tau) = \sum_i v_i$ is a homomorphism $G \rightarrow \mathbb{Z}/3$, with kernel G' . Following Schoen [Sch88, §2], put

$$W_0 = C^h / G', \quad \rho: C^h \rightarrow W_0, \quad \nu: W_0 \rightarrow X^{(h)},$$

and let $\delta = \sigma \times 1 \times \dots \times 1$, so that $\varepsilon(\delta) = 1$ and δ induces an automorphism $\bar{\delta}$ of W_0 over $X^{(h)}$.

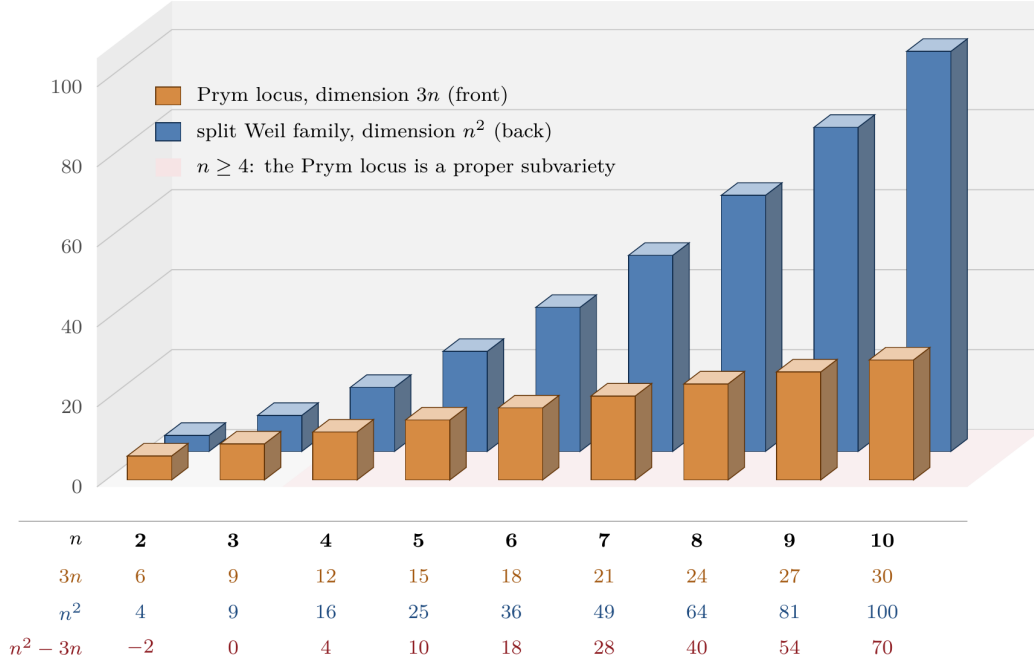


FIGURE 4. The dimension $3n$ of the locus of Prym varieties of étale cyclic triple covers of curves of genus $n + 1$, against the dimension n^2 of the split Weil family of $\mathbb{Q}(\sqrt{-3})$ in dimension $2n$. For $n \geq 4$ the Prym locus is a proper subvariety, and a semiregular Y would have to deform in the $n^2 - 3n$ remaining directions.

Lemma 5.5. (a) *No element of $G \setminus G'$ has a fixed point on C^h . Hence ν is a connected étale Galois cover with group $\mathbb{Z}/3$, generated by $\bar{\delta}$.*

(b) *G acts freely on $\mu^{-1}(D)$ for every reduced divisor $D \in X^{(h)}$.*

(c) *μ is finite and flat, so μ^{-1} of a subvariety of pure dimension r has pure dimension r .*

Proof. Suppose (v, τ) fixes (c_i) , so that $c_{\tau(i)} = \sigma^{v_{\tau(i)}} c_i$ for all i . Going once around a cycle $(i_1, \dots, i_k)$ of τ gives $c_{i_1} = \sigma^{v_{i_1} + \dots + v_{i_k}} c_{i_1}$; since σ has no fixed points, the sum of the v_i over every cycle is 0 in $\mathbb{Z}/3$, and so is $\varepsilon(v, \tau)$. This is (a): W_0 is connected as a quotient of C^h , and $G/G' \cong \mathbb{Z}/3$ acts freely on it with quotient $X^{(h)}$. If moreover the points $\pi(c_i)$ are distinct, then $\pi(c_{\tau(i)}) = \pi(c_i)$ forces $\tau = 1$, and then $v = 0$; this is (b). The map $C^h \rightarrow X^h$ is étale and $X^h \rightarrow X^{(h)}$ is a finite map between smooth varieties of the same dimension, hence flat. By going down for flat maps, every component of the preimage of a subvariety Z dominates a component of Z , and the fibres are finite; this gives (c). $\square$

Let $u: X^{(h)} \rightarrow \text{Pic}^h(X)$ send D to $\mathcal{O}_X(D)$. Its fibre over $[K_X]$ is the canonical system $F = |K_X| \cong \mathbb{P}^n$. Since F is simply connected, the restriction of ν to $\nu^{-1}(F)$ is a trivial cover: $\nu^{-1}(F) = Q_0 \sqcup Q_1 \sqcup Q_2$ with each Q_t mapped isomorphically onto F , and $\bar{\delta}$, which has no fixed points, permutes the Q_t cyclically; number them so that $\bar{\delta}(Q_t) = Q_{t+1}$. These are the three components of Schoen's construction.¹⁶ Put

$$\Lambda_0 = \rho^{-1}(Q_0) \subseteq C^h,$$

¹⁶Patel and Zhang obtain the same three components by base change of the Abel–Jacobi map along an isogeny onto $J(X)$ [PZ27].

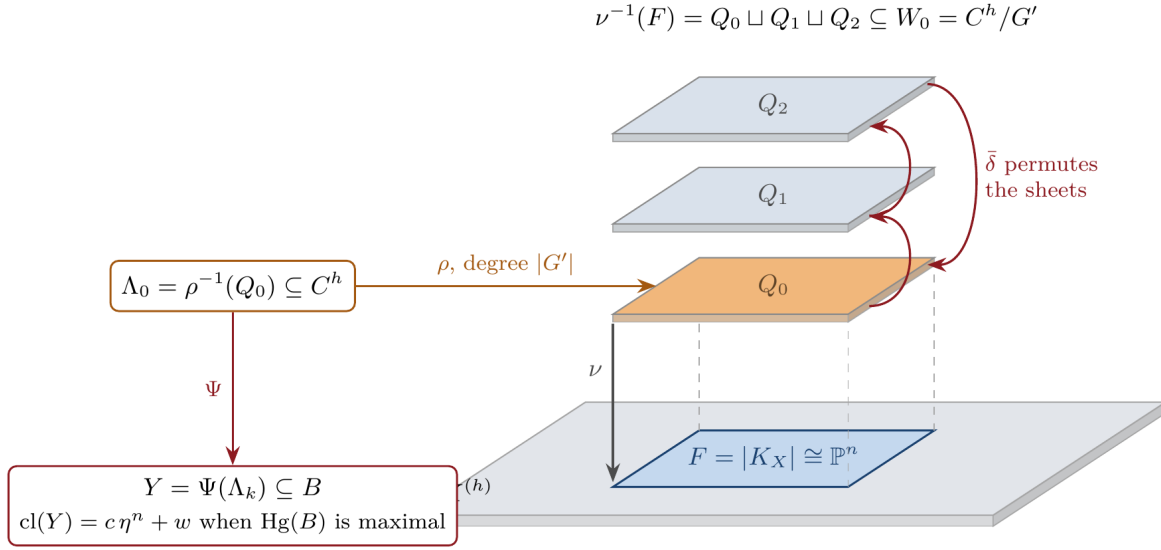


FIGURE 5. Schoen's construction. Over the canonical system $F = |K_X|$ of X , the étale triple cover $\nu: W_0 \rightarrow X^{(h)}$ splits into three sheets, which $\bar{\delta}$ permutes cyclically. $\Lambda_0 \subseteq C^h$ is the preimage of one sheet, and Y is the image under Ψ of one of its components (Theorem E).

with its reduced structure. It has pure dimension n by Lemma 5.5(c), and since the general member of $|K_X|$ is reduced, ρ is étale over a dense open subset of Q_0 by Lemma 5.5(b), so that $\rho_*[\Lambda_0] = |G'|[Q_0]$. The construction is drawn in Figure 5.

Fix $c_0 \in C$ and let

$$\Psi: C^h \rightarrow B, \quad (c_1, \dots, c_h) \mapsto P\left(\sum_i (c_i - c_0)\right).$$

The key input is a computation of cohomology with twisted coefficients.

Lemma 5.6. *Let $\mathcal{L}$ be a nontrivial local system of rank one on $X^{(h)}$. Then restriction $H^h(X^{(h)}, \mathcal{L}) \rightarrow H^h(F, \mathcal{L})$ is an isomorphism, and $H^h(F, \mathcal{L}) \cong H^{2n}(\mathbb{P}^n, \mathbb{C}) \cong \mathbb{C}$.*

Proof. The fibres of u are projective spaces, so $\mathcal{L}$ is trivial on each of them, and $\mathcal{M} = u_*\mathcal{L}$ is a local system of rank one on $T = \text{Pic}^h(X)$ with $\mathcal{L} = u^*\mathcal{M}$.¹⁷ It is nontrivial, because $\mathcal{L}$ is. As a real manifold T is a torus $(S^1)^{2g}$, and $\mathcal{M}$ is an exterior product of local systems on the circles, at least one of them nontrivial; a nontrivial local system of rank one on a circle has no cohomology, so $H^*(T, \mathcal{M}) = 0$ by the Künneth formula. Let $U = T \setminus \{[K_X]\}$. The exact sequence

$$\cdots \rightarrow H_c^k(U, \mathcal{M}) \rightarrow H^k(T, \mathcal{M}) \rightarrow H^k(\{[K_X]\}, \mathcal{M}) \rightarrow H_c^{k+1}(U, \mathcal{M}) \rightarrow \cdots$$

gives $H_c^1(U, \mathcal{M}) \cong \mathbb{C}$ and $H_c^k(U, \mathcal{M}) = 0$ for $k \neq 1$. For $M \in U$ Riemann–Roch gives $h^0(M) = g - 1 = n$, so over U the map u is the projective bundle of a vector bundle of rank n [ACGH85,

¹⁷By proper base change the stalk of $u_*\mathcal{L}$ at y is $H^0(u^{-1}(y), \mathcal{L}) \cong \mathbb{C}$, so a nonzero section over the fibre extends over $u^{-1}(D)$ for a small ball $D \ni y$. This set is connected, since u is proper with connected fibres, and a section of a local system of rank one over a connected set is determined by its value at one point and vanishes nowhere unless it is zero. So restriction from $u^{-1}(D)$ to each fibre over D is an isomorphism, $u_*\mathcal{L}$ is locally constant, and $u^*u_*\mathcal{L} \rightarrow \mathcal{L}$ is an isomorphism on stalks.

Chapter IV], and by the Leray–Hirsch theorem $H_c^*(u^{-1}(U), \mathcal{L}) \cong H_c^*(U, \mathcal{M}) \otimes H^*(\mathbb{P}^{n-1})$, which vanishes outside the degrees $1, 3, \dots, 2n-1$. The exact sequence of the pair $(X^{(h)}, F)$,

$$H_c^h(u^{-1}(U), \mathcal{L}) \rightarrow H^h(X^{(h)}, \mathcal{L}) \rightarrow H^h(F, \mathcal{L}) \rightarrow H_c^{h+1}(u^{-1}(U), \mathcal{L}),$$

has zero outer terms, since $h = 2n$. Finally $\mathcal{L}$ is trivial on the simply connected $F \cong \mathbb{P}^n$. $\square$

Proof of Theorem E. Write $V_\chi, V_{\bar{\chi}} \subseteq H^1(B, \mathbb{C})$ for the eigenspaces of σ^* with eigenvalues $\lambda = \chi(\zeta)$ and $\bar{\lambda}$, and choose a basis $\bar{y}_1, \dots, \bar{y}_h$ of $V_{\bar{\chi}}$; put $\bar{\omega} = \bar{y}_1 \wedge \dots \wedge \bar{y}_h$, which spans $\bigwedge^{2n} V_{\bar{\chi}} \subseteq W(B) \otimes \mathbb{C}$.

Step 1: the pull-back of $\bar{\omega}$. Write $\Psi = P' \circ \Xi$, where $\Xi(c_1, \dots, c_h) = \sum_i \text{aj}(c_i)$ and $\text{aj}(c) = [c - c_0]$ is the Abel–Jacobi map. On H^1 the pull-back by the addition map $J^h \rightarrow J$ is the sum of the pull-backs by the projections, so $\Psi^* \bar{y}_j = \sum_i p_i^* \tilde{y}_j$, where p_i is the i -th projection of C^h and $\tilde{y}_j = \text{aj}^* P'^* \bar{y}_j \in H^1(C, \mathbb{C})$. Since P'^* is injective and aj^* is an isomorphism, both commuting with σ^* , the $\tilde{y}_j$ are independent and lie in the $\bar{\lambda}$ -eigenspace of σ^* on $H^1(C, \mathbb{C})$. That eigenspace has dimension h , because the invariant part of $H^1(C, \mathbb{C})$ is $\pi^* H^1(X, \mathbb{C})$, of dimension $2g$, and the two other eigenspaces are conjugate; so the $\tilde{y}_j$ form a basis of it. Now

$$u := \Psi^* \bar{\omega} = \prod_{j=1}^h \left(\sum_{i=1}^h p_i^* \tilde{y}_j \right).$$

A product $\tilde{y}_j \cup \tilde{y}_k$ lies in the $\bar{\lambda}^2$ -eigenspace of $H^2(C, \mathbb{C})$, which is zero because σ acts trivially on $H^2(C)$ and $\bar{\lambda}^2 \neq 1$. So in the expansion only the terms with one factor on each p_i survive: $u = \sum_{s \in S_h} \pm \prod_i p_i^* \tilde{y}_{s(i)}$. By the Künneth formula these terms are linearly independent, and $u \neq 0$. Each term is multiplied by $\bar{\lambda}^{\sum v_i}$ under $(\sigma^{v_1} \times \dots \times \sigma^{v_h})^*$, and u is symmetric because Ψ is; so u is invariant under G' , and $\delta^* u = \bar{\lambda} u$.

Step 2: restriction to Q_0 . By transfer, ρ^* identifies $H^*(W_0, \mathbb{C})$ with $H^*(C^h, \mathbb{C})^{G'}$, so $u = \rho^* \bar{u}$ for a class $\bar{u} \neq 0$ with $\bar{\delta}^* \bar{u} = \bar{\lambda} \bar{u}$. The decomposition $\nu_* \mathbb{C} = \bigoplus_\psi \mathcal{L}_\psi$ into eigensheaves of $\bar{\delta}$, indexed by the characters ψ of $\mathbb{Z}/3$, identifies the $\psi(1)$ -eigenspace of $\bar{\delta}^*$ on $H^*(W_0, \mathbb{C})$ with $H^*(X^{(h)}, \mathcal{L}_\psi)$, compatibly with restriction to $\nu^{-1}(F)$ and F . For $\psi \neq 1$ the local system $\mathcal{L}_\psi$ is nontrivial, since $H^0(W_0, \mathbb{C})$ is the invariant line. By Lemma 5.6, $\bar{u}$ restricts to a nonzero class on $\nu^{-1}(F) = Q_0 \sqcup Q_1 \sqcup Q_2$. As $\bar{\delta}$ maps Q_0 isomorphically onto Q_t and $\bar{u}$ is an eigenvector, $\bar{u}|_{Q_0} = 0$ would force $\bar{u}|_{Q_t} = 0$ for every t . Hence $\int_{Q_0} \bar{u} \neq 0$, and

$$\int_B \Psi_* [\Lambda_0] \cup \bar{\omega} = \int_{[\Lambda_0]} \rho^* \bar{u} = |G'| \int_{Q_0} \bar{u} \neq 0.$$

Step 3: one component. The class $\Psi_* [\Lambda_0]$ is $\sum_k e_k \text{cl}(\Psi(\Lambda_k))$, the sum over the irreducible components Λ_k of Λ_0 on which Ψ is generically finite, with e_k the degree of $\Lambda_k \rightarrow \Psi(\Lambda_k)$. By Step 2 one of them, say $Y = \Psi(\Lambda_k)$, satisfies $\int_B \text{cl}(Y) \cup \bar{\omega} \neq 0$. It is irreducible of dimension n . Since $W(B) \otimes \mathbb{C}$ is spanned by $\bar{\omega}$ and its conjugate, $\text{cl}(Y)$ pairs nontrivially with $W(B)$.

Step 4: the shape of the class. If $\text{Hg}(B) = \text{SU}(V, H)$, then Proposition 5.1 gives $\text{cl}(Y) = c\eta^n + w$ with $c \in \mathbb{Q}$ and $w \in W(B)$. Since $\eta^n \cup \bar{\omega} = 0$, Step 3 gives $w \neq 0$. Since $\eta^n \cup w = 0$,

$$c \int_B \eta^{2n} = \int_B \text{cl}(Y) \cup \eta^n = \deg_\eta Y > 0,$$

so $c > 0$. The last assertion of Theorem E is Corollary 5.8 below. $\square$

Corollary 5.7 (Schoen [Sch88]). *For every étale cyclic triple cover of a curve of genus at least three, the Prym variety B is Weil-algebraic.*

Proof. If B has maximal Hodge group, $w = \text{cl}(Y) - c\eta^n$ is a nonzero algebraic class in $W(B)$, and B is Weil-algebraic. By Lemma 5.4(c) this covers every (X, L) outside a countable union of proper closed subsets Z_i of a smooth connected base T over which, after a finite étale base change, the Prym varieties form a polarized abelian scheme with a trivial local system of Weil planes.¹⁸ The proof of Lemma 4.6 shows that the locus where the Weil classes have algebraic representatives of degree at most d is Zariski closed in T . These countably many closed sets, together with the Z_i , cover T , so by the Baire category theorem one of them has interior. It is not a Z_i , so it is one of the degree- d loci, and it equals T . $\square$

Corollary 5.8. *Suppose that for one (X, L) of genus $n + 1 \geq 3$ with maximal Hodge group the subvariety Y of Theorem E is semiregular. Then every abelian variety of split Weil type for $\mathbb{Q}(\sqrt{-3})$ of dimension $2n$ is Weil-algebraic.*

Proof. The sheaf $\mathcal{O}_Y$ has $\text{ch}_k(\mathcal{O}_Y) = 0$ for $k < n$ and $\text{ch}_n(\mathcal{O}_Y) = \text{cl}(Y) = c\eta^n + w$ with $w \neq 0$ [Ful98, Chapter 15]; for $k > n$ the class $\text{ch}_k(\mathcal{O}_Y)$ is algebraic, hence a multiple of η^k by Proposition 5.1. By Lemma 5.4(b), B is a member of a split Weil family S as in Section 4.4, and Proposition 5.3 gives $\Sigma = S$. Every abelian variety of split Weil type for K of dimension $2n$ is K -isogenous to a member of S ,¹⁹ and a K -isogeny f carries Weil planes to Weil planes and algebraic classes to algebraic classes in both directions, through f^* and $(\deg f)^{-1}f_*$. $\square$

5.5. Transfer by correspondences. Schoen's cycles reach a family of dimension $3n$ inside one of dimension n^2 . One way to spread them is to carry the Weil classes to other varieties by correspondences. This gives a theorem, but a count of dimensions shows that it cannot reach the general member by itself.

Lemma 5.9. *Let B , A_1 and A_2 be of Weil type for K , of dimensions $2(m_1 + m_2)$, $2m_1$ and $2m_2$, and let $f: B \rightarrow A_1 \times A_2$ be an isogeny that commutes with $\mathcal{O}_K$.*

- (a) *f^* carries a plane in $W(A_1) \otimes W(A_2)$ onto $W(B)$; over $\mathbb{C}$ the plane is spanned by $\omega_1 \otimes \omega_2$ and $\bar{\omega}_1 \otimes \bar{\omega}_2$, where ω_i spans $\bigwedge^{2m_i} V_\chi(A_i)$.*
- (b) *If two of B , A_1 , A_2 are Weil-algebraic, so is the third.*

Proof. As in the proof of Corollary 5.8, we may take $B = A_1 \times A_2$. Then $V_\chi(B) = V_\chi(A_1) \oplus V_\chi(A_2)$, and the top exterior power of a sum is the tensor product of the top exterior powers; this is (a). If A_1 and A_2 are Weil-algebraic, every class in $W(A_1) \otimes W(A_2)$ is a sum of products $\text{pr}_1^* \alpha_1 \cup \text{pr}_2^* \alpha_2$ of algebraic classes, so B is Weil-algebraic by (a). Suppose that B and A_2 are Weil-algebraic. On A_2 we have $\omega_2 \cup \omega_2 = 0$, since $\bigwedge^{4m_2} V_\chi(A_2) = 0$, while $p = \int \omega_2 \cup \bar{\omega}_2 \neq 0$, since $\bigwedge^{2m_2} V_\chi \otimes \bigwedge^{2m_2} V_{\bar{\chi}}$ is the top exterior power of $H^1(A_2, \mathbb{C})$; and p is real, because classes of even degree commute. Nonzero rational classes $x \in W(B)$ and $\gamma \in W(A_2)$ have the form $x = s\omega_1 \otimes \omega_2 + \bar{s}\bar{\omega}_1 \otimes \bar{\omega}_2$ and $\gamma = t\omega_2 + \bar{t}\bar{\omega}_2$ with $s, t \neq 0$. The class

$$\text{pr}_{1*}(x \cup \text{pr}_2^* \gamma) = p(s\bar{t}\omega_1 + \bar{s}t\bar{\omega}_1)$$

is rational, algebraic, nonzero, and lies in $W(A_1)$. $\square$

¹⁸Take for T a connected component of a finite étale cover of the moduli space of curves of genus g with level-3 structure, on which L is the section given by a fixed nonzero vector of $(\mathbb{Z}/3)^{2g}$. The monodromy acts on the Weil planes through the determinant, which takes values in the roots of unity of $\mathcal{O}_K$, so it becomes trivial on a finite cover.

¹⁹Two split hermitian forms of the same dimension are isometric, being orthogonal sums of hyperbolic planes. An isometry carries the complex structure of the first variety to a point of the symmetric domain of the second form, and the two lattices are commensurable in the same K -vector space.

Lemma 5.10. *Let K be any imaginary quadratic field, E an elliptic curve with complex multiplication by $\mathcal{O}_K$, and $\bar{E}$ the same curve with $\mathcal{O}_K$ acting through complex conjugation. For $k \geq 1$ the variety $E^k \times \bar{E}^k$, with the product polarization, is of split Weil type, and its Weil classes are polynomials in divisor classes.*

Proof. On $H_1(E, \mathbb{Q}) = K$ the polarization is $\text{Tr}_{K/\mathbb{Q}}(f x \bar{y})$ for some $f \in K$ with $\bar{f} = -f$, so its hermitian form is $x \bar{y}$. For $\bar{E}$ the K -structure is conjugated, and the same Riemann form is $\text{Tr}_{K/\mathbb{Q}}(f \cdot (-\bar{x} y))$, with hermitian form $-\bar{x} y$. So the form of $E^k \times \bar{E}^k$ is $\text{diag}(1, \dots, 1, -1, \dots, -1)$, for which the vectors $e_i + e_{k+i}$ span an isotropic subspace of dimension k , and both embeddings of K occur k times on $H^{1,0}$. With coordinates $z_1, \dots, z_{2k}$ on the factors, V_χ is spanned by $dz_1, \dots, dz_k$ and $d\bar{z}_{k+1}, \dots, d\bar{z}_{2k}$, so up to sign $\omega_\chi = \bigwedge_{i=1}^k (dz_i \wedge d\bar{z}_{k+i})$. Each factor is a class of type $(1, 1)$ on $E \times \bar{E} \cong E \times E$, whose Néron–Severi group has rank $4 = h^{1,1}$ because E has complex multiplication; so it is a complex combination of divisor classes, and so are ω_χ and its conjugate. $\square$

Theorem 5.11. *Let A be an abelian variety of Weil type for $K = \mathbb{Q}(\sqrt{-3})$, of dimension $2m \geq 2$, on which $\mathcal{O}_K$ acts. There are a smooth curve $C \subseteq A$, stable under ζ and with ζ acting freely on it, so that $C \rightarrow X = C/\zeta$ is an étale cyclic triple cover of a curve of some genus $g \geq m + 1$; an abelian variety A' of Weil type of dimension $2(g - 1 - m)$; and an isogeny commuting with $\mathcal{O}_K$ between the Prym variety of $C \rightarrow X$ and $A \times A'$. For every such curve, A is Weil-algebraic if and only if A' is.*

Proof. The curve. The automorphism ζ has finitely many fixed points, the kernel of the isogeny $1 - \zeta$. Let $\mathcal{N}$ be ample on A . The bundle $\mathcal{N}' = \mathcal{N} \otimes \zeta^* \mathcal{N} \otimes \zeta^{2*} \mathcal{N}$ is isomorphic to its pull-back by ζ , so it carries a linearization for the cyclic group generated by ζ ; ²⁰ its cube descends to an ample line bundle $\mathcal{M}$ on the normal quotient $q: A \rightarrow A/\zeta$, since the stabilizers, of order dividing three, act trivially on its fibres. For k large, $2m - 1$ general members D_i of $|\mathcal{M}^k|$ meet in a smooth irreducible curve X that avoids the images of the fixed points, by Bertini’s theorem. Then $C = q^{-1}(X)$ is étale over X , hence smooth, and it is the complete intersection of the ample divisors $q^* D_i$, hence connected. Schoen observes the existence of such curves in [Sch88, §3].

The surjection. By the Lefschetz hyperplane theorem, restriction $H^1(A, \mathbb{Q}) \rightarrow H^1(C, \mathbb{Q})$ is injective, so the homomorphism $\alpha: J(C) \rightarrow A$ induced by the inclusion is surjective; it commutes with ζ . Since $1 + \zeta + \zeta^2 = 0$ in $\mathcal{O}_K$, $\alpha N = 0$ and $\alpha P = 3\alpha$. So α maps the Prym variety $B = P(J(C))$ onto A , and $g - 1 \geq m$.

The complement. Let A' be the connected component of the kernel of $\alpha|_B$ and A'' its complement in B for the induced polarization. Both are stable under σ , which preserves the polarization, so $\mathcal{O}_K$ acts on them, and $A'' \rightarrow A$ and $A' \times A'' \rightarrow B$ are isogenies commuting with $\mathcal{O}_K$. By Lemma 5.4(a) both embeddings of K occur $g - 1$ times on $H^{1,0}(B)$, and they occur m times on $H^{1,0}(A)$; so they occur $g - 1 - m$ times on $H^{1,0}(A')$, and A' is of Weil type.

The equivalence. B is Weil-algebraic by Corollary 5.7, and Lemma 5.9(b) gives the equivalence. $\square$

Corollary 5.12. *A is Weil-algebraic as soon as, for one curve as in Theorem 5.11, the complement A' is K -isogenous to a product of Weil-algebraic varieties, such as Prym varieties of étale cyclic triple covers (Corollary 5.7), abelian surfaces of Weil type (Theorem 2.3), and the varieties $E^k \times \bar{E}^k$ (Lemma 5.10).*

²⁰Choose an isomorphism $\phi: \zeta^* \mathcal{N}' \rightarrow \mathcal{N}'$. The composite of ϕ , $\zeta^* \phi$ and $\zeta^{2*} \phi$ is an automorphism of $\mathcal{N}'$, hence a scalar c , and $c^{-1/3} \phi$ is a linearization.

Proof. Products of Weil-algebraic varieties are Weil-algebraic by Lemma 5.9(b), and Theorem 5.11 applies. $\square$

When $g - 1 = m$ the complement is zero and A is K -isogenous to a Prym variety. For larger g the theorem trades the Weil classes of A for those of A' , and since curves of higher degree give complements of every size, one might hope to choose the curve so that A' is of a known kind. The count below says that for $m \geq 4$ this hope fails at the general member unless an unexpected intersection occurs.

Remark 5.13. Pryms of étale triple covers of curves of genus $m + k + 1$ form a family of dimension $3(m + k)$ inside the Weil family of dimension $(m + k)^2$. The varieties K -isogenous to $A \times A'$, with A in the family of dimension m^2 and A' in a family of dimension f , form a subvariety of dimension $m^2 + f$. For A to be general the two must meet in dimension at least m^2 , which needs $3(m + k) + f \geq (m + k)^2$ unless they meet in excess dimension. Products of the varieties of Corollary 5.12 of total dimension $2k$ have at most $3k$ moduli, and $3(m + k) + 3k < (m + k)^2$ for every $k \geq 0$ once $m \geq 4$. For $m = 3$ only $k = 0$ passes, the Prym locus itself, and for $m = 2$ exactly $k \leq 2$ pass. So Corollary 5.12 reaches only a meagre part of the family once $m \geq 4$, unless the Prym locus meets these product loci in excess dimension, as Schoen anticipated [Sch88, §3]. This is a count, not a proof that no such intersection exists.

5.6. Where it stops. Everything on the Weil route now rests on one question.

Question 5.14. Let $n \geq 4$ and let (X, L) be very general of genus $n + 1$. Is the subvariety $Y \subseteq B$ of Theorem E semiregular?

A positive answer forces Y to deform with B to first order along all n^2 directions of the Weil family, and not only along the $3n$ directions of the Prym locus; so it asks for $n^2 - 3n$ unexpected first order deformations. We have not been able to decide the question, and the analogues point both ways: the Abel–Jacobi curve, discussed next, and the Abel–Prym curve of Proposition 5.16 are rigid, while the theta divisors discussed after Proposition 5.17 deform.

The Abel–Jacobi curve $C \subseteq J(C)$ of a curve of genus $g \geq 4$ is built from the same map as Y . Its class $\theta^{g-1}/(g-1)!$ is a Hodge class on every principally polarized abelian variety. Yet by the criterion of Matsusaka and Ran [Mat59, Ran81] a curve with this class exists only on Jacobians and products of Jacobians, which form a proper subvariety of the moduli space $\mathcal{A}_g$. So the curve does not deform in the directions normal to the Jacobian locus, and by Theorem 5.2 it is not semiregular. The number of those directions is classical. The tangent space to $\mathcal{A}_g$ at $J(C)$ is dual to $\text{Sym}^2 H^0(K_C)$, the tangent space to the Jacobian locus is dual to $H^0(2K_C)$, and by Max Noether’s theorem the multiplication map between them is surjective for non-hyperelliptic C , with kernel the space I_2 of quadrics containing the canonical curve. So the Jacobian locus has codimension

$$\dim I_2 = \binom{g+1}{2} - (3g-3) = \frac{(g-2)(g-3)}{2}$$

at $J(C)$. Schoen’s subvariety is the analogue one step up, built from the fibre of the Abel–Jacobi map over the canonical class, and the corresponding codimension, that of the Prym locus in the Weil family, is $n^2 - 3n$. Nothing in the construction of Y moves it off the Prym locus. This is an analogy and not a proof.

Nor can a branched cover take the place of the étale one, because branching does not enlarge the Prym locus enough.

Proposition 5.15. *Let $\pi: C \rightarrow X$ be a cyclic triple cover of a curve of genus g , branched at r points, with covering automorphism σ , and let its Prym variety, the image of $3 - N$ on $J(C)$*

with $N = 1 + \sigma + \sigma^2$, have dimension $2n$. Then $2q + r = 2n + 2$, and the covers with these invariants form finitely many families, each of dimension $q + 2n - 1 \leq 3n$, with equality only for étale covers of curves of genus $n + 1$. In particular, for $n \geq 4$ a very general member of the split Weil family of K in dimension $2n$ is not K -isogenous to the Prym variety of any cyclic triple cover, étale or branched.

Proof. Since 3 is prime, every branch point is totally ramified, and the Riemann–Hurwitz formula $2g(C) - 2 = 3(2q - 2) + 2r$ gives $g(C) = 3q - 2 + r$. The invariant part of $H^0(C, K_C)$ under σ^* is $\pi^*H^0(X, K_X)$, of dimension q , and as in the proof of Lemma 5.4(a) the space $H^{1,0}$ of the Prym variety is the sum of the two other eigenspaces. So the Prym variety has dimension $g(C) - q = 2q - 2 + r$, and $2q + r = 2n + 2$. Given X and the r branch points, the cover corresponds to a surjection from the fundamental group of the punctured curve onto $\mathbb{Z}/3$, and there are finitely many of these. So the covers form finitely many families, of dimension $3q - 3 + r$ when $q \geq 2$, r when $q = 1$, and $r - 3$ when $q = 0$; by $r = 2n + 2 - 2q$ each of these equals $q + 2n - 1$. As $r \geq 0$, $q \leq n + 1$, so the dimension is at most $3n$, and it equals $3n$ exactly when $q = n + 1$ and $r = 0$.

The Prym varieties of each family form the image of a variety of dimension at most $3n$ in the moduli space, and the K -isogenies of a given degree carry it to finitely many subvarieties of at most the same dimension. The members of the split Weil family that are K -isogenous to such a Prym variety therefore lie in a countable union of subvarieties of dimension at most $3n$, which for $n \geq 4$ is less than n^2 . $\square$

The analogy with the Abel–Jacobi curve can be tested on B itself, in dimension one. Following Lange and Ortega [LO11], call

$$\psi: C \rightarrow B, \quad \psi(c) = (1 - \sigma)(c - c_0),$$

the *Abel–Prym map*. As $P = (1 - \sigma)(1 - \sigma^2)$, the restriction of Ψ to one factor is $(1 - \sigma^2) \circ \psi$, and $1 - \sigma^2$ is an isogeny of B . Put $V_j = H^0(X, K_X \otimes L^j)$, so that $H^0(C, K_C) = V_0 \oplus V_1 \oplus V_2$ as in the proof of Lemma 5.4(a), and let

$$\mu_L: V_1 \otimes V_2 \longrightarrow H^0(X, K_X^{\otimes 2})$$

be multiplication of sections, through an isomorphism $L^3 \cong \mathcal{O}_X$. A first-order deformation of X with Kodaira–Spencer class $u \in H^1(X, T_X)$ carries the étale cover along with it, uniquely, and with it $J(C)$, σ and B . Write $d\mathcal{P}(u) \in H^1(B, T_B)$ for the Kodaira–Spencer class of the resulting deformation of B . Lange and Ortega identify the dual of $d\mathcal{P}$ with a multiplication map [LO11], and we derive the form of this that we need in the proof below.

Proposition 5.16. *Let $v \in H^1(B, T_B)$ be the Kodaira–Spencer class of a first-order deformation $\mathcal{B}$ of B to which σ extends. Then ψ extends to a morphism into $\mathcal{B}$ from some first-order deformation of C if and only if v lies in the image of $d\mathcal{P}$, and this image has dimension $\text{rk } \mu_L \leq 3n$. Suppose moreover that $n \geq 4$, that B has maximal Hodge group, and that S is a split Weil family containing B (Lemma 5.4(b)). Then the tangent vectors to S at B along which ψ extends form a subspace of codimension at least $n^2 - 3n$, although $\psi_*[C]$ is a rational multiple of η^{2n-1} , whose flat continuation is algebraic on every member of S .*

Proof. All deformations are over $\text{Spec } \mathbb{C}[t]/(t^2)$. Suppose that ψ extends to $\psi_t: \mathcal{C} \rightarrow \mathcal{B}$, where $\mathcal{C}$ is a deformation of C with class $w \in H^1(C, T_C)$. Cover C and B by affine open sets U_i and W_i with $\psi(U_i) \subseteq W_i$ over which $\mathcal{C}$ and $\mathcal{B}$ are trivial, with transition automorphisms $1 + t\theta_{ij}$ and $1 + t\xi_{ij}$, where (θ_{ij}) and (ξ_{ij}) are cocycles of vector fields representing w and v . On U_i the map ψ_t is $\psi + ts_i$ for a section s_i of ψ^*T_B , and compatibility on $U_i \cap U_j$ reads $s_j - s_i = \psi^*\xi_{ij} - d\psi(\theta_{ij})$

[Hor74]. Hence

$$(4) \quad \psi^*v = H^1(d\psi)(w) \quad \text{in } H^1(C, \psi^*T_B),$$

where $d\psi: T_C \rightarrow \psi^*T_B$ is the differential.

We compute both sides by Serre duality. The bundle T_B is trivial with fibre T_0B , and ψ^* identifies $H^0(B, \Omega_B^1)$ with $V_- = V_1 \oplus V_2$. Indeed, since $(1 - \sigma)N = 0$, the pull-back by $1 - \sigma: J \rightarrow B$ maps $H^{1,0}(B)$ injectively into the sum of the nontrivial eigenspaces of σ^* on $H^{1,0}(J)$, which has dimension $2n = \dim B$, and the Abel–Jacobi map identifies this sum with V_- . So $H^1(C, \psi^*T_B) = H^1(C, \mathcal{O}_C) \otimes T_0B$ is dual to $H^0(C, K_C) \otimes V_-$, and $H^1(C, T_C)$ is dual to $H^0(C, K_C^{\otimes 2})$. The dual of $d\psi$ is evaluation $\mathcal{O}_C \otimes V_- \rightarrow K_C$, so the dual of $H^1(d\psi)$ is multiplication

$$m: H^0(C, K_C) \otimes V_- \longrightarrow H^0(C, K_C^{\otimes 2})$$

[LO11]. Thus (4) says that ψ^*v , as a linear form on $H^0(C, K_C) \otimes V_-$, is $w \circ m$.

Now use σ . We have $\psi\sigma = t_a\sigma\psi$, where t_a is the translation by $a = (1 - \sigma)(\sigma c_0 - c_0)$. Translations act trivially on $H^1(B, \mathcal{O}_B)$ and on T_0B , so ψ^* commutes with the actions of σ , and m is equivariant. As σ extends to $\mathcal{B}$, the class v is invariant, and so is the form ψ^*v ; an invariant form vanishes on every eigenspace of σ with an eigenvalue other than 1. The automorphism σ^* acts on V_j by κ^j for a primitive cube root of unity κ , so the invariant part of $H^0(C, K_C) \otimes V_-$ is $E = (V_1 \otimes V_2) \oplus (V_2 \otimes V_1)$. Since $\pi_*K_C^{\otimes 2} = K_X^{\otimes 2} \otimes \pi_*\mathcal{O}_C$, the map m sends E to $H^0(C, K_C^{\otimes 2})^\sigma = \pi^*H^0(X, K_X^{\otimes 2})$, by $(x, y) \mapsto \pi^*M(x, y)$ with $M(x, y) = \mu_L(x + \tau y)$, where $\tau: V_2 \otimes V_1 \rightarrow V_1 \otimes V_2$ is the transposition. So ψ^*v is determined by its restriction to E , which by (4) is $f \circ M$, where f is the linear form $q \mapsto \langle w, \pi^*q \rangle$ on $H^0(X, K_X^{\otimes 2})$, that is, an element of $H^1(X, T_X)$.

Apply this to the deformation of B induced by $u \in H^1(X, T_X)$. The deformation of C has class π^*u , and ψ extends after c_0 is extended to a section. Since π has degree 3, $\langle \pi^*u, \pi^*q \rangle = 3\langle u, q \rangle$, so the restriction of $\psi^*d\mathcal{P}(u)$ to E is $3u \circ M$. Returning to v , the invariant forms ψ^*v and $\psi^*d\mathcal{P}(f/3)$ agree on E , hence everywhere. The map ψ^* on $H^1(B, T_B) = H^1(B, \mathcal{O}_B) \otimes T_0B$ is injective, since on $H^1(B, \mathcal{O}_B)$ it is the complex conjugate of the injective map on $H^0(B, \Omega_B^1)$. So $v = d\mathcal{P}(f/3)$. Conversely, if $v = d\mathcal{P}(u)$, then $\mathcal{B}$ is isomorphic to the deformation induced by u , since first-order deformations are determined by their Kodaira–Spencer classes, and ψ extends. Finally $d\mathcal{P}(u) = 0$ exactly when $u \circ M = 0$, that is, when u vanishes on the image of μ_L ; so $d\mathcal{P}$ has rank $\text{rk } \mu_L \leq h^0(X, K_X^{\otimes 2}) = 3g - 3 = 3n$.

For the last assertion, the element $\zeta \in \mathcal{O}_K$ acts on the abelian scheme over S as σ on B , so the Kodaira–Spencer map $T_{S,B} \rightarrow H^1(B, T_B)$ has invariant image. It is injective, because S is locally isomorphic to the symmetric domain $\mathcal{D}$, the period map of the family lifts to the inclusion of $\mathcal{D}$ in the period domain, and the differential of the period map is the Kodaira–Spencer map followed by the isomorphism $H^1(B, T_B) \cong \text{Hom}(H^{1,0}(B), H^{0,1}(B))$ [Voi02, Chapter 10]. So the tangent vectors along which ψ extends form a subspace of dimension at most $3n$ in the n^2 -dimensional space $T_{S,B}$. By Proposition 5.1, $\psi_*[C] \in \text{Hdg}^{2n-1}(B) = \mathbb{Q}\eta^{2n-1}$. $\square$

So the curve $\psi(C)$, whose class stays algebraic on the whole family, moves with B only along the Prym locus. When X has Clifford index at least 2, as a very general curve of genus at least 5 does [ACGH85, Chapter V], the map ψ is a closed embedding [LO11], and then, for $n \geq 4$ and B with maximal Hodge group, the curve $\psi(C)$ is not semiregular. Otherwise Theorem 5.2 would extend $\mathcal{O}_{\psi(C)}$ along every tangent vector of S ; a first-order deformation of $\mathcal{O}_{\psi(C)}$ is locally generated by one section, so it is a line bundle on the subscheme defined by its annihilator, which is a first-order deformation of $\psi(C)$ and so carries an extension of ψ , against Proposition 5.16.

This makes the analogy precise on B , but it does not decide Question 5.14. For Y itself we have the following description, which does not mention Λ_k : up to an étale map, Y is a Brill–Noether locus of the Prym variety. Let $\text{Nm}: \text{Pic}^{2n}(C) \rightarrow \text{Pic}^{2n}(X)$ be the norm map, let $N_K = \text{Nm}^{-1}([K_X])$, and let $V \subseteq N_K$ be the closed subset of the line bundles M with $h^0(C, M) > 0$. On J the endomorphism N is $\pi^* \circ \text{Nm}$. Put

$$\phi: N_K \rightarrow B, \quad \phi(M) = P(M \otimes \mathcal{O}_C(-2nc_0)).$$

Proposition 5.17. (a) *On each connected component of N_K the map ϕ is a translate of multiplication by 3, an étale map onto B of degree 3^{4n} , and $Y = \phi(V_k)$ for an irreducible component V_k of V of dimension n .*

(b) *Let M be a general point of V_k . Then $h^0(C, M) = 1$, and the divisor E of a nonzero section s of M contains no two points of one fibre of π . The varieties V_k and Y are smooth of dimension n at M and at $\phi(M)$, and $T_M V_k \subseteq H^1(C, \mathcal{O}_C)$ is the annihilator of the subspace*

$$\pi^* H^0(X, K_X) + s \cdot H^0(C, K_C(-E))$$

of $H^0(C, K_C)$, which has dimension $2n + 1$: the two summands have dimension $n + 1$ and meet in the line spanned by $\pi^ \omega$, where $\text{div } \omega = \pi_* E$.*

Proof. (a) Since $\pi^* \circ \text{Nm} = N$ on J and $NP = 0$, the norm maps B into the kernel of π^* on $J(X)$, which is finite; as B is connected, $B \subseteq \ker \text{Nm}$. This kernel has dimension $(3n + 1) - (n + 1) = 2n = \dim B$, so B is its identity component, and every connected component of N_K is a coset $M_1 + B$. For $b \in B$ we have $\phi(M_1 + b) = \phi(M_1) + Pb = \phi(M_1) + 3b$, and $\phi(M_1) \in P(J) = B$. For $(c_1, \dots, c_h) \in \Lambda_0$ put $E = \sum_i c_i$; then $\pi_* E = \mu(c_1, \dots, c_h)$ lies in $F = |K_X|$, so $\mathcal{O}_C(E) \in V$, and $\Psi(c_1, \dots, c_h) = \phi(\mathcal{O}_C(E))$. Let V_k be the image of Λ_k under $(c_i) \mapsto \mathcal{O}_C(\sum_i c_i)$. It is irreducible and closed, and $Y = \phi(V_k)$. Its dimension is at most $\dim \Lambda_k = n$ and at least $\dim Y = n$. On the other hand, if $M \in V$ and E is the divisor of a nonzero section, then $\mathcal{O}_X(\pi_* E) \cong \text{Nm}(M) \cong K_X$; so V is the image of $Z = \{E \in C^{(2n)} : \pi_* E \in F\}$, the preimage of F under the finite map $C^{(2n)} \rightarrow X^{(2n)}$, of dimension at most n . Hence V_k is an irreducible component of V .

(b) Let $Z_k \subseteq Z$ be the image of Λ_k ; it has dimension n , so it is an irreducible component of Z , and it maps onto V_k . The map $\Lambda_k \rightarrow Q_0 \cong F$ is finite between irreducible varieties of the same dimension, hence surjective, so for E general in Z_k the divisor $\pi_* E$ is a general member of $|K_X|$. It is reduced, because $|K_X|$ has no base points, so the $2n$ points of E lie over distinct points of X . If $h^0(\mathcal{O}_C(E)) \geq 2$ for general $E \in Z_k$, the linear system $|E|$ lies in Z , since $\mathcal{O}_X(\pi_* E') \cong \text{Nm } \mathcal{O}_C(E)$ for $E' \in |E|$; being irreducible, it lies in Z_k , the only component through a general point. Then the fibres of $Z_k \rightarrow V_k$ have positive dimension and $\dim V_k < n$, which is false. So $h^0(M) = 1$ for general $M = \mathcal{O}_C(E)$ in V_k .

Let $W \subseteq \text{Pic}^{2n}(C)$ be the image of $C^{(2n)}$. Since $h^0(M) = 1$, W is smooth at M and $T_M W$ is the annihilator of $s \cdot H^0(C, K_C(-E))$ [ACGH85, Chapter IV]. The differential of the norm is the trace $H^1(C, \mathcal{O}_C) \rightarrow H^1(X, \mathcal{O}_X)$, which is Serre dual to $\pi^*: H^0(X, K_X) \rightarrow H^0(C, K_C)$; so N_K , a coset of $\ker \text{Nm}$, is smooth with $T_M N_K$ the annihilator of $\pi^* H^0(X, K_X)$. As $V_k \subseteq W \cap N_K$, the tangent space of V_k at M lies in the annihilator of the sum. By Riemann–Roch $h^0(C, K_C(-E)) = h^0(C, \mathcal{O}_C(E)) - \deg E + g(C) - 1 = n + 1$, and $h^0(X, K_X) = n + 1$. A form $\pi^* \omega$ in both summands vanishes on E ; since the points of E lie over distinct points of X , ω vanishes on $\pi_* E$, which has degree $2n = \deg K_X$, so $\text{div } \omega = \pi_* E$. Conversely this ω lies in both, since $\text{div } \pi^* \omega = E + \sigma E + \sigma^2 E$. So the sum has dimension $2n + 1$ and its annihilator in the space $H^1(C, \mathcal{O}_C)$ of dimension $3n + 1$ has dimension $n = \dim V_k$. Hence V_k is smooth at M with this tangent space. Finally $\phi^{-1}(Y)$ meets the component of N_K containing V_k in the union of the translates $V_k + t$, $t \in B[3]$. Those different from V_k meet it in a proper closed

subset, which the general point M avoids, so the étale map ϕ carries a neighbourhood of M in V_k isomorphically onto a neighbourhood of $\phi(M)$ in Y . $\square$

So Y is the analogue, for cyclic triple covers, of a Prym–Brill–Noether locus, and the comparison with étale double covers points the other way from the Abel–Prym curve. For an étale double cover of a curve of genus g , Mumford shows that $\mathrm{Nm}^{-1}([K_X])$ has two components, on which h^0 is even and odd, and that the locus $h^0 \geq 2$ in the even one is a theta divisor Ξ of the principally polarized Prym variety [Mum74]. So there V is the union of the odd component and Ξ , and Ξ deforms with every principally polarized abelian variety of dimension $g - 1$, although for $g \geq 7$ the Prym varieties of double covers, with $3g - 3$ moduli, form a proper subvariety of that moduli space. The analogy is loose in one respect: Ξ is a divisor whose class is a principal polarization, while Y has codimension n . Likewise, the theta divisor of a Jacobian is the image of the $(g - 1)$ -st symmetric product of the curve under the Abel–Jacobi map, and it deforms with every principally polarized abelian variety, while the Abel–Jacobi curve does not.

Proposition 5.17 also shows what a direct computation would need. On the open subset of V_k where (b) holds, the normal space of V_k at M is dual to $H^0(C, K_C(-E))/\mathbb{C}t$, where t is the section with divisor $\sigma E + \sigma^2 E$; so there the normal bundle comes from the bundle with fibres $H^0(C, K_C \otimes M^{-1})$. The obstruction to extending $\mathcal{O}_Y$ along a tangent vector of S is a class in a global cohomology group, and computing it needs the normal sheaf on all of V_k , including the locus where $h^0 \geq 2$ and the divisors E that meet a fibre of π twice. We have not determined it.

A weaker hypothesis would do if the semiregularity theorem were sharper. Markman asks whether Theorem 5.2 holds when $\sigma_{\mathcal{E}}$ is injective only on the image of the evaluation map from the second Hochschild cohomology $HH^2(Z)$ to $\mathrm{Ext}^2(\mathcal{E}, \mathcal{E})$ [Mar25b, Question 11.4]. A positive answer would replace the semiregularity of $\mathcal{O}_Y$ in Corollary 5.8 by injectivity of $\sigma_{\mathcal{O}_Y}$ on that image, a condition that involves only the deformations of B and of its category of sheaves. For a general imaginary quadratic field, cycles representing Weil classes at members with maximal Hodge group are known in dimension four, and for split forms in dimension six [Mar25a]. For nonsplit forms in dimension six, and in dimension $2n \geq 8$ for fields other than $\mathbb{Q}(\sqrt{-3})$ and $\mathbb{Q}(\sqrt{-1})$, we know of none, so the route has nothing to deform.

6. THE ROUTES

We now treat the statements as propositional letters and the results of the paper as rules between them. Besides the seven statements of Definition 2.7 and (HC) there are three more: (SR), that the subvariety Y of Theorem E is semiregular for some (X, L) of each genus $n + 1 \geq 5$ with maximal Hodge group; (W3), that the Weil classes are algebraic on every abelian variety of split Weil type for $\mathbb{Q}(\sqrt{-3})$; and (Fer), the Hodge conjecture for every Fermat variety. The rules are listed in Table 1.

For $2n \leq 6$ the conclusion of (SR) is already known, which is why (SR) only concerns the genera $n + 1 \geq 5$. A set of statements is *closed* if it contains the conclusion of every rule whose premises it contains. The statements implied by a set S form the smallest closed set containing S , so a closed set that contains S and not (HC) shows that (HC) does not follow from S .

Lemma 6.1. *Let $\mathcal{U}$ be the set of all eleven statements. The four sets*

$$\begin{aligned} T_1 &= \mathcal{U} \setminus \{(\mathrm{HC}), (\mathrm{M}), (\mathrm{F3}')\}, & T_2 &= \mathcal{U} \setminus \{(\mathrm{HC}), (\mathrm{L}), (\mathrm{F3}')\}, \\ T_3 &= \mathcal{U} \setminus \{(\mathrm{HC}), (\mathrm{F2}), (\mathrm{L}), (\mathrm{V}), (\mathrm{IP})\}, & T_4 &= \mathcal{U} \setminus \{(\mathrm{HC}), (\mathrm{F2}), (\mathrm{L}), (\mathrm{V}), (\mathrm{CM})\} \end{aligned}$$

are closed, and so is $\{(\mathrm{CM}), (\mathrm{Fer})\}$.

Implication	Source
$(F2) \wedge (F3') \Rightarrow (HC)$	Section 4.3: if (F2) holds then $\text{Ab}^p(X) = \text{Alg}^p(X)$
$(HC) \Rightarrow (F2), (F3'), (L), (M), (V)$	Lemma 2.11
$(L) \wedge (M) \Rightarrow (HC)$	Proposition 2.8, after André [And96]
$(L) \Rightarrow (F2)$	Abdulali, André [Mil20, Theorem 4]
$(V) \Rightarrow (F2)$	[Mil20, Theorem 3 and Remark 2]
$(V) \Rightarrow (IP)$	(IP) is a case of (V)
$(F3') \Rightarrow (M)$	Proposition 2.9
$(CM) \wedge (IP) \Rightarrow (F2)$	Proposition 4.4
$(F2) \Rightarrow (CM), (IP)$	Proposition 4.4
$(CM) \Rightarrow (\text{Fer}), (HC) \Rightarrow (\text{Fer})$	Theorem C
$(SR) \Rightarrow (W3)$	Corollary 5.8 for $2n \geq 8$; Theorem 2.3 and [Sch88, Sch98] for $2n \leq 6$
$(F2) \Rightarrow (W3)$	(W3) is a case of (F2)

TABLE 1. The implications between the statements. A set of statements *implies* a statement if the statement is reached from the set by repeated use of these rules.

Proof. A rule can fail to be respected by a set only if its conclusion is missing from the set while all its premises are present. We check the rules whose conclusions are missing. In T_1 the missing conclusions are (HC), (M) and (F3'); every rule concluding one of them has a premise among (HC), (F3') and (M), which are missing. In T_2 the rules concluding (HC), (L) or (F3') need one of (F3'), (HC) or (L). In T_3 every rule concluding (HC), (F2), (L), (V) or (IP) has a premise among (F2), (HC), (L), (V) and (IP); the rule $(CM) \wedge (IP) \Rightarrow (F2)$ is blocked by (IP). T_4 is the same with (CM) in place of (IP). In $\{(CM), (\text{Fer})\}$ the only rule with all premises present is $(CM) \Rightarrow (\text{Fer})$. $\square$

Proof of Theorem B. Let S be a set of statements, not containing (HC), that implies (HC). Then S is contained in none of $T_1, \dots, T_4$, so it meets each of

$$\{(M), (F3')\}, \quad \{(L), (F3')\}, \quad \{(F2), (L), (V), (IP)\}, \quad \{(F2), (L), (V), (CM)\}.$$

If $(F3') \notin S$, the first two give $(M), (L) \in S$, which is (b). If $(F3') \in S$ and S contains none of $(F2), (L), (V)$, the last two give $(IP), (CM) \in S$. This is (a).

Conversely, $\{(F2), (F3')\}$ and $\{(L), (M)\}$ imply (HC) directly, and $\{(L), (F3')\}, \{(V), (F3')\}$ and $\{(CM), (IP), (F3')\}$ imply (F2) and then (HC). From (HC) the rules give (F2), (F3'), (L), (M), (V), then (CM), (IP), (W3) and (Fer): every statement except (SR). No minimal set contains (SR), because (SR) is a premise only of the rule that concludes (W3), and (W3) is a premise of no rule. $\square$

Corollary 6.2. (i) *A set of statements that implies (HC) and contains neither (F2) nor (F3') contains (L) and (M); by Theorem A this pair is equivalent to (HC).*
(ii) *Every set that implies (HC) also implies (F2) and (F3').*
(iii) *A set that implies (HC) still implies it when (SR) is removed.*
(iv) *(CM) implies (Fer) and nothing else; neither $\{(CM), (F3')\}$ nor $\{(CM), (SR)\}$ implies (HC).*

Proof. (i) is Theorem B(b), since (a) requires (F3'). (ii) holds because (HC) implies (F2) and (F3'). (iii) is the last sentence of the proof of Theorem B. For (iv), $\{(\text{CM}), (\text{Fer})\}$ is closed by Lemma 6.1, and $\{(\text{CM}), (\text{F3}')\}$ and $\{(\text{CM}), (\text{SR})\}$ lie in the closed set T_3 , which does not contain (HC). $\square$

So the conjecture can be reached without assuming (F2) or (F3'), but only through (L) and (M), which carry the whole conjecture between them; and it cannot be reached without proving (F2) and (F3') along the way. Schoen's subvariety, if it is semiregular, settles a family of Weil classes in every dimension, but it lies on no route to (HC). The closed set $\{(\text{CM}), (\text{Fer})\}$ is a valuation of the eleven statements in which every rule of Table 1 holds, (CM) holds, and (HC) fails: the rules alone cannot derive the conjecture from the preprint [Ope26b].

7. WHAT IS PROVED AND WHAT IS NOT

Proved. The following hold without any unproved hypothesis.

- (1) The equivalences of Theorem A, and the description in Theorem B and Corollary 6.2 of every set of the statements of Table 1 from which the conjecture follows by those implications.
- (2) The statement (F3') for every Fermat variety, every product of Fermat varieties and every variety onto which such a product maps surjectively (Theorem C), together with the description of the Hodge classes of Fermat varieties by characters (Proposition 3.6).
- (3) The closedness of the bounded-degree loci Σ_d in a split Weil family, and the criterion of Theorem D for $\Sigma = S$.
- (4) For every étale cyclic triple cover of a curve of genus $n + 1 \geq 3$: its Prym variety is of split Weil type (Lemma 5.4); Schoen's construction gives an irreducible n -dimensional subvariety Y whose class pairs nontrivially with the Weil plane; at a member with maximal Hodge group $\text{cl}(Y) = c\eta^n + w$ with $c > 0$ and $w \neq 0$ (Theorem E); and the Prym variety is Weil-algebraic (Corollary 5.7, first proved by Schoen).
- (5) The transfer of Theorem 5.11: an abelian variety of Weil type for $\mathbb{Q}(\sqrt{-3})$ is, up to isogeny, a factor $A \times A'$ of a Prym variety, and A is Weil-algebraic if and only if A' is.
- (6) The count of Proposition 5.15: Prym varieties of cyclic triple covers, étale or branched, form families of dimension at most $3n$, so for $n \geq 4$ they miss the very general member of the split Weil family.
- (7) The rigidity of Proposition 5.16: the Abel–Prym curve of an étale cyclic triple cover deforms with its Prym variety only along the Prym locus, although its class stays algebraic on the whole split Weil family; so for $n \geq 4$ and maximal Hodge group, when it is embedded, it is not semiregular.
- (8) The description of Y in Proposition 5.17: up to an étale map, it is a component of the locus of line bundles with a nonzero section in the fibre of the norm map over $[K_X]$. It is smooth of dimension n at its general point, with an explicit tangent space.
- (9) The Hodge conjecture for the Fermat fourfold X_m^4 of every odd degree m (Theorem F); for m prime to 6, the classification of the Hodge characters behind it: each contains a pair or is 5-standard (Theorem 3.9); and for $3 \mid m$, the reduction of every Hodge character without a pair, by moves, to three exceptional orbits (Theorem 3.27 and lemma 3.26). Besides the results of this paper, the proof uses the inductive structure of Shioda and Ran, Aoki's cycles [Aok87], and the classical nonvanishing of $B_{1,\chi}$ for odd primitive Dirichlet characters [Was97].

Proved under a stated hypothesis.

- (10) Under (CM): the Hodge conjecture for every Fermat variety, every product of Fermat varieties and every variety such a product dominates (Theorem C), and the algebraicity of the Weil classes at every CM point of a split Weil family (Theorem D).
- (11) Under the semiregularity of one subvariety Y of genus $n + 1$ with maximal Hodge group: the Weil classes of every abelian variety of split Weil type for $\mathbb{Q}(\sqrt{-3})$ of dimension $2n$ (Corollary 5.8).

Not proved. The Hodge conjecture; each of (F2), (F3'), (L), (M), (V), (IP); the semiregularity of Y for any $n \geq 4$ (Question 5.14); and (CM), which is claimed in the unrefered preprint [Ope26b] and used here only as a hypothesis.

The first open cases. By Theorem B every route passes through (F2) and (F3'), and each has a first case that no known method reaches.

Weil classes in dimensions six and eight. (F2) holds in dimension at most five by [MZ99] and the Weil fourfolds of the preprint [Mar25a] (Section 5.2). In dimension six the Weil classes of split forms are algebraic, for $\mathbb{Q}(\sqrt{-3})$ by [Sch88, Sch98] and in general by [Mar25a], and those of nonsplit forms are open. For $2n = 8$ we know of no family of Weil type whose general member has maximal Hodge group and which carries algebraic Weil classes at every member. For the split family of $\mathbb{Q}(\sqrt{-3})$ this is Question 5.14 with $n = 4$, where Y would have to deform in $n^2 - 3n = 4$ directions beyond the Prym locus. Proposition 5.15 shows that Prym varieties of branched cyclic triple covers do not reach the general member either, and Remark 5.13 shows the same for Corollary 5.12. The Abel–Prym curve, built from the same map as Y , does not deform in those directions (Proposition 5.16), while the Prym theta divisor of an étale double cover, the analogue of Y by Proposition 5.17, deforms with every principally polarized abelian variety; neither decides the question for Y . Preprints announcing the split eightfolds are cited in [BMB26]; we could not locate them in the public release of [Ope26b], and we do not use them.

Beyond Weil classes. On the square $A \times A$ of a Mumford fourfold A [Mum69] there are Hodge classes that are not generated by divisor classes and Weil classes. They are the first instance of (F2) that the Weil route does not cover. In [BMB26, Theorem 21.4] they are reduced to the Kuga–Satake class of one K3 surface, which is not known to be algebraic.

The first case of (F3'). For a K3 surface S whose transcendental lattice has a totally real field of degree at least two as its algebra of Hodge endomorphisms [vG08], the classes on $S \times S$ that induce these endomorphisms are not known to be algebraic in general [BFvGS24]; $S \times S$ lies in the reach of abelian varieties by algebraic correspondences once the Kuga–Satake correspondence of S is algebraic [BMB26, Remark 20.36]. They are algebraic on several explicit families, where the real multiplication comes from double planes or from isogenies [Sch10, vGS25], and on the general member of the first four-dimensional families with real multiplication by a real quadratic field, through symplectic automorphisms [Var23]. For real multiplication by $\mathbb{Q}(\sqrt{d})$ it would suffice to find an algebraic correspondence from S to some K3 surface S' that induces a Hodge similarity $T(S) \rightarrow T(S')$ of multiplier d : composing its inverse with $\sqrt{d}$ gives a rational Hodge isometry, which is algebraic by the theorem of Buskin and Huybrechts [Bus19, Huy19]. For a real quadratic field the maximal families have Picard number 2 and dimension 8 [vG08], and their very general member is the first open case.

So the conjecture is equivalent to two statements, and each has an explicit first open case. What this paper adds to the Weil case is that, for $\mathbb{Q}(\sqrt{-3})$ in every dimension, it now depends on a single, explicitly constructed subvariety of a Prym variety and the question whether it is semiregular.

A note on preparation. Parts of this paper and of the code that checks its finite steps were drafted with the help of an AI assistant, working under the author’s direction. Responsibility for the content rests with the author.

APPENDIX A. MACHINE CHECKS

No proof in this paper depends on a computer. The finite steps have been repeated by machine, in several independent implementations, and the files are in the author’s repository <https://github.com/creelie/e84hc> under `verification/`; each release of the repository is archived on Zenodo under the concept DOI 10.5281/zenodo.23234362, and releases v1.2.0 and v1.3.0 under 10.5281/zenodo.23272739 and 10.5281/zenodo.23272887. One script, `shell/run_all.sh`, runs them all and compares their outputs.

Exact computations in three languages. The program `closure_checks` exists in Python, Julia and C, all in exact integer or rational arithmetic, and the three versions print the same 218 lines; the shell script compares them line by line and against a stored copy. The lines fall into six groups.

- F Fermat varieties X_m^n for $n = 2$, $m \leq 12$ and for $n = 4$, $m \leq 8$: the characters of Proposition 3.6, the primitive Betti number against $((m-1)^{n+2} + (-1)^n(m-1))/m$, the middle Hodge number, the Hodge characters and those of pair type; for surfaces, $h^{1,1} = (2m^3 - 6m^2 + 7m)/3$, the Picard number ρ , and Aoki and Shioda’s formula $\rho = 3(m-1)(m-2) + 1$ for $(m, 6) = 1$ [AS83]. The values $\rho = 20, 37, 86$ for $m = 4, 5, 6$ appear among them.
- J Fermat curves of degree $m \leq 30$: every orbit of characters under $(\mathbb{Z}/m)^\times$ has an odd type function, as in the proof of Lemma 3.3.
- S The switch lemma (Lemma 4.1) for all pairs of lists with $p \leq 4$ rows and $s \leq 3$ columns: the column correction reaches the target by two-row switches, each preserving the column sums, in at most $s\lfloor p/2 \rfloor$ steps.
- G The signs $(-1)^{p(p-1)/2}$ of the permutations that regroup a product of $2p$ classes of degree one between interleaved and blocked order, for $p \leq 10$.
- W The dimensions $3n$ and n^2 of the Prym locus and the Weil family, and the count of Remark 5.13: the pairs (m, k) with $3(m+k) + 3k \geq (m+k)^2$, which are $k \leq 2$ for $m = 2$, $k = 0$ for $m = 3$, and none for $4 \leq m < 60$, $k < 200$; and the count of Proposition 5.15: for $2 \leq n \leq 12$ and every genus g , the Riemann–Hurwitz relation, the dimension $q + 2n - 1$ of the family, and its unique maximum $3n$ at $q = n + 1$, $r = 0$; and for Proposition 5.16, the pairs (a, b) with $0 \leq a \leq 2$, $1 \leq b \leq 2$ and $a + b \equiv 0 \pmod{3}$, which are $(1, 2)$ and $(2, 1)$, the dimensions $g, g - 1, g - 1$ of V_0, V_1, V_2 adding up to the genus $3g - 2$ of C , and the dimensions $2n^2$ of E and $3n$ of $H^0(X, K_X^{\otimes 2})$.
- L The rules of Table 1: the closed sets of Lemma 6.1; the valuation in which (CM) holds and (HC) fails; the five minimal sets of Theorem B, found by enumerating all sets of statements; and the fact that (L) and (M) form the only minimal set avoiding (F2), (F3’), (IP), (CM) and (SR).

Lean. The file `lean/Closure.lean` is checked by the kernel of Lean 4 with the core library only, without `sorry` and without `native_decide`; every theorem in it depends only on the standard axioms. It states the rules of Table 1 as an inductive derivability relation and proves: that the four sets of Lemma 6.1 and the set $\{(CM), (Fer)\}$ are closed; Theorem B itself, for an arbitrary set of statements; Corollary 6.2(i); and the derivations of (HC) from each of the five minimal sets. It also proves, for all sizes at once, the counting identity behind the switch lemma, the equivalence $3n < n^2 \iff n \geq 4$, the inequality of Remark 5.13 for every $m \geq 4$

and $k \geq 0$, and the count of Proposition 5.15 for all n , q and r , and the eigenspace bookkeeping of Proposition 5.16. By evaluation in the kernel, it computes the Picard numbers 20, 37 and 86 of the Fermat quartic, quintic and sextic surfaces from the characters, and it checks that the sets O_{21} , O_{33} and O_{39} of Section 3.7 are balanced, contain no pair, generate, and have no direct or lowering move, and that the three identities in the proof of Lemma 3.26 hold with every part balanced.

Lean with Mathlib. The directory `lean-mathlib` holds a formalization of Theorem 3.9 in Lean 4 with Mathlib, about 3600 lines in 15 files. It follows the proof of Section 3.6 step by step: the counting function, Dirichlet characters and their conductors, Fourier inversion on $(\mathbb{Z}/e)^\times$, the subgroups K_p and their orders, split functions and Lemmas 3.15 to 3.17, Proposition 3.18, Lemmas 3.19 and 3.20, and Step 7. The theorems `hodge_quadruple` and `hodge_sextuple` state Theorem 3.9 for every m prime to 6, and `hodge_fermat_fourfold` derives the conclusion of Theorem F for an abstract predicate “the line $V(\alpha)$ is algebraic” from the two geometric inputs, algebraicity for characters that contain a pair and for 5-standard characters. Theorem 3.27, for m divisible by 3, is not formalized there; it is checked by the searches below for $m \leq 105$.

The nonvanishing of $B_{1,\chi}$ for odd primitive χ is proved there as well, from the nonvanishing of $L(1, \bar{\chi})$ in Mathlib, along the classical route of Step 2. Expanding $\bar{\chi}(n)$ by the Gauss sum $\tau(\chi)$ gives

$$\tau(\chi) \sum_{n \geq 1} \frac{\bar{\chi}(n)}{n} = - \sum_{s=1}^{e-1} \chi(s) \log(1 - e^{2\pi i s/e}) = - \frac{\pi i}{e} \sum_{s=1}^{e-1} \chi(s) s,$$

where the logarithmic series converges on the unit circle away from 1 by Dirichlet’s test and Abel’s limit theorem, the real parts cancel because χ is odd, and $\arg(1 - e^{i\theta}) = \theta/2 - \pi/2$ for $0 < \theta < 2\pi$. A summation by parts, uniform in $\sigma \in [1, 2]$, shows that the value at $s = 1$ of the analytic continuation of $L(s, \bar{\chi})$ is the sum of the conditionally convergent series on the left. The geometric inputs and the description of the Hodge classes by characters remain hypotheses of the formal statements, since Mathlib has no Hodge theory. With them, the main theorems, among them the classification and the conclusion of Theorem F with $B_{1,\chi} \neq 0$ discharged, depend only on the standard axioms, and no file uses `sorry`.

Fermat fourfolds by exhaustive search. The program `fermat_fourfolds` exists in Python, Julia and C. For every m prime to 6 up to a bound it lists the balanced multisets of four and of six nonzero residues modulo m and checks Theorem 3.9: every quadruple contains a pair, and every sextuple without a pair is 5-standard. The three versions print the same lines for $m \leq 55$, and the C version is run up to $m = 125$ and compared with a stored copy of its output; for $m = 125$, for instance, it finds 41688 balanced sextuples, of which 24 contain no pair, all of them 5-standard.

The program `fermat_odd`, also in Python, Julia and C, treats the odd m divisible by 3. It lists the balanced sextuples, and for each one that contains no pair and generates $\mathbb{Z}/m$ it finds a direct or a lowering move (Definition 3.23), or checks that tA is one of O_{21} , O_{33} , O_{39} for a unit t , as Theorem 3.27 asserts. It also checks the identities of Lemma 3.26 and that the three exceptional sets have no move. The three versions print the same lines for $m \leq 63$, and the C version is run up to $m = 105$ and compared with a stored copy of its output; for $m = 105$ it finds 27690 balanced sextuples, of which 78 contain no pair and 54 of these generate $\mathbb{Z}/105$, each with a direct move. These searches check the theorems only for the degrees listed; the proofs cover every m .

Macaulay2. Two scripts use Macaulay2. The first computes the primitive Hodge numbers of the Fermat surfaces and fourfolds from the Jacobian ring, $h_{\text{prim}}^{n-q,q} = \dim R_{(q+1)m-(n+2)}$ with $R = \mathbb{Q}[x_0, \dots, x_{n+1}]/(x_0^{m-1}, \dots, x_{n+1}^{m-1})$ [Gri69], and checks that they agree with the character counts of item F. The second takes random smooth plane curves of degree $4 \leq d \leq 8$ modulo the prime 32003 and checks Max Noether’s theorem in the form used in Section 5.6: the multiplication map $\text{Sym}^2 H^0(K) \rightarrow H^0(2K)$ has rank $3g - 3$, so $\dim I_2 = (g - 2)(g - 3)/2$. A smooth curve of full rank modulo p is smooth and of full rank over $\mathbb{Q}$, so these are checks over $\mathbb{Q}$.

References. The script `references/check_references.py` looks up every entry of the bibliography online, through its DOI where it has one and through Crossref, DataCite, arXiv or the publisher’s page otherwise, and compares title, year, volume and pages; its report is stored next to it.

None of these checks touches the geometry the paper relies on, and none of them bears on Question 5.14.

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